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Characterisation of the Atomic Hydrogen beam for Precision Spectroscopy Experiments

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*“I’m smart enough now to know I’m stupid.
That’s progress.”*

*– Ryland Grace
Project Hail Mary*

Characterisation of the Atomic Hydrogen beam for Precision Spectroscopy Experiments

ABSTRACT

Due to its very simple structure, the Hydrogen atom has emerged as an ideal candidate for determining values of fundamental constants, since all QED terms for its energy levels can be calculated with great precision. The transition energies of atomic hydrogen are very sensitive to the Rydberg constant (R_∞) and the proton charge radius (r_p); thus, measuring two such transition frequencies yields precise values of these two fundamental constants. Additionally, comparison with a third transition serves as a test of bound-state QED theory itself [1]. To this end, the 1S-3S Hydrogen experiment at MPQ [2] in Garching measures the precise value of the 1S-3S transition frequency with direct frequency-comb spectroscopy.

Since the 1S-3S measurement of hydrogen in Garching in 2020 [3], various modifications have been planned for the experiment to improve on the previous measurement. One of these improvements is the characterisation and stabilisation of our atomic hydrogen source. From the operational experience of all hydrogen experiments at MPQ, a reliable and stabilised source of atomic hydrogen, with a means to quantify the atomic hydrogen population independent of the spectroscopic laser systems, was deemed highly beneficial. In this effort, this thesis presents work on Optical Emission Spectroscopy (OES) of the hydrogen plasma source used in our experiments to produce atomic hydrogen. This method involves comparing the atomic and molecular hydrogen populations in the plasma by analysing their respective emissions in the collected plasma fluorescence, specifically the Balmer lines and the Fulcher Bands [4]. Due to a large scale of uncertainties in the plasma dynamics, as well as a lack of knowledge of the parameters of our plasma, a quantitative determination of the degree of dissociation was beyond the scope of this thesis. Hence, an informed estimate of the degree of dissociation was made, under certain assumptions, to qualitatively analyse the dissociation fraction in the plasma and investigate its dependence on different operational parameters. Additionally, for in situ detection of hydrogen near the spectroscopic region, a Calorimetric Wire Detector is being developed and analysed. The detector is a very thin tungsten wire that detects temperature changes caused by the heat of atomic hydrogen recombination on its surface, measured by changes in its resistance. This thesis will present preliminary results in the search of an atomic hydrogen signal in the wire resistance signal by estimating and decoupling the various heat loads affecting the wire. Eventually, a comparative measurement with both these methods will provide a way to confirm a proper operation of the hydrogen discharge for maximum atomic hydrogen population, as well as a way to measure the fraction of that population that reaches the spectroscopic region.

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Chapter 1

Precision Spectroscopy of Hydrogen

1.1 Motivation

The hydrogen atom, being a simple two-level system, has played a crucial role in the development of spectroscopy and modern quantum mechanics. A very nice interplay can be seen in the measurement of transitions and the development of quantum theory in the history of the hydrogen atom [5]. This dates back to 1852, when the first detailed measured spectra of atomic hydrogen were published by Angstrom, even before Thomson's discovery of the electron in 1897. Before the development of theory, further spectra were measured by Balmer in 1885 to provide an empirical formula for the discrete transitions, which was generalised by Rydberg in 1888 as,

$$\frac{1}{\lambda} = R_{\infty} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (1.1)$$

The Bohr model was the first to provide insight into the structure of the atom, explaining previously measured spectra. The Bohr model assumed a heavy, positively charged nucleus with light, negatively charged electrons orbiting it, bound by Coulombic interactions, with only stable orbits defined by discrete radii. These discrete orbitals were explained to be an effect of the angular momentum of the electron being only integer multiples of Planck's constant. This was the first instance of quantisation, which served as the basis for Schrödinger's description of a matter-wave in 1926, extending the simple structure of Balmer's formula from the principal quantum number n to the angular-momentum quantum numbers l and m . Further measurements of transitions and theoretical developments led to Dirac's theory, which explained the electron's spin-orbit coupling, lifting the degeneracy of the states defined by angular momentum and essentially giving rise to the fine structure. Further discoveries of the Lamb shift and hyperfine structure eventually led to the description of energy levels by the theory of Quantum electrodynamics (QED). The QED theory, though it has undergone many corrections to include higher-order perturbations, has stood the test of time for more than 70 years and is often referred to as the most accurate theory known. The energy levels described by QED today are represented as

$$E_{n,l,j} = hcR_{\infty} \left(-\frac{1}{n^2} + f_{n,l,j}(\alpha, \frac{m_e}{m_p}, \dots) + \delta_{l,0} \frac{C_{NS}}{n^3} \mathbf{r}_{\mathbf{p}}^2 \right) \quad (1.2)$$

where, n , l , j and the principal, orbital angular, and total angular momentum quantum number, $f_{n,l,j}$, is the QED series in the fine structure constant α , and the correction due to the root mean square proton charge radius r_p is represented separately in the last term, including the effect of spatial overlap of the s wave functions with the finite size of the proton. Detailed description of these formalisms can be found in [1].

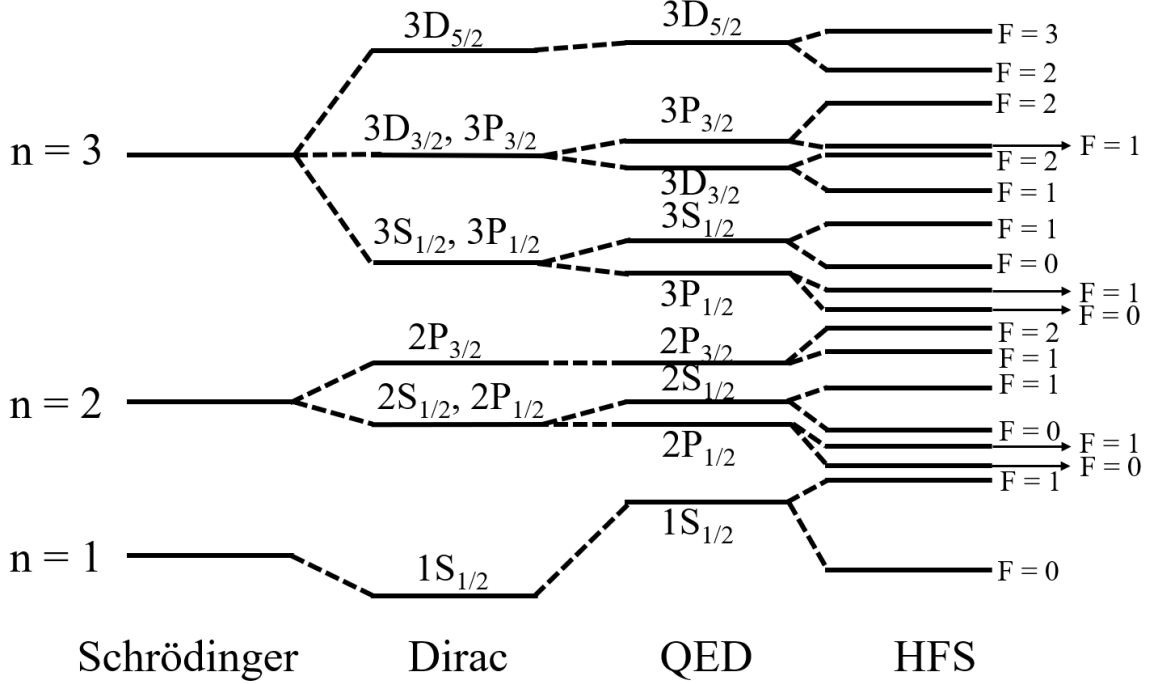


Figure 1.1: Energy levels of atomic hydrogen starting from the simple picture by Schrödinger, with only the principle quantum numbers n , to corrections of fine structure, lamb shift and hyperfine structure to the full QED description as of today. Figure adapted from [2]. Energy levels are not to scale.

In the above expression, whereas the fine-structure constant and the electron-to-proton mass ratio are more precisely determined by Penning-trap and atom-interferometer experiments ([6],[7]), respectively, spectroscopy remains the primary method for calculating the precise values of the Rydberg constant (R_∞) and proton charge radius (r_p). Measuring two transitions in atomic hydrogen gives us precise values for the R_∞ and the proton charge radius (r_p) [1]. Additionally, comparison with a third transition can serve as a test of QED itself. For hydrogen, due to the low lifetime of the metastable $2S$ state, the $1S$ - $2S$ transition [8] has the narrowest linewidth and the most precisely known transition frequency. Hence, different hydrogen transitions are compared with the most precise $1S$ - $2S$ frequency value to solve for R_∞ and r_p . The plot in Fig. 1.2 shows the calculated proton charge radius from the transition frequencies of different measured transitions, compared with that of the $1S$ - $2S$ transition. In 2016, Lamb shift measurement in muonic hydrogen [9] produced a value of the proton radius that was discrepant with the then

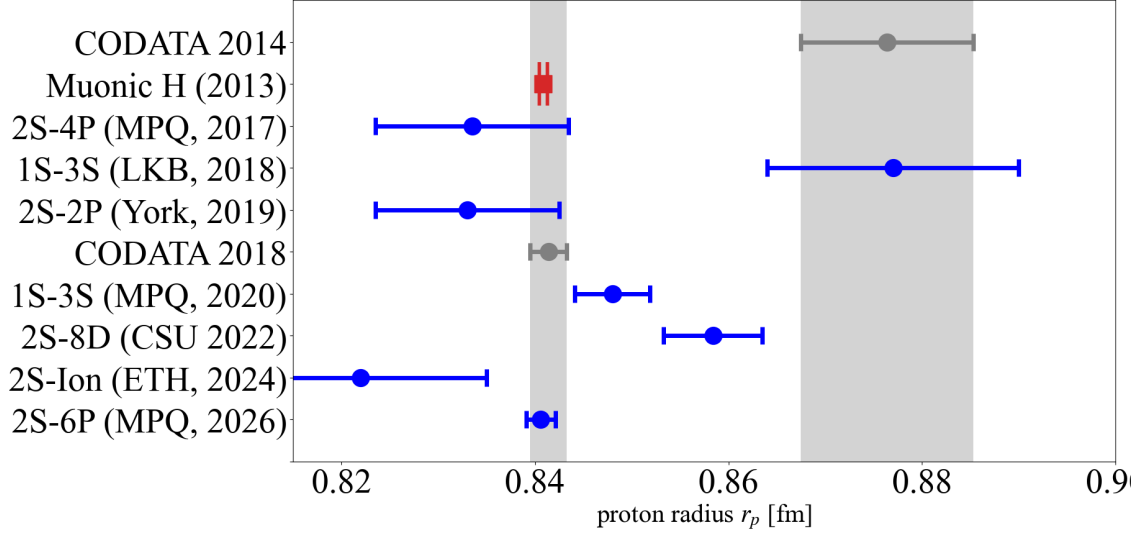


Figure 1.2: Plot showing the value of RMS proton charge radius when different transitions in atomic hydrogen are compared with the 1S-2S transition [8]. The red marker represents the muonic hydrogen measurement [9], which started the proton radius puzzle. Grey values represent the CODATA 2014 [10] and 2018 [1] measurements. Recent measurement of 2S-6P transition [11] solves the proton radius puzzle.

combined world data from both spectroscopy and charge radius determination from electron scattering, represented by the recommended value from CODATA 2014 [10], and hence, started the “proton radius puzzle”. This limited the test of QED to the level of experimental uncertainty. In efforts to resolve this puzzle or search for new physics, many groups have improved upon the previous value by producing results closer to the muonic hydrogen value. The result from the 1S-3S hydrogen spectroscopy experiment at MPQ [3] reported a transition frequency of

$$f_{1S-3S(F=1)} = 2922742936716.68(72) \text{ kHz} \quad (1.3)$$

with a proton radius in agreement with the muonic hydrogen value within 1.9σ . Additionally, the new CODATA 2018 [1] value also agrees with the muonic hydrogen value.

The recently published 2S-6P measurement of hydrogen at MPQ [11] provides a measurement of the proton radius, in excellent agreement with the muonic hydrogen value and with sufficient precision to perform a sub-parts-per-trillion test of the standard model. This measurement resolved the proton radius puzzle and, again, enables a test of QED with hydrogen spectroscopy.

Since the last measurement, several improvements have been planned for the 1S-3S experiment to enable a new, improved measurement. From experience with various hydrogen experiments at MPQ, the general conclusion was that a stable, reliable atomic hydrogen source and a method to quantify the atomic hydrogen population indepen-

dently of the laser systems would be beneficial. To this end, this thesis will present various methods tested to quantify the atomic hydrogen population.

1.2 1S-3S Hydrogen Spectroscopy at MPQ

In the aforementioned efforts, the 1S-3S hydrogen spectroscopy experiment at MPQ [3] measures the precise frequency of the 1S-3S transition using novel direct frequency comb spectroscopy [12]. For many years, the precision of spectroscopy of various optical transitions was constrained by limitations in frequency reference systems. This limitation was overcome in the 1990s with the discovery of frequency combs, which enabled the conversion of optical frequencies to radio frequencies, effectively providing an extremely accurate “frequency ruler.” Owing to their broad spectral coverage and narrow spacing between comb lines, frequency combs have become the standard for frequency calibration and precision spectroscopy. In recognition of this groundbreaking discovery, Theodor W. Hänsch and J. H. Hall were awarded the Nobel Prize in Physics in 2005 [13].

In the 1S-3S hydrogen spectroscopy experiment, a frequency comb is employed not only for referencing and final frequency measurement, but also to directly drive the transition. The 1S-3S transition is dipole-forbidden and therefore cannot be driven by a single photon; instead, two photons at 205 nm are required. Unlike a one-photon transition, which is driven only by a single resonant comb line, a two-photon transition with a frequency comb utilises all pairs of comb lines whose frequencies sum to the 1S-3S transition frequency (Fig. 1.3). As a result, the same excitation rate as with a continuous-wave (CW) laser of equivalent average power can be achieved using a two-photon frequency comb excitation. Since the AC Stark shift is dependent on the average power rather than the peak power of pulses, it remains the same as that of the CW case. Furthermore, the transition linewidth is determined by the linewidth of individual comb modes rather than the broad spectral envelope of the comb. Higher peak powers of frequency combs, compared to CW lasers, also enable more efficient nonlinear frequency conversion, thereby generating sufficient power at deep-UV wavelengths. In principle, a one-photon transition experiences a significant Doppler shift due to the relative motion of the pulses and atoms; however, this effect can be efficiently suppressed in two-photon transitions by using two counter-propagating photons for excitation. It is still essential to suppress the broad excitation signal from two co-propagating pulses to ensure the signal remains inherently Doppler-free. Additionally, in two-photon frequency comb spectroscopy, all available transitions within the broad spectrum of the comb are folded into a window half of the repetition rate ($f_{rep}/2$). Therefore, a specific repetition rate of the frequency comb used for excitation is required to avoid excitation of unintended transitions and to prevent distortion of the 1S-3S spectral line.

1.2.1 Laser System

The schematic of the laser system used for excitation is shown in Fig. 1.4. Our fundamental excitation laser is a Spectra Physics Tsunami mode-locked Ti:Sa laser with a

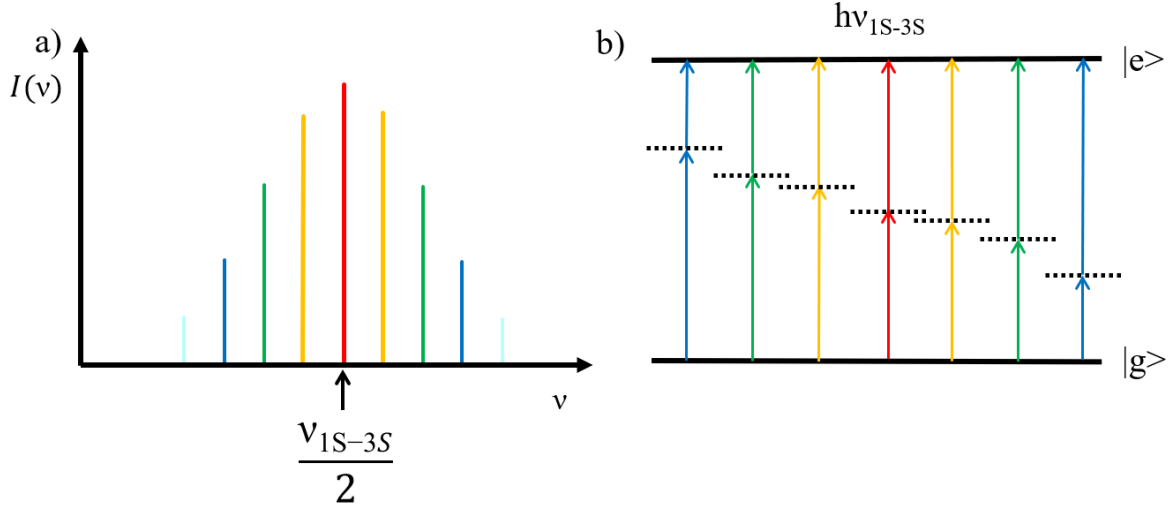


Figure 1.3: In two photon direct frequency comb spectroscopy, since one of the modes is set to half the frequency of the transition, all comb modes whose frequency pairwise adds up to the transition frequency contribute to excitation.

1.3 ps pulse length at 820 nm. The laser is locked to an Extended Cavity Diode Laser (ECDL), which is locked to an Ultra-Low Expansion (ULE) cavity, and the ECDL frequency is measured with a commercial fs-frequency comb [2]. The laser is mode-locked both actively via an AOM driven by a feedback signal from the laser output and by the Kerr lens in the Ti:Sa crystal. This 820 nm output from the comb is fed into the bow-tie Second Harmonic Generation (SHG) cavity with a BBO crystal to generate a 410 nm comb. The bow tie cavity output is stabilised to the laser by use of the Hänsch-Couillaud locking scheme [14]. This 410 nm comb is fed into the second SHG cavity with an LBO crystal, cooled to -40°C to maximise conversion efficiency. In this process, the 2.8 W at 820 nm gets converted to a 850 mW at 410 nm and eventually to 60 mW at 205 nm.¹ This 205 nm comb first passes through a delay line consisting of a half-wave plate and a PBS to split the light into two orthogonal polarisations, and an adjustable delay that can create an optical path length of $L = c/2f_{rep}$ to precisely insert the orthogonal split pulses between the transmitted pulses. This is finally fed through a quarter-wave plate to convert the polarisation from linear to circular, creating a train of σ^+/σ^- pulses, delayed precisely to meet twice per round trip at the centre of the enhancement cavity inside the vacuum chamber, which is labelled the pulse collision volume. The alternating

¹For the measurement in 2020, the fundamental Ti:Sa laser used was a commercial system at the repetition rate of 80 MHz, in contrast to the free spectral range of the SHG and the enhancement cavity of 160 MHz. This effectively doubled the repetition rate at the first SHG cavity, resulting in additional losses. To remove these losses and obtain more power for excitation, the repetition rate of the Tsunami laser was doubled to match the free spectral range of the SHGs. This major modification, along with the exchange of several components, rendered the system unreliable and led to the development of a new in-house mode-locked Ti:Sa laser with a 160 MHz repetition rate [15].

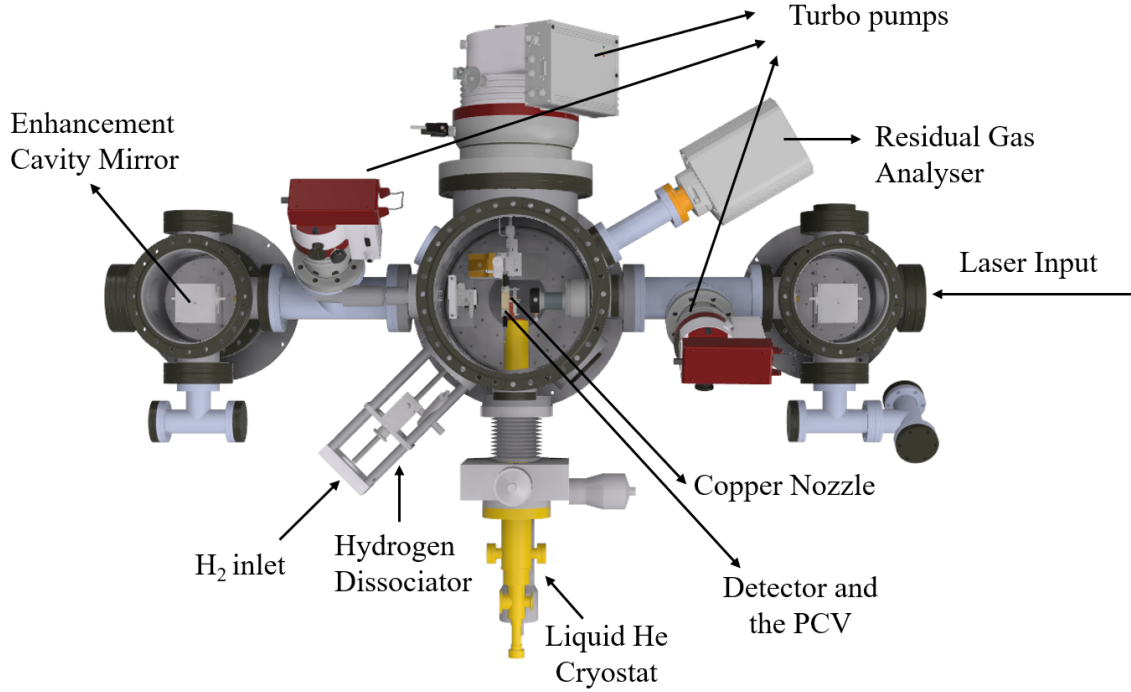


Figure 1.5: Schematic of the Spectroscopic Chamber of the 1S-3S Experiment. The chamber consists of 3 main regions: the left and right pods, which house the enhancement cavity mirrors, and the large central region, which contains the cryogenic nozzle with the incoming hydrogen, the PCV beside it, and the fluorescence detector. The chamber is differentially pumped; hence, the segments joining the two smaller pods to the central one contain orifices to maintain a lower pressure in the central chamber.

multimode fibres. This is the "Doppler-free"² detector, which collects the excitation from two counterpropagating photons. Additionally, a normalisation signal is obtained by collecting fluorescence from the opposite side of the cryogenic nozzle. This is the "Doppler-Broadened" detector since two copropagating photons essentially excite the transition outside the PCV. Additionally, the transition excited is 1S-3D, not 1S-3S, since the latter cannot be excited by two σ^+ (or σ^-) photons.

1.3 Steps towards characterisation of the Atomic Hydrogen beam

One of the improvements planned for the 1S-3S hydrogen spectroscopy experiment was the characterisation of the plasma discharge to ensure a reliably high atomic hydrogen

²This signal is not completely free of first order doppler shift; the chirped laser pulses cause a residual frequency shift due to the divergence of the atomic beam [16].

population and the development of a way to quantify and detect this population. Essentially, once the 1S-3S transition is excited, the measured count rate provides the best measure of the atomic hydrogen population. However, in all hydrogen experiments at MPQ, it was observed that a reliable way to confirm the good operation of the discharge, independent of the laser system, would be highly beneficial. Previously, in the 1S-3S spectroscopy experiment, efforts have been made to estimate the degree of dissociation using different methods [2]. Firstly, the H^+ signal from the background gas was analysed with a Residual Gas Analyser (RGA), and a comparison was made with and without the discharge while keeping the same input molecular hydrogen flux. Unfortunately, this proved to be a very unreliable signal because H^+ ions were produced by a bath of organic compounds in the chamber at the ioniser stage of the RGA, which could have overestimated the atomic hydrogen signal. An alternative to this was to observe the reduction in H_2^+ signal after turning on the discharge- a reduction of 5 % to 10 % was observed, which estimates the degree of dissociation to be about 5%. But due to physical constraints, the RGA was installed quite far from the hydrogen beam and not directly in the path, and hence much of the atomic hydrogen may have already recombined on its way to the RGA, likely leading to an underestimation of the atomic hydrogen signal. Another method used to estimate the degree of dissociation was to compare the temperature dependence of the measured count rates on resonance with Monte Carlo simulations assuming 100% dissociation. The discrepancy in the value was assumed to be attributable to the reduced degree of dissociation in reality. But the unknown temperature dependence of the recombination coefficient for atomic hydrogen on copper and Teflon makes it difficult to obtain a quantitative estimate.

This thesis mainly presents two methods tested in this effort. Chapter 2 entails the analysis of the hydrogen discharge with Optical Emission Spectroscopy of the collected plasma fluorescence. This method, described in [17], is a non-invasive tool for comparing the populations of atomic and molecular hydrogen in the plasma by comparing the intensities of their respective emissions of the Balmer- γ line and the Fulcher- α band. For this, Sec. 2.1 gives a description of our hydrogen plasma discharge used for dissociating molecules. Sections 2.2 and 2.3 provide a brief introduction to molecular physics and plasma dynamics concepts. Sections 2.4 and 2.5 explain the analyses required to extract accurate rate coefficients and intensities, respectively. This will also explain the limitations to a quantitative analysis of the plasma, arising from uncertainties in both intensities and rate coefficients, and will motivate a qualitative optimisation, with trials of such optimisation across different operational parameters given in Sec. 2.6.

Chap 3 will introduce the in situ measurement device for atomic hydrogen- the Calorimetric Wire Detector, based on resistance change of a wire from recombinative energy release. Sec. 3.4 provides a detailed description of all loads on the wire and a description of the simulations followed in Sec. 3.5. Lastly, Sections 3.6 and 3.7 present measurements and analysis that conclusively demonstrate the atomic hydrogen signal.

It is important to note that these efforts began with an analysis of the discharge setup in the main spectroscopy chamber and led to the development of a new vacuum chamber setup for effective parameter control and analysis of the wire detector. Additionally, after realising the need for a lower chamber pressure for the wire detector analysis, the

test setup vacuum chamber was replaced to achieve a lower pressure. All data will include the important operational parameters at which it was taken; however, most of the discharge and chamber parameters were kept identical across all three setups, except for the background vacuum pressure improvement from the first test setup to the second.

Chapter 2

Optical Emission Spectroscopy of plasma

To measure the atomic hydrogen population produced in the first step at the dissociator, the non-invasive method of optical emission spectroscopy was used to correlate plasma emissions with species populations. The populations of atomic and molecular hydrogen, which are of interest in this thesis, are related to their respective emissions of Balmer lines and Fulcher band, with a direct proportionality between them given by the proportionality constants- the rate coefficients. Additionally, observing emissions, especially in the visible spectrum, is a good way to detect impurities in the plasma through their characteristic lines and to confirm proper discharge operation. Since the first introduction of optical emission spectroscopy of plasma for calculating the degree of dissociation in [17], different population models and formalisms have been proposed to better understand plasma dynamics and achieve greater accuracy in the rate coefficients. To this end, the next sections will first provide a description of our hydrogen dissociator, then give a general introduction to the plasma physics formalism for low-temperature hydrogen plasmas and a description of the population model used for our hydrogen plasma.

2.1 The Hydrogen Dissociator

As mentioned before, the atomic hydrogen beam for the spectroscopy is derived from molecular hydrogen by igniting a hydrogen plasma. Fig. 2.1 shows the schematic of the discharge. The H_2 gas is fed into a Pyrex discharge tube with a dosing valve, on which a Evenson¹ Microwave (MW) cavity from Ophos Instruments is installed. With 30 W of 2.45 GHz Microwave applied to the cavity, a few electrons are produced by a brief spark from a Tesla coil applied to the discharge tube, which, when accelerated inside the microwave, start dissociating molecular hydrogen in an avalanche process, creating hydrogen plasma. The reaction dissociating the hydrogen looks as follows:



The discharge was typically operated at a hydrogen flow rate of 0.37 mln/min, measured with a Bronkhorst EL-Flow Select Mass Flowmeter before the dosing valve, and at a

¹Cavity 5 from [18].

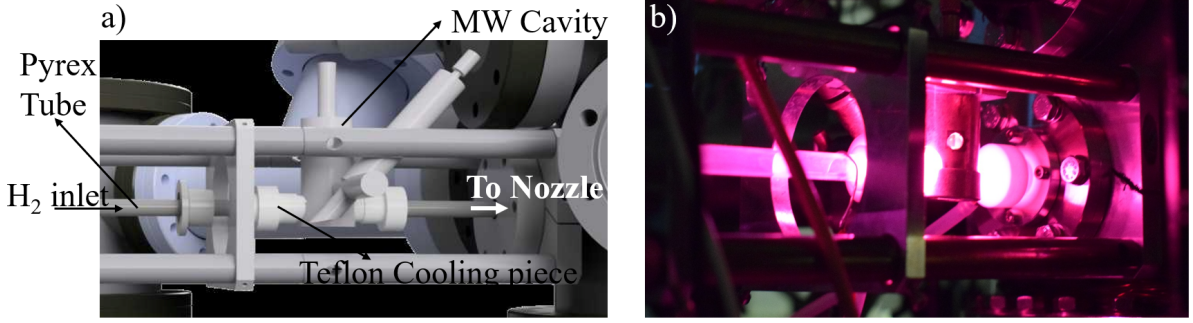


Figure 2.1: a) Schematic of the hydrogen dissociator used in all hydrogen experiments at MPQ. Hydrogen is dissociated in a Pyrex tube using a 30 W MW Cavity at 2.45 GHz. b) Bright pink glow of the Hydrogen discharge when in good operation, characteristic of the Balmer- α emission.

pressure of 1-3 mbar inside the discharge tube, measured with an MKS 925 micropirani gauge. The cavity was precisely impedance-matched after ignition using tuning stubs on the cavity to achieve less than 1% of the incident power reflected. The discharge tube has an orifice of 200-400 μm right after the cavity for developing enough pressure in the discharge tube to sustain the plasma. The discharge tube was actively cooled by circulating air cooled with liquid nitrogen around it via the Teflon cooling piece, as shown in Fig. 2.1. The temperature of the discharge tube was assumed to be that of the outgoing cooling air, which had thermalised with the tube, and was therefore measured using a temperature sensor AD590 hanging in the outgoing cooling air. The flow of this cooling air was controlled by a valve, which helped stabilise the discharge temperature. For a good magenta glow of the plasma, the temperature of the discharge tube was stabilised to -30°C to -10°C [19]. A Teflon tube right after the orifice carries the dissociated hydrogen to the cryogenic copper nozzle inside the vacuum chamber, from which the atomic hydrogen beam is formed at the PCV.

The dominant factor that determined the presence of a stable atomic hydrogen beam was the plasma discharge itself. A good operation of the discharge required handling a wide range of coupled parameters, namely the applied MW power, impedance matching of the MW cavity, pressure and flow of hydrogen in the discharge tube, temperature of the discharge tube, etc. The only metric used to assess the discharge's proper operation was the characteristic magenta plasma glow (from Balmer- α emission), indicating a high concentration of atomic hydrogen, but no systematic method was used to quantify it. The degree of dissociation in the plasma was previously assumed to be 90%, assuming the discharge operation to be similar to that in [20], but the two discharge configurations are quite different to be directly compared. Also, controlling all these parameters simultaneously to reproducibly achieve a sufficiently high degree of dissociation was a source of inefficiency. Additionally, physical factors such as the material and diameter of the discharge tube, the orifice size, the Teflon tube geometry and temperature, the nozzle temperature, and recombination rates on each surface directly influenced the total population of atomic hydrogen reaching the Pulse Collision Volume. Hence, there

was a need to analyse the dependence of the parameters with a quantifiable method and optimise them.

2.2 Introduction to Hydrogen Plasmas

This section will introduce plasma and molecular physics concepts and terminology required to understand the analysis of hydrogen plasma emission spectra for population retrieval. In principle, a plasma is characterised by the temperature and the densities of the species (atoms/ions) in the plasma, most important of which is the electron temperature (T_e) and the electron density (n_e). Most laboratory plasmas, especially gas discharges, are assumed to be low-temperature, low-pressure plasmas with hydrogen pressures of 0.001 to 10 mbar in the plasma region. As a rule of thumb, due to quasi-neutrality of low temperature plasmas, the ion density can be considered to be equal to the electron density $n_i = n_e \ll n_n$, and due to mass ratios, the electron temperature is much higher than the temperature of ions and neutral particles ($T_n \approx T_i \ll T_e$) [21]. Low temperature-low pressure plasmas are usually singly ionised, since the mean energy of electrons in such plasmas is lower than the ionisation threshold of the neutral species. For our purposes, we will mostly be concerned with such low-temperature, low-density plasmas, meaning that $10^{17} \text{ m}^{-3} < n_e < 10^{24} \text{ m}^{-3}$ and $1 \text{ eV} < T_e < 10 \text{ eV}$ ². In these, due to the higher frequency of microwaves, MW discharges generally have higher electron temperatures than RF discharges [21].

2.2.1 Optical emissions from a low temperature, low pressure Hydrogen Plasma

As shown in Fig. 2.1, our MW hydrogen discharge glows bright magenta colour during good operation, predominantly due to the atomic hydrogen Balmer- α emission. The most important emissions of atomic hydrogen in the plasma are the Balmer emission lines, given as: H_α at 656.28 nm (pink), H_β at 486.14 nm (cyan), H_γ at 434.05 nm (blue), H_δ at 410.17 nm (violet), and so on. The counterpart emission of the molecular hydrogen population in the plasma is the Fulcher- α band, spanning 520 nm to 770 nm, which arises from vibrational and rotational transitions of the hydrogen molecule. These emissions can be easily observed from the fluorescence of the plasma using a wide-range spectrometer. Fig. 2.3 shows an example of such an emission spectrum from our MW discharge recorded by the Thorlabs CCS200 wide range - low resolution spectrometer, where the first few Balmer lines and the Fulcher Band can be clearly identified. The spectrum was recorded by simply holding a multimode high-NA fibre very close to the plasma emission, and the light was then carried to the handheld spectrometer. To better understand these transitions, the following section briefly describes the molecular hydrogen energy-level diagram and corresponding terminology.

²For reference, 1 eV corresponds to a temperature of about 11,604.5 K.

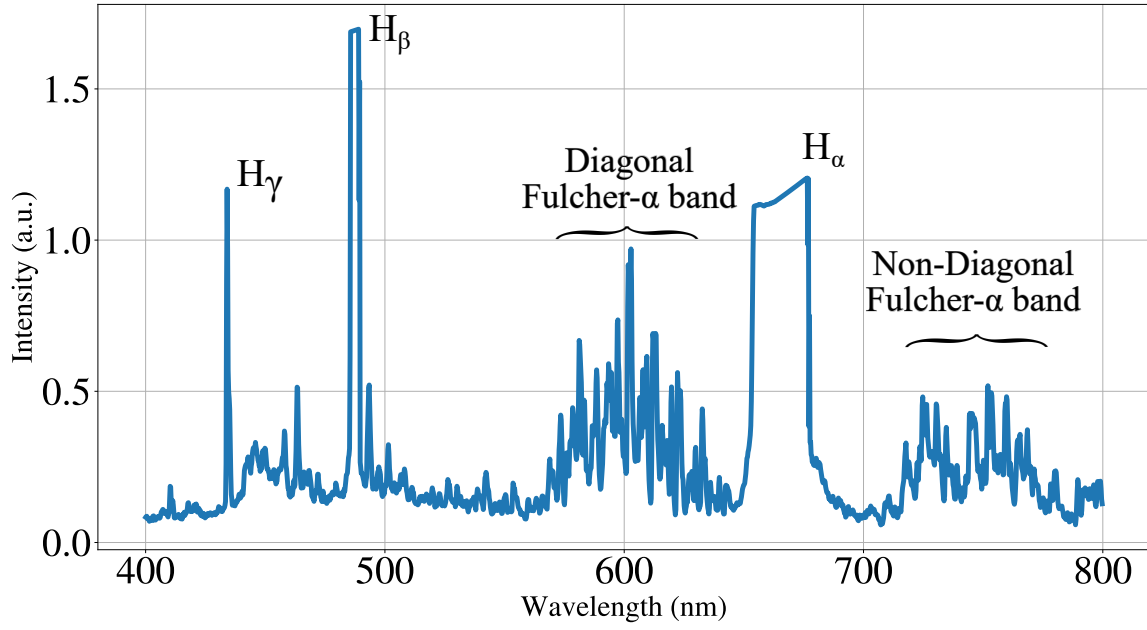


Figure 2.2: An example spectrum of the plasma fluorescence with the wide-range spectrometer Thorlabs CCS200, from simply collecting the fluorescence with multimode $NA = 0.6$ fibre. The Spectrum shows the atomic hydrogen Balmer- α , β and γ emissions and the molecular hydrogen Fulcher- α band. The Balmer- α and β lines have to be highly saturated in the spectrometer to see the Balmer- γ and Fulcher bands. A spectrum at very low dissociation was chosen to have a much stronger Fulcher band for representative purposes.

2.2.2 The Fulcher Band

Molecular Terminologies

Similar to the notation used for atomic electronic levels ($^{2S+1}L_{L+S}$), an electronic level for a diatomic molecule is given by: $^{2S+1}\Lambda_{g,u}^{+,-}$, where the $2S+1$ corresponds to the electronic spin multiplicity of the state (1 or 3 for singlet or triplet state), '+' and '-' represent the reflection symmetry of the total electronic wavefunction about the plane containing the internuclear axis³, and 'g' and 'u' describe the inversion symmetry (even or odd parity) of the spatial electronic wavefunction about the centre of the molecule. The symbol Λ denotes the projection of the coupled electronic orbital angular momentum L on the internuclear axis of the diatomic molecule, denoted by $\Sigma, \Pi, \Delta, \dots$ in the state's notation corresponding to projection values 0, 1, 2, ... (analogous to s, p, d...). In addition, the ground state is labelled with a Roman letter (X for hydrogen), higher singlet states with an uppercase letter (A, B, C...), and higher triplet states with a lowercase letter (a, b, c...). A vibrational level within the electronic level is denoted by v , and a rotational level within a particular vibrational level is denoted by J .

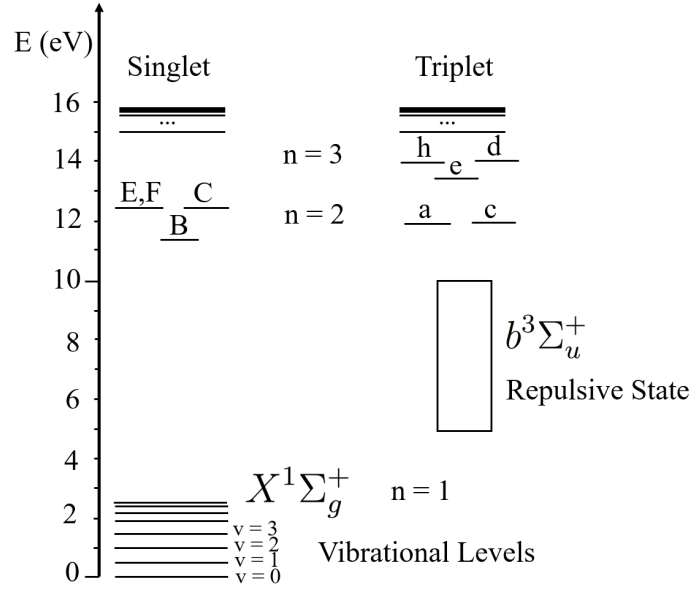


Figure 2.3: Energy level diagram of molecular hydrogen, with singlet and triplet state system. Figure adapted from [4].

Examples of states mainly relevant for the analysis in this thesis are, $X^1\Sigma_g^+$: singlet ground state, $\Lambda = 0$, symmetric with even parity. $d^3\Pi_u$: triplet excited state, $\Lambda = 1$, odd parity, $a^3\Sigma_g^+$: triplet excited state, $\Lambda = 0$, symmetric with even parity. In a hydrogen

³The reflection symmetry meaning of the '+' and '-' symbols is only valid for the Σ states, with a zero orbital angular momentum projection along the internuclear axis. When speaking about the $\Lambda \neq 0$ states ($\Pi, \Delta, \Phi, \dots$), the '+' and '-' represent the splitting of the state due the coupling of the electronic motion and rotational (nuclear) motion, labeled as " Λ -Doubling" [22].

plasma, $X^1\Sigma_g^+$ to $d^3\Pi_u$ transition, since not spin preserving, is optically forbidden, and hence occurs via collisional excitation, which in turn leads to the spontaneous emission from the ro-vibrational levels of $d^3\Pi_u^{+/-}$ to the ro-vibrational levels $a^3\Sigma_g^+$. These transitions are referred to as the Fulcher band⁴. Henceforth, an X state corresponds to the $X^1\Sigma_g^+$ ground state, any mention of Π or d^3 state corresponds to $d^3\Pi_u$ and any mention of Σ or a^3 state corresponds to $a^3\Sigma_g^+$, and all irrelevant subscripts and superscripts will be omitted.

Transition Probabilities of the Fulcher emissions

The Fulcher transitions from the rotational state J' of $d^3\Pi_u$ to rotational state J'' of $a^3\Sigma_g^+$ are split into in 3 branches according to $\Delta J = J' - J''$ [22]

- $\Delta J = -1$: P branch
- $\Delta J = 0$: Q branch ($J = 0$ to $J = 0$ is forbidden)
- $\Delta J = +1$: R branch

For each of these transitions, the emission intensity is proportional to the population in the initial rotational state via the transition probability. Within the Born-Oppenheimer approximation⁵, the line strengths of these transitions are give by the Hönl-London Factors.⁶ This factor essentially gives the distribution of transition probabilities across different rotational branches. The Hönl-London factors based on ΔJ are given by,

Branch	$S_{J'}$
P ($\Delta J = -1$)	$\frac{J'}{4}$
Q ($\Delta J = 0$)	$\frac{2J' + 1}{4}$
R ($\Delta J = 1$)	$\frac{J' + 1}{4}$

Table 2.1: Hönl-London factors for the P, Q, and R branches of the Fulcher band.

⁴It is important to note that, since H_2 is an unpolar molecule, an electronic transition has to happen when ro-vib state of the molecule changes, since, ro-vib states' energy difference is very small to create a sufficient transition dipole moment for EM radiation to interact (when no permanent dipole moment exists). Hence, each Fulcher emission is strictly a transition between the two electronic levels and not a transition between the ro-vib levels within the same electronic level, and hence is a band in the visible wavelength range.

⁵Born-Oppenheimer approximation assumes no coupling of electronic and nuclear motion in a molecule and hence, in the dipole matrix element for electronic transition, we can separate the rotational and electronic wavefunctions [22].

⁶The square modulus of the overlap integral of the rotational parts of the wavefunctions of the respective states in an electronic transition is called the Hönl-London factor [22].

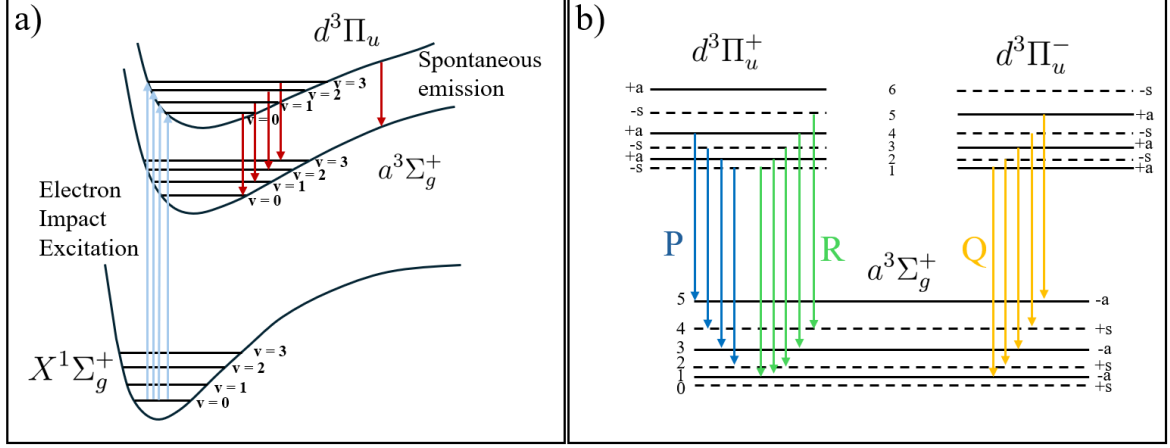


Figure 2.4: a) Energy level diagram of Molecular hydrogen, mainly showing the singlet ground state $X^1\Sigma_g^+$, and the two triplet states $d^3\Pi_u$ and $a^3\Sigma_g^+$. Electron impact excitations from the $X^1\Sigma_g^+$ state populate the $d^3\Pi_u$ state, from which the radiative decay to $a^3\Sigma_g^+$ is called the Fulcher- α band. b) Emissions from the Fulcher band split into P, Q and R lines depending upon ΔJ , and the coupling of rotational branches with the states split by "Λ-Doubling" and the parity [23]. Figure adapted from [24].

and hence, the transition probability of each of these transitions in the Fulcher band is proportional to the Hönl-London factor of the branch, normalised by the statistical weight

$$A_{J'J''} \propto \frac{S'_{J'}}{2J' + 1} \quad (2.1)$$

As we can see from the Hönl-London factors, the transition probability of the Q branch emission for a particular J' in a particular vibrational band v is the highest and hence, the Q lines are dominant emissions in the Fulcher band. In addition to the theoretical line strength, two weighting factors have to be considered with the transition probability: namely the statistical weight of the rotational state J' , given by, $(2J' + 1)$, and specifically for hydrogen: the nuclear spin degeneracy factor ($\gamma'_{J'} = 2I + 1$) of the initial state J' of the transition. Since atomic H nuclei are fermions with spin $1/2$, we get two variants in molecular hydrogen, $I = 1$ called the ortho hydrogen, and $I = 0$ called the para hydrogen, from the parallel and anti-parallel nuclear spin alignments. For the total molecular wave function to be antisymmetric, the odd rotational quantum numbers in $d^3\Pi_u^-$ are occupied by ortho hydrogen and even by para hydrogen (and vice versa for $d^3\Pi_u^+$, See Table 2.2.). This creates an alternating intensity ratio of 3:1 in a particular rotational branch, due to the corresponding abundance ratio of ortho to para hydrogen at room temperature. A concise summary of the relevant energy levels, transitions and couplings in the H_2 Fulcher band can be found in [23], and a detailed quantum mechanical picture of the same under the Hund's Coupling Case b for diatomic

molecules can be found in [22].

	ortho H_2 ($I = 1$) (sym.)	para H_2 ($I = 0$) (antisym.)
$d^3\Pi_u^-$ (Q branch)	odd J	Even J
$d^3\Pi_u^+$ (P and R branch)	Even J	odd J

Table 2.2: Coupling between nuclear degeneracy and rotational branches

And hence, the intensity of a particular rotational transition in the Fulcher band becomes

$$I_{J'J''} \propto n_{J'} \gamma_{J'} (2J' + 1) A_{J'J''} = n_{J'} \gamma_{J'} S_{J'} \quad (2.2)$$

2.2.3 Emission Intensities to Populations

To analyse the fraction of ground state molecular hydrogen dissociated to ground state atomic hydrogen via optical emission spectroscopy, the atomic Balmer- γ emission and the molecular Fulcher- α emissions are compared. These emissions are commonly used to correlate populations because they have comparable intensities and emission rates, which are least affected by changes in plasma dynamics. Essentially, a simple proportionality exists between the emission intensity and the population, with the proportionality constant given by the effective rate coefficients.

For the Balmer- γ emission, this translates to,

$$I_\gamma \propto n_H(n=1) \cdot X_\gamma^{eff} \quad (2.3)$$

whereas, for the Fulcher emissions,

$$I_{Fulcher} \propto n_{H_2}(X^1\Sigma_g^+) \cdot X_{Fulcher}^{eff} \quad (2.4)$$

where, X_γ^{eff} and $X_{Fulcher}^{eff}$ are the effective rate coefficients [17]. It is important to note that, in contrast to Eq. 2.2, here we correlate to the population of the ground state of the atom/molecule and not that of the initial state of transition, hence the rate coefficients are not only the transition probabilities of the emissions, but also include the plasma dynamics that populate the states.

Taking the ratio of the two, we get the ratio of populations from the ratio of intensities,

$$\frac{I_\gamma}{I_{Fulcher}} = \frac{n_H(n=1)}{n_{H_2}(X^1\Sigma_g^+)} \frac{X_\gamma^{eff}}{X_{Fulcher}^{eff}} \quad (2.5)$$

and hence the Degree of Dissociation α , is given by

$$\alpha = \frac{0.5n_H/n_{H_2}}{1 + 0.5n_H/n_{H_2}} \quad (2.6)$$

This has been a established way of investigating the populations in a low pressure low temperature hydrogen plasma [4], which started out as a simple method for measuring the dissociation fraction, but advancements in plasma population models have since made modifications to the plasma description which required further analysis for obtaining the degree of dissociation.

2.3 Plasma dynamics

In general, there are two classes of transitions occurring inside a plasma, namely, radiative and collisional transitions. The probability of radiative transitions is simply given by Einstein's transition probability (A in s^{-1}), and the rate of collisional transitions from electron impact is given by rate coefficients (generally X in units m^3s^{-1}). These transitions are further divided into excitation, recombination and ionisation processes. Due to the higher density of the neutral particles inside a low-temperature, low-pressure plasma, most of them thermalise with each other with elastic collisions, and hence the energy distribution of the neutral particles can be considered to be Maxwellian, and a thermal Boltzmann population distribution. Since, in first approximation, the neutral particles and ions in the plasma have similar mass, which is much larger compared to the electron mass, the temperature of all heavy particle species can be considered to be the same, which is usually slightly above room temperature, and much lower than that of electrons ($T_H \approx T_{H_2} \approx T_i \approx 300 - 1000K \ll T_e$) [21].

The rate coefficient for electron-impact excitation is the rate of collisions with electrons at a given electron density. We only consider binary collisions in a plasma with low enough n_e . Hence, the rate coefficients are obtained by convolution of the collisional cross-section $\sigma(v)$ and velocity v , over the complete velocity distribution $f_T(v)$ of electrons, which is assumed to be Maxwellian [25]

$$\begin{aligned} X(T_e) &= \langle v\sigma(v) \rangle_{T_e} = \int d^3v f_T(v) v\sigma(v) \\ &= \int_{E_{thr}}^{\infty} dE f(E) (2/m_e)^{1/2} \sqrt{E} \sigma(E) \end{aligned} \quad (2.7)$$

To extract population densities from emission intensities, theoretical population models are used to determine the population densities of different species in different states based on plasma parameters. For the most general description, the population density of an excited state (n) of a plasma species can be determined from the rate equation of all transitions that affect the population, which includes summing over the product of all sink or source reactions using the rate coefficient for the particular reaction [26]:

$$\frac{dn}{dt} = \sum_{sources} [(\prod_{\alpha} n_{\alpha}) \cdot X_{sources}] - \sum_{sinks} [(\prod_{\beta} n_{\beta}) \cdot X_{sinks}] \quad (2.8)$$

where α and β represent each species participating in each source and sink transition.

Various population models exist depending on the electron temperature and density regime to which the plasma belongs. We are primarily concerned with low- to intermediate-electron-density plasmas, for which we use the Collisional Radiative (CR) Model.

2.3.1 Yet Another Collisional Radiative Model (Yacora)

Yacora is a collisional radiative model for the "partially recombining" plasmas (i.e., typically, $1 \text{ eV} \leq T_e \leq 10 \text{ eV}$) developed by Max Planck Institute for Plasma Physics (IPP), which solves for the excited state populations for a given set of plasma parameters. This model is available for atomic H, molecular H, and He, and is directly accessible on the web for plasma diagnostic purposes [27].

To determine the excited-state population of a species in the CR model, Yacora essentially solves rate equations for the population density, using the source and sink populations and their corresponding rate coefficients 2.8, for a particular channel. The excited states of any species are mainly populated by electron collisions, while the ground state is mainly populated from heavy particle collisions, since the lifetimes of the former are much shorter than (ns) compared to those of the latter (μs or ms). Hence, on the timescales of excited-state lifetimes, the ground-state populations can be considered virtually constant. For each such quasi-constant population, a "channel" is defined. A channel consists of all processes that eventually originated from the same quasi-constant species. E.g., all electron-impact excitation, electron-impact de-excitation, and spontaneous emission from an excited state p of the Hydrogen atom, started by an excitation from the ground state of atomic H; hence, this is called the "Direct Excitation channel", and its quasi-constant species is the ground state of H. This way, a steady-state solution is found by treating the ground-state populations as quasi-constant and feeding them into the CR model. Essentially, this decouples each channel from the others, and we can calculate the contribution of a channel to the population of a particular excited state. A typical rate equation of an excited state p looks like the following:

$$\begin{aligned} \frac{dn_p}{dt} = & \underbrace{\sum_{q>p} A_{qp} n_q - \sum_{q<p} A_{pq} n_p}_{\text{Radiative Transitions}} + n_e \underbrace{\left(\sum_{q \neq p} X_{qp} n_q - \sum_{q \neq p} X_{pq} n_p \right)}_{\text{Electron Impact Transitions}} \\ & + \underbrace{n_e [(\alpha + \beta n_e) n_+]_{\text{Recombination}}}_{\text{Recombination}} - \underbrace{n_e (S_p n_p)}_{\text{Ionization}} \end{aligned} \quad (2.9)$$

where, first two terms are spontaneous emission processes to p from higher states (populating) and from p to lower states (depopulating); next two terms are electron

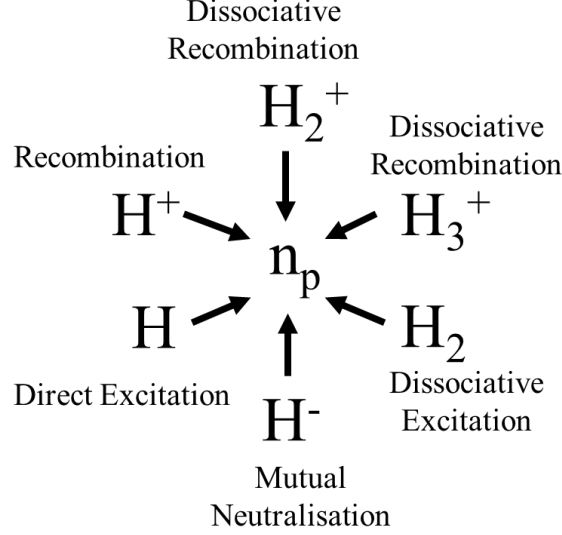


Figure 2.5: Channels included in the Yacora model for atomic hydrogen. Of these, we consider the atomic hydrogen excited states to be populated only by the Direct Excitation Channel. Figure adapted from [28]. A detailed description of the reactions in each channel can be found in Appendix B.2.

impact excitation to p and de-excitation from p ; α and β are the radiative and three body recombination rates when positive ion density is n_+ and the last term is ionization from state p .

In a typical low-pressure, low-temperature hydrogen plasma, a total of 6 different plasma species are present with quasi-constant densities, namely: $H, H^-, H_2, H_3^+, H_2^+, H^+$ [28], i.e., we can assume their population density to be constant on the general timescale of plasma excitation channels. The main excitation channels for atomic hydrogen are shown in the Fig. 2.5. For molecular Hydrogen, in addition to the direct excitation and ionisation channel, 3 additional channels can optionally be added to the CR model yacora: Quenching, Dissociative attachment, and Charge Exchange. A description of all the reactions can be found in Sec. 2.4.2, and additional information can be found in [28] and [25].

The steady state population density (n_p) from solving the rate equation for a particular channel (S) is expressed in terms of a "population coefficient" (R) for that channel (with a quasi-constant state density n_0) populating or depopulating a particular state (p):

$$R_{0p} = \frac{n_p}{n_e \cdot n_0} \quad (2.10)$$

Each population coefficient gives the excited-state population, normalised to the population of quasi-constant species (n_0) in that channel and to the electron density (n_e), thereby essentially removing the dependence on these parameters, at least to first order. Combining all such channels, the population density of state p turns out to be

$$n_p = n_e \cdot \sum_S R_{Sp} \cdot n_S \quad (2.11)$$

In Yacora, all aforementioned channels in the Fig. 2.5 are considered and can be solved separately to get the population coefficient. So, the excited state population becomes

$$n_p = n_e(R_{H_p}n_H + R_{H_p^+}n_{H^+} + R_{H_{2p}}n_{H_2} + R_{H_{2p}^+}n_{H_2^+} + R_{H_{3p}^+}n_{H_3^+} + R_{H_p^-}n_{H^-}) \quad (2.12)$$

and hence, the effective rate coefficient for a transition from state k to m becomes

$$X_{k \rightarrow m}^{eff} = \frac{n_k}{n_0 n_e} A_{km} = \frac{\sum_{s=1}^N n_s \cdot R_s(k)}{n_0} A_{km} \quad (2.13)$$

2.3.2 Degree of Dissociation

Now, for simplicity, we can assume our plasma discharge operates in the corona regime. Within the corona approximation, we assume that the electron impact excitation is balanced by the spontaneous emission. This means that only the direct excitation channel is assumed to populate the excited states of both molecular and atomic hydrogen when calculating rate coefficients. Implications of this assumption, as well as the effect of adding additional channels, will be discussed in Sec. 2.4.2 and Sec. 2.6.3.

Using population coefficients from Yacora, we can extract effective rate coefficients to correlate the intensity ratio with the population ratio. For the Balmer- γ line, considering the direct excitation (DE) of atomic H to be the only channel contributing to the population of $n = 5$, we get an expression relating the intensity of the gamma line to the population of the quasi-constant ground state population

$$\begin{aligned} I_\gamma &\propto \epsilon_\gamma = \mathbf{n}_H(n=5)A_\gamma \\ &= \mathbf{n}_H(n=1)n_e \underbrace{R^{DE}(n=5)A_\gamma}_{X_\gamma^{eff}} \\ &= \mathbf{n}_H(n=1)n_e X_\gamma^{eff} \end{aligned} \quad (2.14)$$

Similarly, for the Fulcher band, considering only the direct excitation channel from the ground state (X^1), the intensity of the Fulcher band can be given by

$$\begin{aligned} I_{Fulcher} &\propto \epsilon_{Fulcher} = n_{H_2}(d^3\Pi)A_{Fulcher}^{eff} \\ &= n_e n_{H_2}(X^1) \underbrace{R^{DE}(d^3\Pi)A_{Fulcher}^{eff}}_{X_{Fulcher}^{eff}} \\ &= n_e n_{H_2}(X^1) X_{Fulcher}^{eff} \end{aligned} \quad (2.15)$$

Taking the ratio of intensities, the dependence of n_e drops out and we can extract the ratio of populations of molecular and atomic hydrogen using the ratio of effective rate coefficients⁷ [17],

$$\frac{I_\gamma}{I_{Fulcher}} = \frac{n_H(n=1)}{n_{H_2}(X^1\Sigma_g^+)} \frac{X_\gamma^{eff}}{X_{Fulcher}^{eff}} \quad (2.16)$$

⁷For consistency, henceforth, all mentions of ratio of effective rate coefficients will always correspond to the ratio $X_\gamma^{eff}/X_{Fulcher}^{eff}$, and not the other way around.

and find the degree of dissociation (α) as

$$\alpha = \frac{0.5n_H/n_{H_2}}{1 + 0.5n_H/n_{H_2}} \quad (2.17)$$

The main challenge for this analysis was determining suitable values for the effective rate coefficient ratio and the Fulcher intensity to ensure at least consistent approximations, if not complete accuracy, throughout the formula.

2.4 The Effective rate coefficients

This method of calculating degree of dissociation hydrogen plasma started with use of constant value of the effective rate coefficients ratio $X_\gamma^{eff}/X_{Fulcher}^{eff} = 1.3$ in a low pressure low temperature hydrogen plasma, which was derived from the plot in Fig. 5 from [17], based on the plasma model ADAS [29]. But the approximations under which the value could be used were uncertain; hence, there was a need to better understand the dependence of rate coefficients on plasma parameters and to calculate their values for our particular case.

To calculate the effective rate coefficients, as mentioned in the previous section, we need to find the population coefficients for the direct excitation channel and the transition probabilities. For the Balmer- γ emission, this is very easily achieved since the transition probability of Balmer- γ is a known measured value $A_\gamma = 2.4551 \times 10^7 s^{-1}$ [30], and the population coefficient of the $n = 5$ state ($R_{n=5}$) can be obtained from Yacora on the web. For the molecular emissions, the population coefficient for the $d^3\Pi$ state can be obtained from Yacora. But since Yacora only provides population coefficients resolved at the electronic level, not at the vibrational level, an effective transition probability for the entire Fulcher band at the electronic level was used, obtained by averaging over the vibrational states [31].

2.4.1 Vibrational Population

Vibrational states of the $d^3\Pi_u$ state are very short-lived, so we cannot consider a Thermal Boltzmann distribution for its populations. In contrast, the electronic ground state (X^1) vibrational levels are very long-lived; hence, their population distribution can be expressed as a Thermal distribution with a $T_{vib}(X^1)$. Hence, a projection of the ground state vibrational population to the that of the $d^3\Pi_u$ state is carried out using the Franck Condon Factors, to be used in calculation of the effective rate coefficients ⁸ [32]. The Franck-Condon matrix determines how a population distribution changes during a transition from one electronic state to another. In contrast to [33], we ignore the change in population of $d^3\Pi_u$ vibrational states from the spontaneous emission to $a^3\Sigma_g^+$ vibrational levels (i.e. the Fulcher band transitions), assuming that it is a very small correction,

⁸Analogous to the Hönl-London factors for the Rotational transitions, the Franck Condon factors are the square modulus of the overlap integrals for vibrational transitional states, within the Born Oppenheimer approximation, see Appendix B.3.

and only include the Franck-Condon factors for the X^1 to $d^3\Pi_u$ transition. The Franck-Condon factors used were taken from [31], and are listed in Appendix B.3. Hence, by simple matrix multiplication, we get the vibrational population distribution of the $d^3\Pi$ state

$$n_{d^3\Pi_u}(v') = \sum_{v''} FC F_{v''v'}^{X^1 \rightarrow d^3\Pi} n_{X^1}(v'') \quad (2.18)$$

Hence, for a given $T_{vib}(X^1)$, we get a vibrational distribution of the ground state and hence through the Franck-Condon projection, of the $d^3\Pi_u$ state. Fig. 2.6 shows such a projection for different $T_{vib}(X^1)$.

Within the Fulcher transition, the transition probability from a particular vibrational state of $d^3\Pi_u$ can be calculated by summing over transition probabilities from the particular state to all vibrational states of the lower state $a^3\Sigma_g^+$

$$A_{v'}^{d^3\Pi \rightarrow a^3\Sigma} = \sum_{v''} A_{v' \rightarrow v''}^{d^3\Pi \rightarrow a^3\Sigma} \quad (2.19)$$

To get the effective transition probability from all states v' of $d^3\Pi_u$, the transition probabilities are summed, weighted over the normalised population of each vibrational state [31], calculated from the Franck-Condon projection.

$$A_{eff}^{d^3\Pi \rightarrow a^3\Sigma} = \sum_{v'} A_{v'}^{d^3\Pi \rightarrow a^3\Sigma} \cdot n_{v'}^{d^3\Pi}(T_{vib}(X^1));$$

$$\sum_{v'} n_{v'}^{d^3\Pi} = 1 \quad (2.20)$$

and hence, with this effective transition probability, we can get the effective rate coefficients for the whole Fulcher band as

$$X_{Fulcher}^{eff} = R_{d^3\Pi} \cdot A_{eff}^{d^3\Pi \rightarrow a^3\Sigma} \quad (2.21)$$

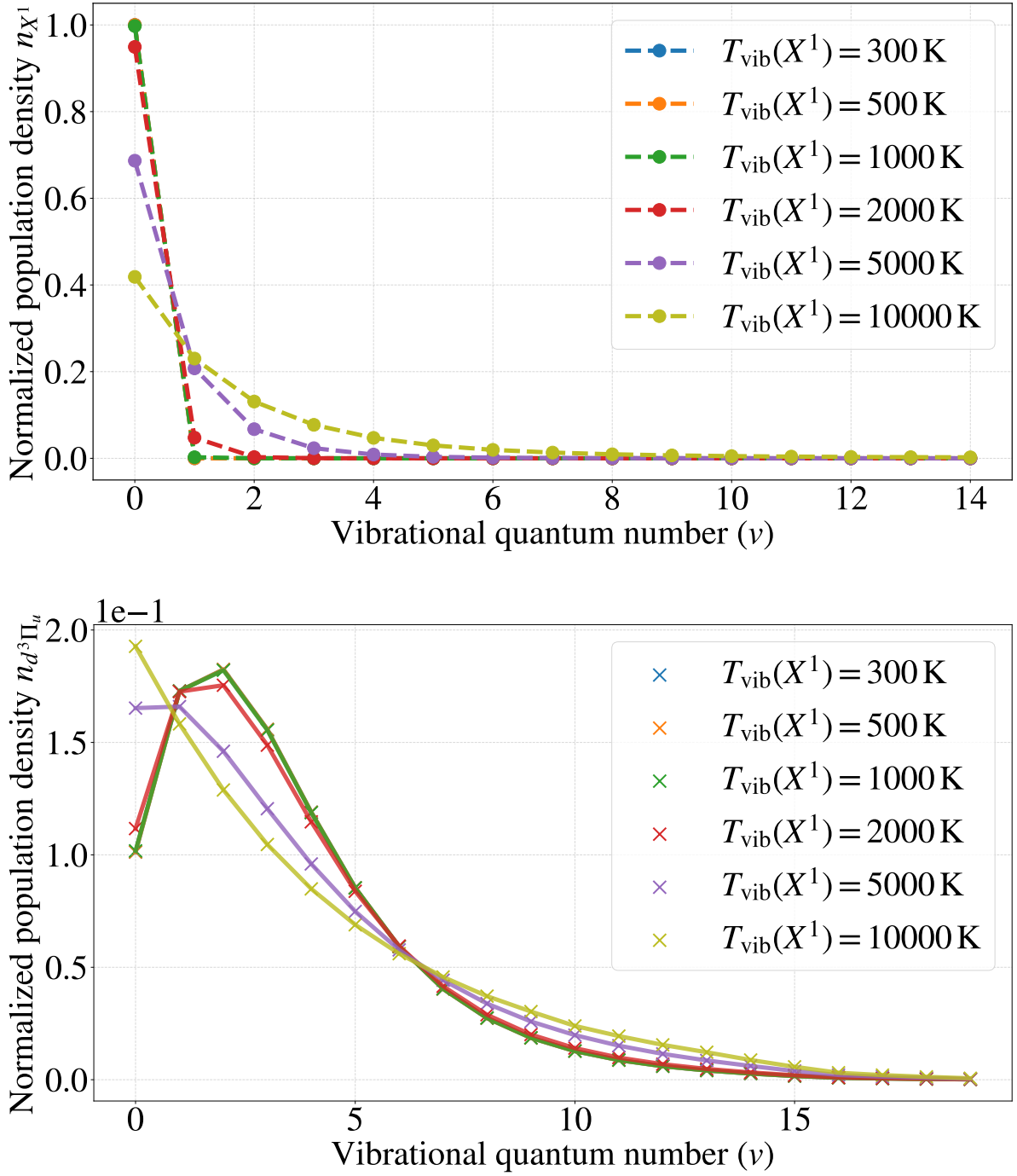


Figure 2.6: Franck Condon Projection of $n_{X^1}(v'')$ population distribution of a given $T_{vib}(X^1)$ in a) to $n_{d^3\Pi}(v')$ in b). On account of the Franck-Condon matrix being asymmetric, we can see that even though increasing $T_{vib}(X^1)$ increases the population of higher v levels of X^1 , it shifts the population distribution of $d^3\Pi$ towards lower v states.

2.4.2 Sources of uncertainty in the Effective Rate Coefficients

The largest source of uncertainty in the degree of dissociation was induced from the effective rate coefficients ratio ($X_{\gamma}^{eff}/X_{Fulcher}^{eff}$). Due to a lack of knowledge of plasma dynamics, we highly relied on the Yacora model for acquiring the population coefficients, but a lot of approximations had to be made to directly use Yacora for our plasma with ease. Firstly, the largest error arises from the assumption that only the direct excitation channel is dominant in determining the populations of the excited states of both atomic and molecular hydrogen. This, in fact, is not a good approximation since the rate coefficients from some of the other reactions involved in the plasma dynamics are comparable to that of the direct excitation from the ground state, for the expected range of plasma parameters that apply to our plasma⁹ [23]. In addition, from the definition of the population coefficients, they should be inherently (at least to first order) independent of the absolute quasi-static state population ($n_H(n = 1)$ and $n_{H_2}(X^1)$). While this held true for the direct excitation channel, it was no longer valid once the additional channels from the Yacora model for H_2 were included in the calculation. Fig. 2.7 shows the dependence of the rate coefficients ratio on the population of the quasi static state ($n(X^1\Sigma) = n_{H_2}$), depending on the additional channels added to the calculation, namely: Charge Exchange (C)¹⁰, Dissociative Attachment (D)¹¹ and the Quenching (Q)¹².

Additionally, extra channels that affect the atomic hydrogen population in plasma were also not included in calculations¹³. Most importantly, we expect the dominant correction to arise from the re-absorption of Lyman emission lines into the plasma, called "Radiation Trapping", leading to an increased population of the atomic excited states [34]. Hence, the opacity of the Lyman lines directly affects the Balmer emissions and their ratios, which is the probable cause for the discrepancy seen in 2.4.3. The effect of radiation trapping on Balmer- γ emission is considerably small compared to Balmer- α and Balmer- β , which is one of the reasons behind its use for diagnostics of the atomic hydrogen population, but nonetheless it was observed to be significant¹⁴.

⁹General observation from reaction-wise rate coefficients obtained from Yacora

¹⁰Charge exchange between the excited state of H_2 with positive ion of H, given as $H_2 + H^+ \rightarrow H_2^+ + H$

¹¹Unstable vibrationally excited H_2 collides with electrons, forms H_2^- , which being an unstable ion, dissociates to H and H^- , written as $H_2(v) + e^- \rightarrow H_2^- \rightarrow H + H^-$

¹²De-excitation from the triplet states a^3 and c^3 to ground state X^1 from heavy particle collisions is labeled as Quenching.

¹³Dissociative Attachment should also affect the atomic hydrogen population, but [34] shows this effect to contribute to a correction $< 10\%$, hence it was ignored. Additionally, the correction from additional channels of H_2 to Balmer emissions was shown $< 5\%$ in [34] and hence ignored.

¹⁴This discrepancy can potentially be resolved by use of correction factors for the Balmer- γ emission, including the effect of optically thick Lyman lines. But the unknown dependence of these correction factors on the unknowns T_e , n_e , and n_H in our plasma made this calculation infeasible.

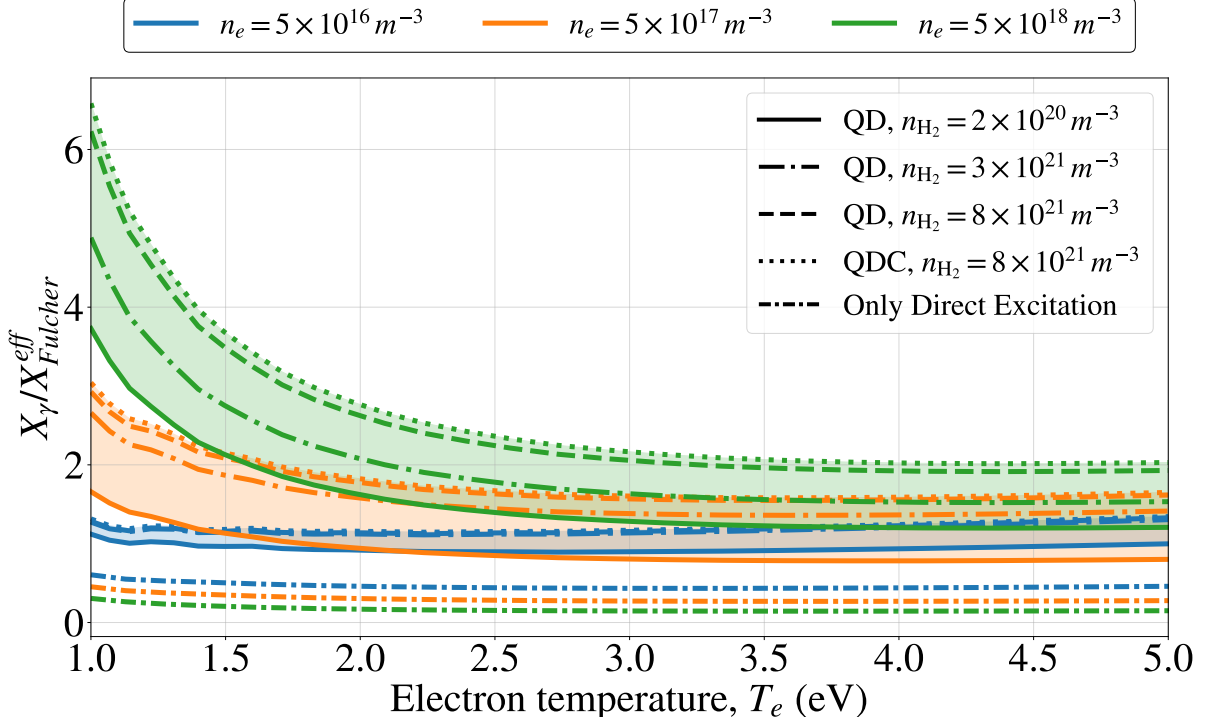


Figure 2.7: Plot depicting the uncertainties introduced to the effective rate coefficients ratio by inclusion of additional channels in the Yacora H_2 model. The bottom 3 lines show the ratio by including only the Direct Excitation channel for molecular hydrogen, at different electron densities and temperatures. Additionally, when Quenching (Q) and Dissociative Attachment (D) Channels are added, the ratio progressively increases with increasing electron density (in contrast to the case of Direct Excitation Channel) and additionally depends on the absolute value of molecular hydrogen population (n_{H_2}), and hence for representative purposes, the spread due to n_{H_2} is plotted in terms of 3 values of n_{H_2} (2×10^{20} , 3×10^{21} and $8 \times 10^{21} \text{ m}^{-3}$) corresponding to 3 values of degree of dissociation (95 %, 50 % and 5 %), keeping the n_H value constant. Note that the value does not correspond to the true value of n_{H_2} from our defined flow rate, but is only used for representation. All values were calculated with $T_{vib}(X^1) = 5000 \text{ K}$.

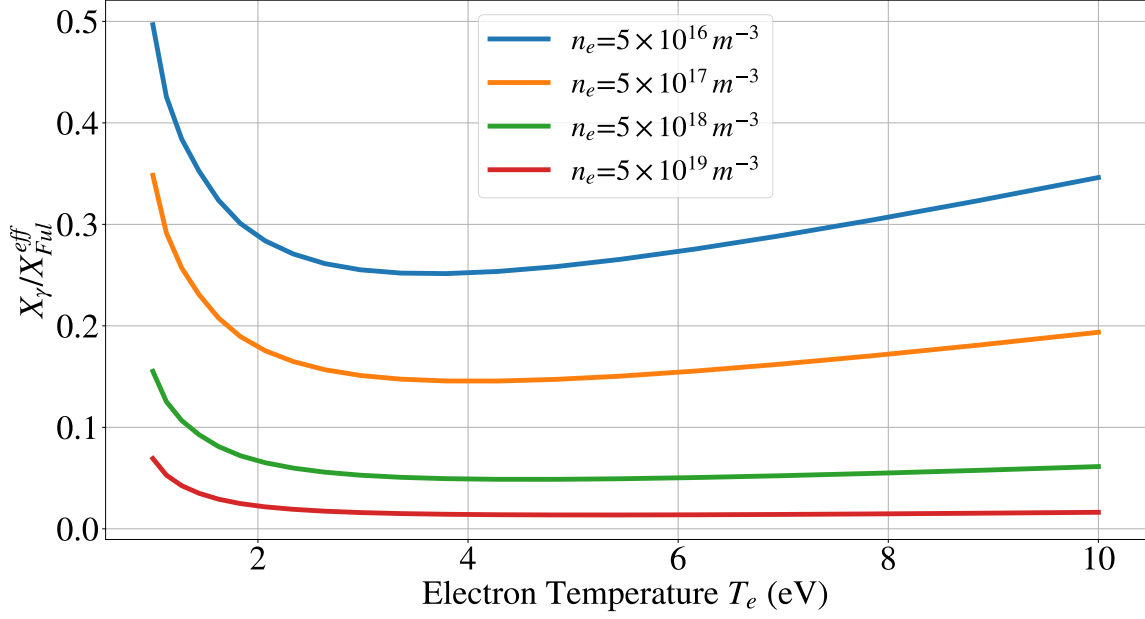


Figure 2.8: Dependence of the ratio of rate coefficients on the plasma parameters of electron temperature T_e and electron density n_e , when using a particular collisional cross-sectional database is in the molecular hydrogen Yacora model, here the MCCC Database [23]. Dummy values of $T_{H_2} = 500 \text{ K}$ and $T_{vib}(X^1) = 5000 \text{ K}$ used.

In addition to uncertainties in the validity of the Yacora model for our plasma, another uncertainty in the effective rate coefficients stems from the lack of knowledge of the plasma parameters for our discharge. Even in the approximation of the direct excitation channel, the ratio of rate coefficients still largely depends on the order of magnitude of the electron temperature and density in our plasma, as shown in Fig. 2.8. Additionally, in the Yacora model for H_2 , the population coefficients can be calculated using one of the three databases for the collisional cross sections- Miles [35] and Janev [36] and MCCC (Molecular Convergent Close-Coupling) [23] databases. As seen in Fig. 2.9, the choice of the cross-sectional database largely affected the value of the ratio of effective rate coefficients and, hence, the degree of dissociation.¹⁵

¹⁵Although Janev database consists of the more recent collisional cross sections based on measurements, [37] showed that the older database, Miles, led to a better agreement with the measured population densities for the d^3 state. The newly added vibrationally resolved MCCC cross sections are recommended by Yacora now, and hence are used in the calculation of the degree of dissociation in Sec. 2.6.

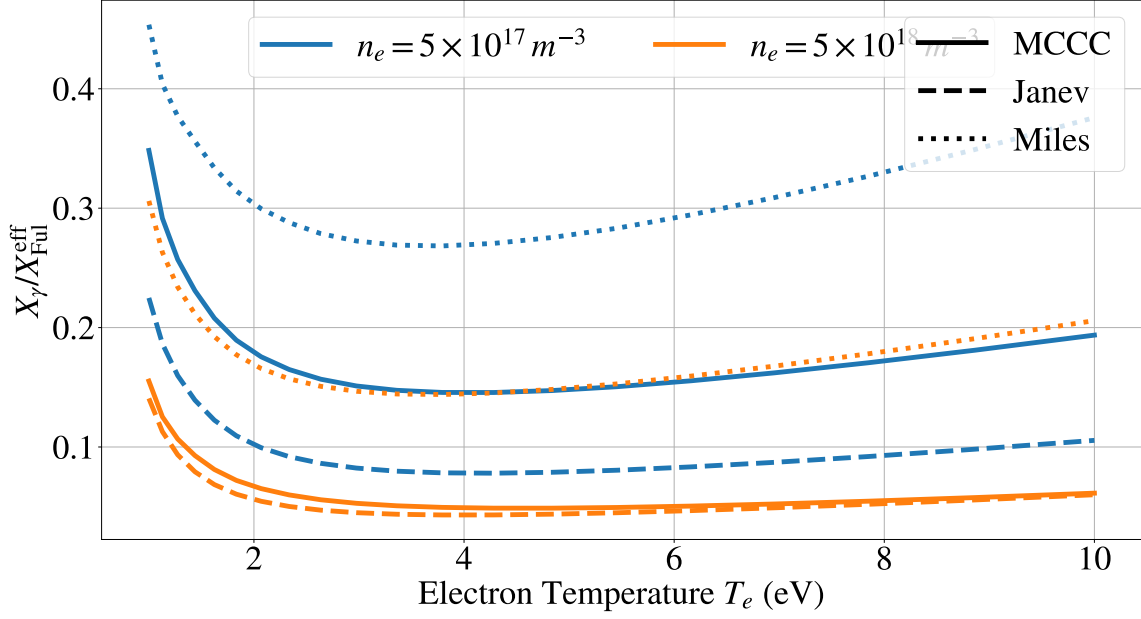


Figure 2.9: Plot shows the dependence of the ratio of effective rate coefficients on the electron temperature T_e and electron density n_e , in different collisional cross sectional databases used in Yacora model for H_2 , namely Janev [36], Miles [35] and MCCC [23]. Dummy values of $T_{H_2} = 500\text{ K}$ and $T_{vib}(X^1) = 5000\text{ K}$ used.

Finally, to have a relative measure of the degree of dissociation in the plasma discharge in different operational conditions, the rate coefficients used were based on the following assumptions:

- Only the Direct Excitation channel was assumed to populate the molecular excited states in Yacora; the effects of extra channels of charge exchange, dissociative attachment and quenching, and hence of the absolute value of the molecular hydrogen n_{H_2} were neglected.
- Only the Direct Excitation channel was assumed to populate the atomic excited states in Yacora; additional channels, as well as the change in population arising from the radiation trapping of Lyman emissions, were neglected.
- The MCCC cross-sectional database was used for the molecular hydrogen population coefficients from Yacora
- The plasma parameters were assumed to be on the higher end of the low temperature and pressure plasmas, considering MW will impart more energy to the electrons (T_e)[21]. Hence, the errorbars from the uncertainty of the plasma parameters correspond to the effective rate coefficients for the range of plasma parameters

$3.7927 \text{ eV} \leq T_e \leq 10 \text{ eV}$ and $5 \times 10^{16} \text{ m}^{-3} \leq n_e \leq 5 \times 10^{19} \text{ m}^{-3}$, neglecting the high effective rate coefficients at very low T_e .

2.4.3 Discrepancy of Plasma Parameters

An effort was made to determine the electron temperature and density of our plasma discharge by comparing the measured ratio of Balmer intensities (and hence populations) to the ones obtained from Yacora direct excitation channel [38]. Fig. 2.10 shows the measured Balmer line emission ratios for our plasma discharge as the discharge temperature is swept. We do not expect the plasma parameters to change significantly with changes in discharge temperature; hence, the slight variation observed here could be attributed to an intensity calibration error. An average of ratios for all discharge temperatures was considered as the measured line emission ratio.

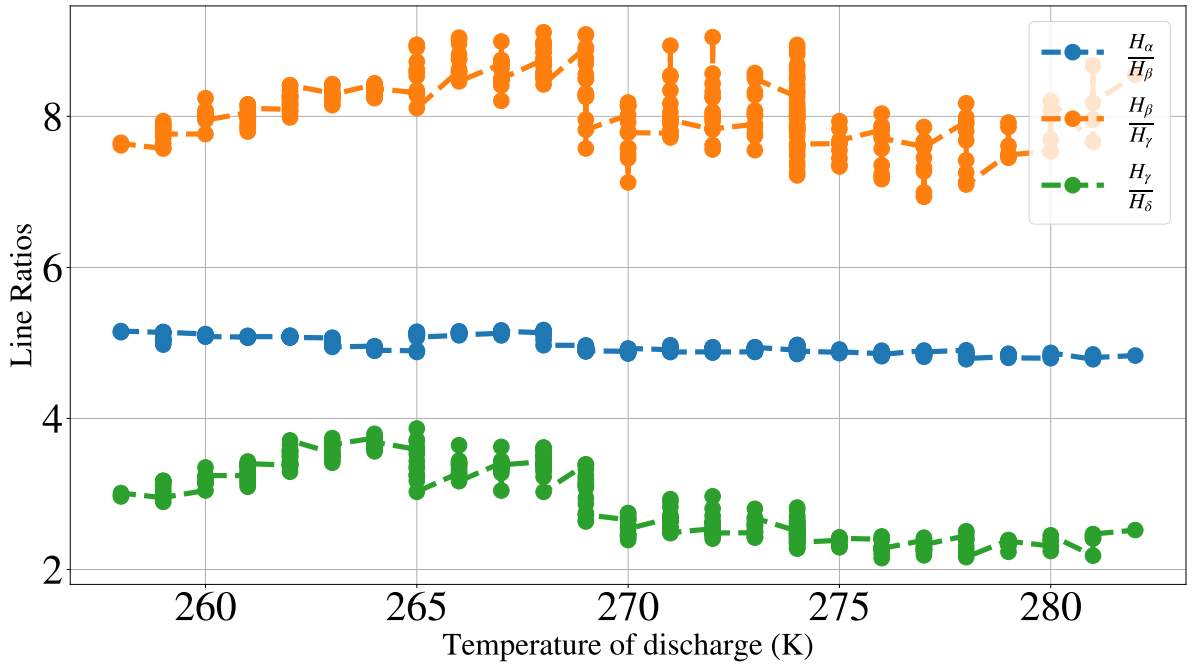


Figure 2.10: Measured values of intensity ratios of first four Balmer emissions (H_α/H_β , H_β/H_γ and H_γ/H_δ) in our plasma. The values are plotted for spectra as the discharge tube temperature was swept, to represent the variance of values across different discharge operations. In principle, we expect the ratios to remain constant across different discharge temperatures, and the slight variation observed is probably due to poor calibration of the spectrometer. An average of all values across different discharge temperatures was used for diagnostics. The data was acquired at an H_2 flow of $\approx 0.37 \text{ mln/min}$, a discharge pressure of $\approx 4.2 \text{ mbar}$, and 25 W of MW power applied.

Fig. 2.11 shows the plot of Balmer line ratios from the direct excitation channel of the atomic hydrogen Yacora model for different T_e and n_e values, with the horizontal

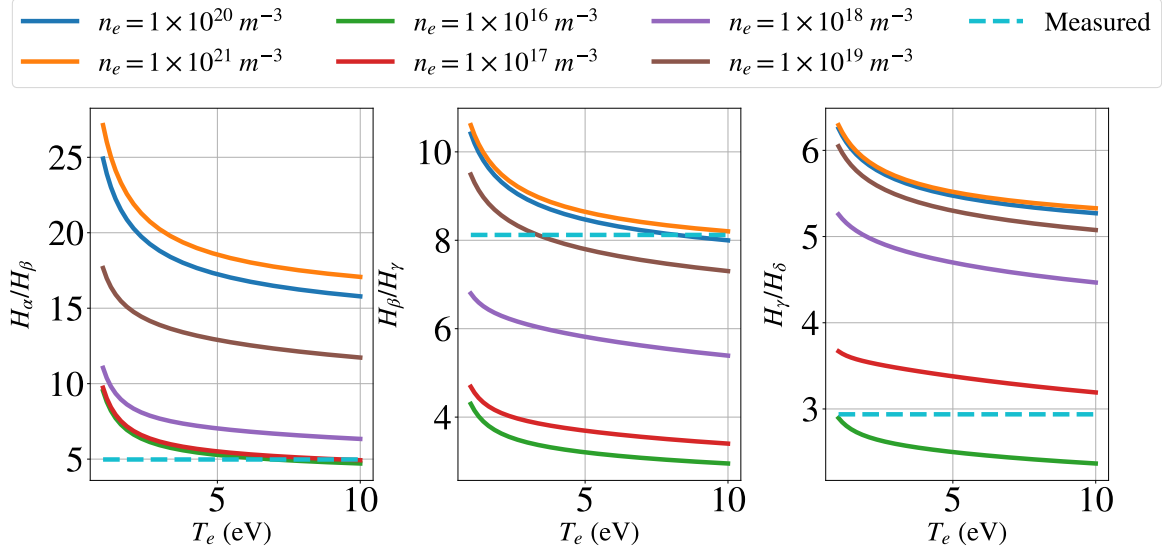


Figure 2.11: Calculated values of the Balmer line emission ratios (H_α/H_β , H_β/H_γ and H_γ/H_δ) from the direct excitation channel of the Yacora model for atomic hydrogen for varying electron temperature (T_e) and electron density (n_e) with the mean measured value in our plasma for each ratio, in an attempt to estimate the plasma parameters. A consistent value for plasma parameters was not obtained from comparing different ratios.

dotted lines corresponding to the measured ratios from spectra of our plasma discharge. A consistent value for plasma parameters was not obtained from comparing different ratios. We expect this discrepancy to have originated from the neglected change in the populations of the atomic hydrogen states due to radiation trapping of the Lyman emissions, which are not included in the Yacora.

2.5 Required Fulcher Intensities

An important caveat in obtaining the total Fulcher intensity was uncertainty about which emissions to use and how to determine the accurate value. Firstly, the use of just Q emission intensity as Fulcher intensity turned out to be a gross underestimation of Fulcher intensity since the Q lines, although the dominant emission, made up only $\approx 50\%$ of the total emission of the Fulcher band (See Fig. 2.14). Accurately obtaining the whole Fulcher band by integration was not feasible because of the undetected non-diagonal¹⁶ Fulcher band in the measured spectra, as well as the saturated and hence broadened Balmer- α line overlapping with some of the Fulcher emission. Even an extrapolation of the visibly integrable Fulcher band, as done in [26], required knowledge of plasma parameters, which were not available from the broadened, unresolved Fulcher spectra obtained with the low-resolution spectrometer. Additionally, a consistency was desired across the equation of degree of dissociation and hence just integrating and including the anomalous P and R emissions¹⁷ with the rate coefficients from Yacora, which did not include the anomalies, was deemed incorrect. Hence, the following methods were employed to retrieve information on intensities within the Born-Oppenheimer approximation and to gain insight into the range of plasma parameter values.

2.5.1 Q lines fitting and extrapolation

To retrieve the full Fulcher emission intensity and obtain information about the plasma parameters, a fitting procedure was performed on the low-resolution spectra, as in [40]. For this, we can fit a Gaussian corresponding to the first 5 Q lines ($J = 1$ to 5) within a vibrational band, which corresponds to the measured Gaussian instrumental broadening by the spectrometer (See Fig. C.3). The intensity of the Q lines can be obtained from the transition probabilities, given by the Hönl-London factors and the population of the rotational state, given by a thermal distribution at a temperature T_{rot} .

$$A_{J,J} \propto S_{J,J} \gamma_J$$

$$n_J \propto \exp\left(-\frac{hc}{kT_{rot}^v} F_v(J)\right) \quad (2.22)$$

where, $F_v(J)$ are the rotational energy levels (See App. B.1). Hence, the intensities become,

¹⁶Any mentions of "Non-diagonal" Fulcher emission always refer to a vibrational non-diagonal transition from v_1 in $d^3\Pi$ to v_2 in $a^3\Sigma$, $v_1 \neq v_2$.

¹⁷In molecular hydrogen, the rotational coupling between electronic states $d^3\Pi_u^f$ and the $e^3\Sigma_u^+$ causes anomalous emissions in the P and R branches, so that the emissions cannot be predicted from Hönl London factors. The effect is large enough that for $v > 3$, the P and R branches are completely absent. Various studies have sought to explain and model these anomalous intensities, but no definitive formalism is available yet [39].

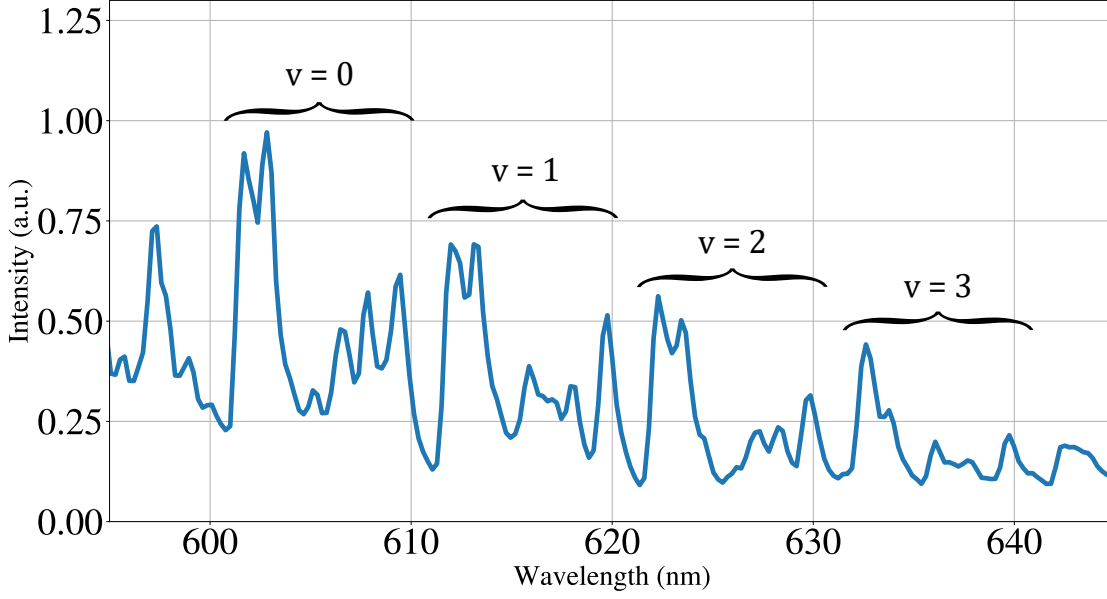


Figure 2.12: A closer look at the first four diagonal vibrational branches of the Fulcher band, with the low resolution CCS200 spectrometer.

$$I_{v,J}^{v,J} \propto S_{J,J} \gamma_J n_J \propto S_{J,J} \gamma_J \exp \left(-\frac{hc}{k_B T_{rot}^v} F_v(J) + C \right); \text{ for each } v \quad (2.23)$$

Rotational Population

In the simplest approximation, we can consider that the $d^3\Pi$ state is populated only from electron impact excitation from the $v = 0$ of the ground state (X^1). The ground state is mainly populated by heavy particle collisions with neutral ground state species, and hence a thermal Boltzmann distribution with temperature $T_{gas} = T_{rot}(X_{v=0}^1)$ is considered. Due to the selection rule of $\Delta J = 0$ in an electron-impact excitation, the rotational quantum number (J) is not altered, and hence we can preserve the rotational population distribution. Hence, a Boltzmann distribution with a T_{rot} can be considered for the rotational population of each vibrational branch of $d^3\Pi$ [33]. The rotational populations ($n_{J'}$) within each vibrational state are given by

$$\frac{n_{J'}}{n_{J''}} = \frac{g_{J'}}{g_{J''}} \exp \left(\frac{\Delta E_{J'J''}}{k_B T_{rot}^v} \right); \text{ in a given } v \quad (2.24)$$

where, T_{rot}^v is the rotational temperature for that vibrational branch, $\Delta E_{J'J''}$ is obtained from the vibrational energy levels (See. Appendix A.1 and $g_{J'}$ are statistical weights.

Knowing the transition wavelengths of each Q line and their relative intensities, we can fit a curve to each of the first 4 vibrational bands to determine the rotational

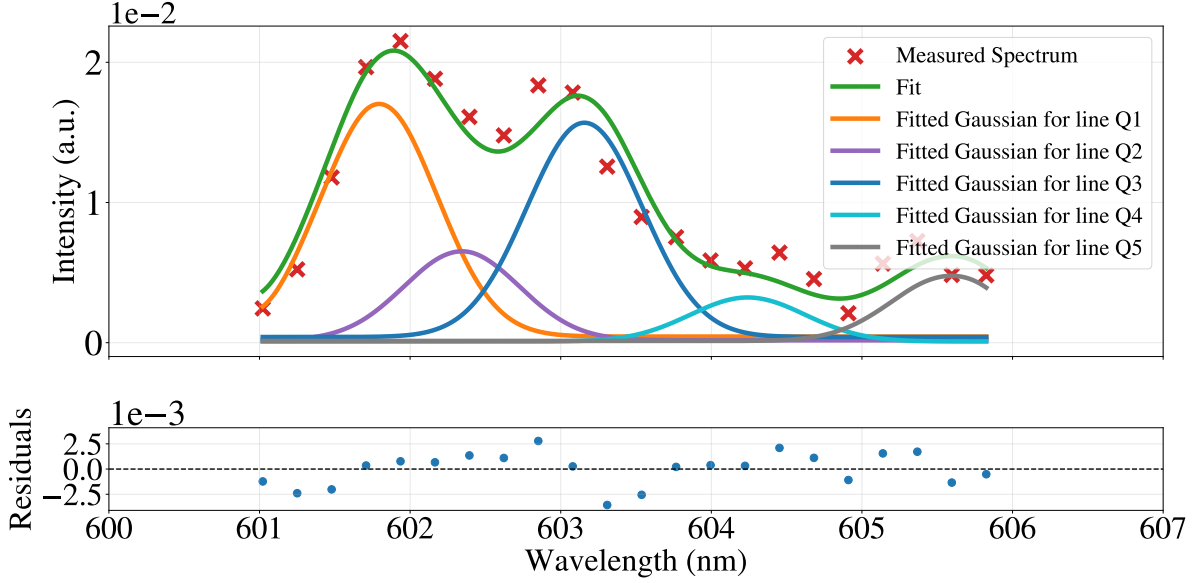


Figure 2.13: Fitting to extract T_{rot} and intensities of Q lines for the $v = 0$ branch of the Fulcher- α diagonal band. A similar fitting was done for $v = 1$, $v = 2$ and $v = 3$, the prominent diagonal vibrational bands. These intensities are used in methods "Only Q peaks (fits)" and with extrapolation, in "Extrapolate from Q-peak fit".

temperature and, hence, the intensity of each Q line. The transition wavelengths were calculated from the known energy differences between the energy levels, as explained in the App. B.1.

So, for each of the first 4 vibrational bands, the following model was fit, to

$$\mathcal{F}(\lambda, T_{rot}^v, A, \lambda_{cen}, wid, offset, c) = A \sum_{J=1-5} (I_v(J, T_{rot}^v) \cdot G(\lambda, \lambda_{cen}, wid, offset)) + c \quad (2.25)$$

where, intensities $I_v(J, T_{rot}^v)$ are from Eq. 2.23 and the Gaussian point spread function at the transition wavelength $\lambda_{cen}^{J,v}$ of each Q line is given by

$$G_{J,v}(\lambda, \lambda_{cen}^{J=1,v}, wid, offset) = \exp \left[- \left(\frac{\lambda - (\lambda_{cen}^{J=1,v} + \Delta\lambda^{J,v})}{\sqrt{2}wid} \right)^2 \right] + offset \quad (2.26)$$

Here, the transition wavelengths of each Q line $\lambda_{cen}^{J,v}$ are replaced by $\lambda_{cen}^{J=1,v} + \Delta\lambda^{J,v}$, fixing the $\Delta\lambda^{J,v}$ from calculated theoretical values. An example of such a fit for the first vibrational band of a Fulcher Spectrum recorded from our discharge is shown in Fig. 2.13. All spectra were intensity-corrected for grating and CCD efficiencies, and wavelength calibration was performed ex post facto using the Balmer- γ line as a reference. More Information can be found in the Appendix C.

In addition, due the conserved rotational population distribution in $X^1 \rightarrow d^3\Pi$ transition, we can directly project the optimized rotational temperature of $d^3\Pi_v$ from the fit

to ground state using ratio of respective rotational constants and find the T_{gas} ¹⁸. Best agreement has been found to be from projecting from $T_{rot}(d^3\Pi_u^-, v' = 2)$ [33], and hence was used for projection. The ratio of rotational constants for the corresponding states was taken from [26].

$$T_{gas} = T_{rot}(X^1\Sigma_g^+, v = 0) = \frac{B_v(X^1\Sigma_g^+, v = 0)}{B_v(d^3\Pi_u^-, v' = 2)} T_{rot}(d^3\Pi_u^-, v' = 2)$$

$$\frac{B_v(X^1\Sigma_g^+, v = 0)}{B_v(d^3\Pi_u^-, v' = 2)} = 2.11 \quad (2.27)$$

To calculate the full Fulcher intensity, the intensity of the dominant Q line in each vibrational band was extrapolated to the intensity of all P, Q, R lines in the vibrational band within the Born Oppenheimer Approximation, as a function of the rotational temperature, as done in [26]. Knowing the rotational population distribution (Eq. 2.24) and the line emissivity (Eq. 2.23), we can calculate the intensity of all P, Q, R lines in a vibrational band. Summing over the emissivities of the first five P, Q, R rotational lines in the first four vibrational bands gives us the theoretical value of total $I_v^{P,Q,R}$ for each v . Comparing this sum of calculated intensities with the intensity of a reference line in the corresponding vibrational band leads us to the factor $f_v(T_{rot})$, which we can use for the extrapolation of measured intensities. For our case, the most dominant emission line in a vibrational band is the Q1 line and hence is used as the reference line for extrapolation.

$$f_v(T_{rot}) = \frac{\sum_{J', J''=1}^5 \dot{N}_{J' \rightarrow J''}^v}{I_v^{Q1}}$$

$$= \frac{\sum_{J', J''=1}^5 \dot{N}_P^v + \dot{N}_Q^v + \dot{N}_R^v}{I_v^{Q1}} \quad (2.28)$$

where, $\dot{N}$ is the calculated emissivity of the transition.

This factor scales the calculated emissivities to match the measured absolute values and scales the intensities according to the rotational temperature. Hence, knowing the rotational temperature and the resolved intensity optimised from the fit, we get

$$I_{Fulcher}^{v \rightarrow v} = \sum_{J', J''=1}^5 I_{J'', J''}^{v \rightarrow v}(P, Q, R)$$

$$= I_{v=0}^{Q1, measured} \cdot f_v(T_{rot}) \quad (2.29)$$

and hence,

$$I_{Fulcher}^{diagonal} = \sum_{v=0}^3 I_{Fulcher}^{v \rightarrow v} \quad (2.30)$$

¹⁸Doppler broadening of Balmer lines at typical plasma gas temperatures is much smaller than the spectrometer broadening, hence projection from $T_{rot}(d^3\Pi_u^-)$ was the method used to estimate the gas temperature.

This fitting of Q lines and extrapolation to the diagonal Fulcher- α band will henceforth be labelled as "Extrapolation from Q-peak fit". Additionally, the degree of dissociation was analysed using only the summed Q-line intensities from the fit for comparison, labelled "Only Q peaks (fits)".

2.5.2 Fit from all P, Q, R lines

Now, using the rotational population distribution and the Hönl-London factors, we can calculate the relative intensities of the P, Q, and R lines within a vibrational band, as done in the previous section. Additionally, knowledge of the vibrational population (See Sec. 2.4.1) gives us the relative intensities of the vibrational bands, and hence the intensities of all rotational P, Q, R transitions in all diagonal and non-diagonal vibrational branches can be calculated, and we can essentially simulate the whole Fulcher- α , given some plasma parameters. The information on the calculation of the transition wavelength of each transition can be found in B.1.

So, the intensity of each transition is:

$$I_{v_1, J}^{v_2, J'} \propto S_{J, J'} \gamma_{Jn_J n_{d^3}(v_1)} A_{v_1, v_2} \quad (2.31)$$

where, $n_{d^3}(v_1)$ is the population for the v_1 state from the normalized vibrational population distribution from the Franck-Condon Projection from ground state X^1 (Sec. 2.18), and A_{v_1, v_2} is the transition probability for a vibrational transition $v_1 \rightarrow v_2$, taken from [31].

An example of the calculated Fulcher- α spectrum at a dummy rotational temperature ($T_{rot}(d^3\Pi)$) and vibrational temperature ($T_{vib}(X^1)$) is shown in Fig. 2.14. It also shows the percentage of each emission band (diagonal P, Q, and R bands and non-diagonal vibrational bands) in the total Fulcher emission. It can be very clearly seen that even though the Q branch is dominant, P and R branches have a non-negligible contribution to the total emission and hence neglecting these would be a gross underestimation of the Fulcher emission and hence overestimation of the degree of dissociation.

Now, with the full P, Q, R emission intensities, we can carry out a fit similar to the one in Sec. 2.5.1, to calculate the diagonal Fulcher intensity. In principle, a fit of first 5 rotational P, Q, R lines in the first four vibrational bands was carried out to extract the total diagonal Fulcher intensity and extract the values of $T_{vib}(X^1)$, $T_{rot}^{v=0}$, $T_{rot}^{v=1}$, $T_{rot}^{v=2}$, $T_{rot}^{v=3}$.

$$\begin{aligned} \mathcal{F}(\lambda, T_{vib}(X^1), T_{rot}^0, T_{rot}^1, T_{rot}^2, T_{rot}^3, A, wid, c) = \\ A \cdot (\sum_{v=0}^3 \sum_{J=1}^5 (I_{v_1, J}^{v_2, J'} \cdot G(\lambda, wid, 0))) + c \end{aligned} \quad (2.32)$$

where, $I_{v_1, J}^{v_2, J'}$ is the intensity from Eq. 2.31 and $G(\lambda, wid, 0))$ is a Gaussian function from Eq. 2.26 whose centre wavelengths are transition wavelengths obtained from energy levels in Appendix C.

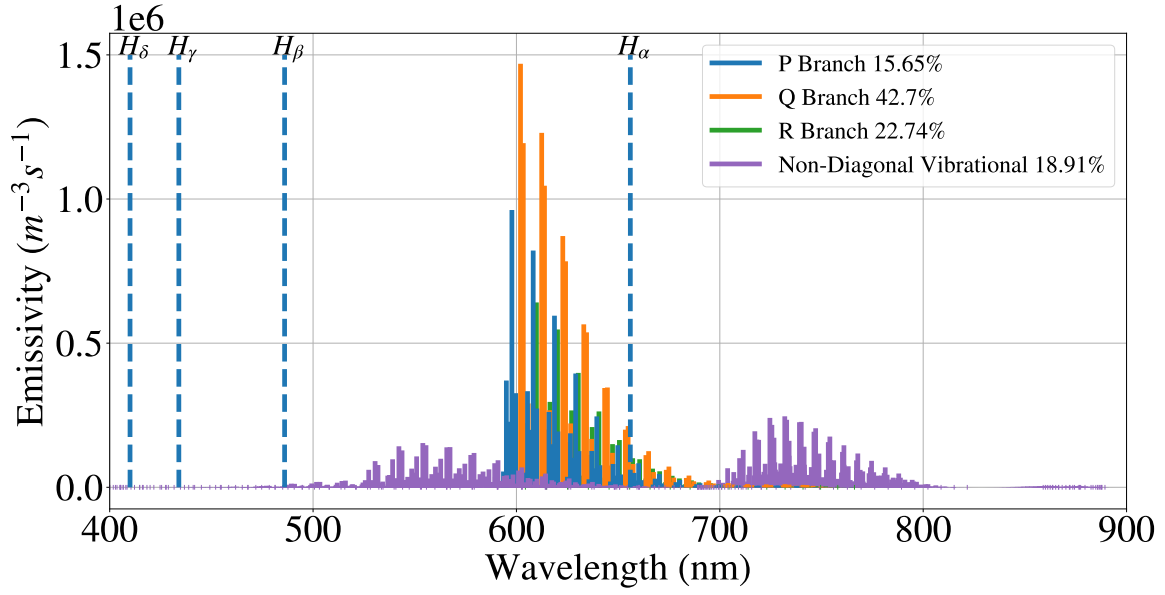


Figure 2.14: Simulated intensities of all rotational in vibrational bands of the Fulcher- α band, for a dummy value of $T_{vib}(X^1) = 5000\text{ K}$ and $T_{rot}(d^3) = 400\text{ K}$. We can clearly see that the P and R emissions form a significant part of the total Fulcher emission. The position in wavelength of the first four Balmer- α lines is shown for reference.

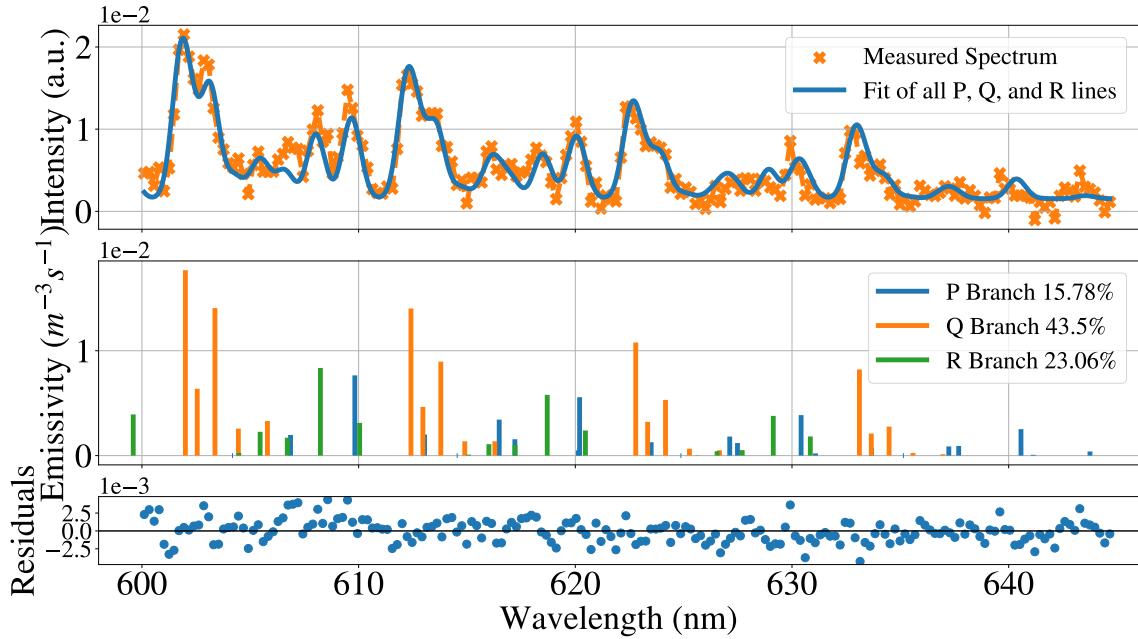


Figure 2.15: Fit of all P, Q, and R emissions for the first four vibrational bands of the measured Fulcher spectrum, and the corresponding calculated lines used for fitting. The method, after extrapolation, is termed as "Integrate fit (P, Q, R)".

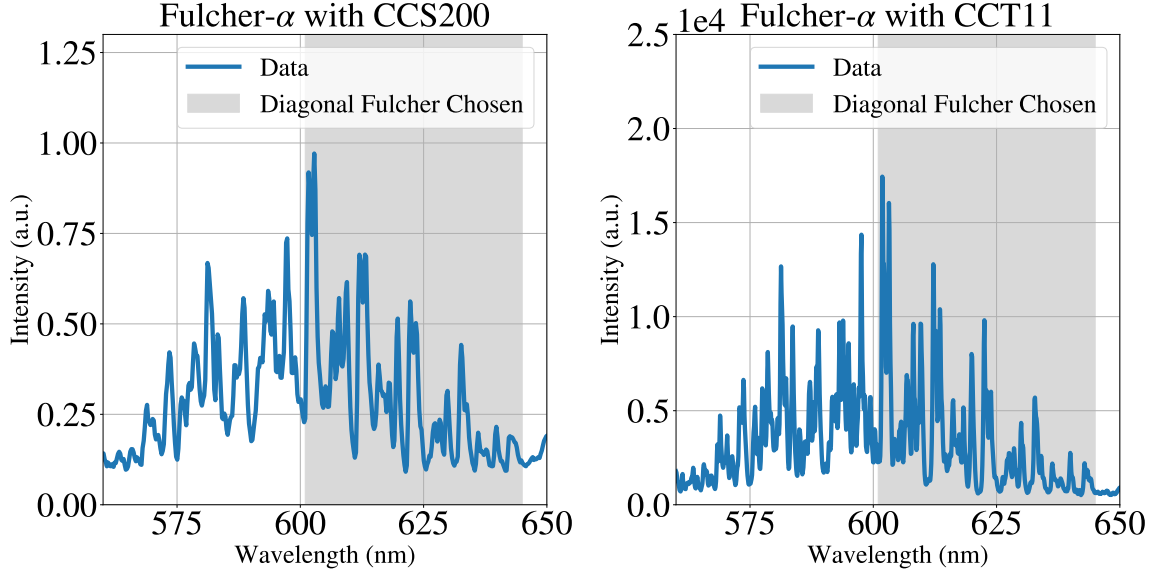


Figure 2.16: Fulcher- α band with the CCS200 and CCT11 spectrometers, and the section chosen for Fulcher intensity extraction, in the "Integrating" method.

Such a fit, and the corresponding lines are shown in Fig. 2.15. The fit was then integrated to calculate the intensity of the diagonal Fulcher band, and this method will henceforth be referred to as "Integrate fit (P, Q, R)".

In addition, the integrated measured diagonal Fulcher intensity over the wavelength range 600-645 nm was also used as the diagonal Fulcher intensity, henceforth labelled "Integrating". A representation of the chosen wavelength range for integration to get the Fulcher intensity is shown in Fig. 2.16, and the same for the Balmer- γ line is shown in Fig. 2.17. The analysis was done with two different spectrometers, Thorlabs: CCS200 and Thorlabs: CCT11, and two Fulcher and Balmer- γ emissions are shown for comparison.

Extrapolation to non-diagonal emissions

Obtaining the full Fulcher intensity directly from the spectrum was not possible because the non-diagonal vibrational bands were not consistently observed, and a large part of these emissions was crowded by the highly saturated and broadened Balmer- α line. For all the aforementioned methods of obtaining intensities (except the "Only Q peaks (fits)"), only the diagonal Fulcher intensity was extracted, and hence an extrapolation from the diagonal to the Full Fulcher emission was carried out from the calculated ratio of diagonal to full Fulcher intensities, by defining an extrapolation factor [26].

$$f_{vib}(T_{vib}) = \frac{I_{Fulcher}^{Diag+nondiag}}{I_{Fulcher}^{diag}} \quad (2.33)$$

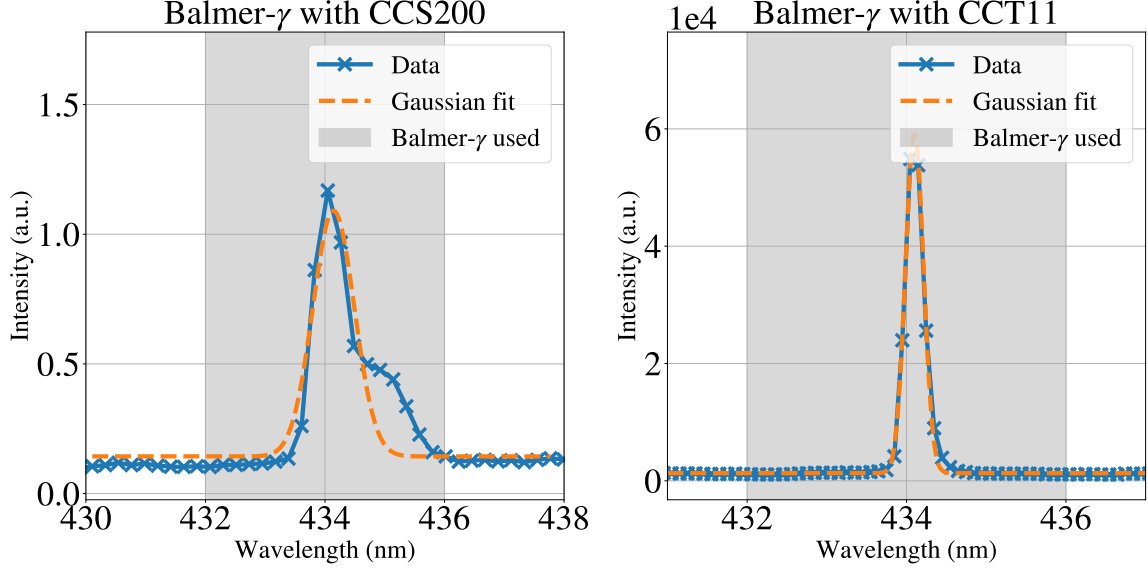


Figure 2.17: Balmer- γ line with both spectrometers, the Gaussian fit to it and the chosen section for intensity in analysis. The source of the non-Gaussian profile of Balmer- γ with the CCS200 spectrometer, even though the measured point spread function was Gaussian (Fig. C.3), is unknown.

And hence, for all methods, the final Fulcher intensity used for the calculation of the degree of dissociation is the one after the extrapolation.

$$I_{Fulcher}^{Full} = I_{Fulcher,data}^{diagonal} \cdot f_{vib}(T_{vib}) \quad (2.34)$$

where $I_{Fulcher,data}^{diagonal}$ is the diagonal Fulcher intensity from one of the different fitting methods.

2.5.3 Comparison of Intensity extraction methods

On comparison of the degree of dissociation using all different intensity extraction methods 2.18, it was observed that using only the Q emissions for the Fulcher intensity (Only Q peaks (fits)", as expected, resulted in a gross overestimation of the degree of dissociation. On the other hand, the other methods, namely "Integrating", "Extrapolating from Q peak fits" and "Integrate fit (P, Q, R) agreed very well in terms of degree of dissociation.

Even though the intensity fitting proved futile for extracting intensity and only "Integrating" proved to be enough, it provided useful information about the plasma parameters from the optimised fit parameters. Firstly, the rotational temperatures for vibrational levels of $d^3\Pi$ state (T_{rot}^v) were used to calculate the gas temperature via Eq. 2.27, which was then used to compute the effective rate coefficients using Yacora. Both

the Q-lines fit and the P, Q, R fit methods provided optimised rotational temperatures for the first four vibrational bands, but unfortunately, these values, although always decreasing with increasing vibrational number as expected, did not agree between the two fitting methods for the same spectrum. Since the P, Q, R fit had more unknown parameters, higher residuals were observed for the spectrum with lower Fulcher intensity. Hence, the rotational temperatures from the Q lines fitting should be trusted more for further calculations. Additionally, since no clear trend was observed in rotational temperatures across different data iterations and discharge spectra, and since the deviations were not significant in a particular iteration, a constant value of $T_{rot} = 500K$ was used for all data¹⁹. In addition, the P, Q, R fit provides us with the value of $T_{vib}(X^1)$ which is used not only in the calculation of the effective rate coefficients of the Fulcher transition ($X_{Fulcher}^{eff}$) through the effective transition probabilities ($A_{Fulcher}^{eff}$) (Sec. 2.4.1), but also in the extrapolation of Fulcher intensity (Sec. 2.5.2). The latter effect was much more dominant than the former. A list of the general optimised parameters obtained from the fits is provided in Appendix D.1.

2.6 Parameter Sweeps

Now, with at least a relative measure for the dissociation fraction, some discharge parameters were varied to check their effect on the dissociation fraction. The discharge has been notoriously difficult to handle and unreliable for reproducibility due to the presence of too many unknown parameters. Mainly, the pressure of hydrogen inside the discharge tube in the plasma region, the hydrogen flow, the MW power applied to the cavity, the cavity's impedance matching, and the Temperature of the discharge tube were expected to play a significant roles. The goal of this analysis was to understand how these different parameters affect the atomic hydrogen population, either to find an optimised set of parameter values or to improve the discharge design to control reproducibility. The sections below show the effect of the discharge temperature and the applied MW power on the dissociation fraction, efforts to vary the flow/pressure of hydrogen in the discharge to check dependence of the dissociation fraction did not produce reproducible curves.

2.6.1 Discharge Temperature

It was observed that the discharge temperature was the most influential parameter on the degree of dissociation. As mentioned previously, the temperature of the outgoing cooling air primarily reflects the overall heating of the discharge tube caused by the plasma inside. This temperature is a hard parameter to control and extremely irreproducible. Nevertheless, an effort was made to assess a qualitative dependence of the dissociation fraction on this temperature. The discharge temperature was periodically stabilised at

¹⁹In principle, a fit can be done for each iteration of discharge operation for a rough calculation of the gas temperature, but in the assumption of Direct Excitation Channel, T_{gas} does not affect the effective rate coefficients ratio.

different values, and the calculated degree of dissociation from plasma spectra, obtained using different intensity extraction methods, was plotted against temperature, as shown in Fig. 2.18. A very consistent improvement in the degree of dissociation was observed as the temperature was swept from cold regions to room temperature across different iterations, but the temperature measurement introduced uncertainty in the temperature at which this peak occurred. It is important to note that, due to a lack of knowledge of the actual plasma parameters, the effective rate coefficients for this sweep were calculated using a dummy set of plasma parameters (average value over the known range), and error bars were added based on the two extremes (See Fig. 2.18).

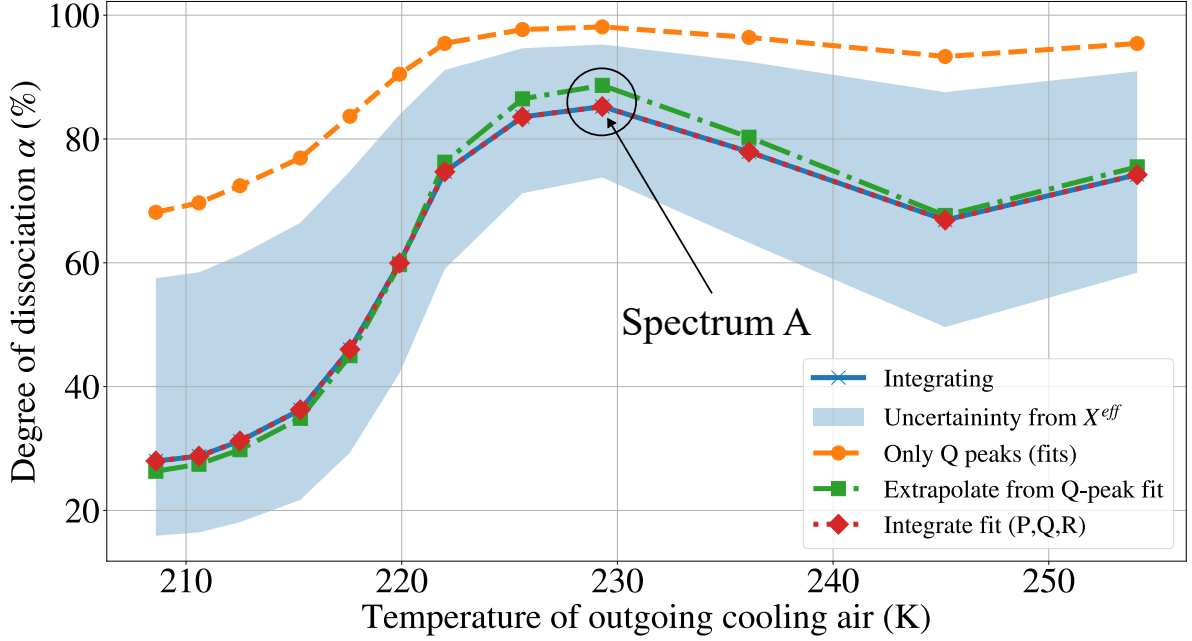


Figure 2.18: Plot shows the change in dissociation fraction in our hydrogen plasma when the discharge temperature is varied. All data taken at H_2 flow of $\approx 0.37 \text{ mln/min}$ with 30 W of MW power applied. The degree of dissociation was calculated from Fulcher intensities from all four intensity methods (Sec. 2.5). It is clear here that estimating the degree of dissociation using only the Q-line emission overestimates it by a large margin. Additionally, due to unknown values of plasma parameters, the rate coefficients used for calculations were as follows: the curves for each method use $T_e = 7.85 \text{ eV}$, $n_e = 5 \times 10^{17} \text{ m}^{-3}$ and $T_{vib}(X^1) = 5000 \text{ K}$, whereas the shaded region represents the spread of the curves when the plasma parameters vary. In particular, the upper boundary corresponds to $T_e = 3.79 \text{ eV}$, $n_e = 5 \times 10^{18} \text{ m}^{-3}$ and $T_{vib}(X^1) = 1000 \text{ K}$, while the lower boundary corresponds to $T_e = 10 \text{ eV}$, $n_e = 5 \times 10^{16} \text{ m}^{-3}$ and $T_{vib}(X^1) = 12000 \text{ K}$. The MCCC cross-sectional database with only direct excitation channels was used for the calculation in Yacora. Spectrum A is the spectrum used for the calculation of uncertainties in the degree of dissociation, in Sec. 2.6.3

2.6.2 Applied MW power

Another important parameter for the hydrogen discharge dissociation is the applied microwave power to the cavity, and its impedance matching. Similar to the temperature sweep, a sweep of the applied power was carried out by sweeping the input power from 10 W to 45 W, available range from the MW generator after suitable amplification²⁰. For each applied power, the cavity was impedance-matched by monitoring the reflected power, and the data with the best total coupled power in each case were used to calculate the degree of dissociation, which was then plotted against the coupled power to the cavity. For data across different applied powers, an effort was made to keep the cooling gas temperature at a similar value to have a negligible effect from temperature change.

As shown in Fig. 2.19, the dissociation fraction increases with increasing coupled MW power, but above ≈ 23 W, it saturates. One explanation for this saturation could be that any further increase in MW power is essentially radiated away and does not couple to the gas load at all. For lower applied powers, it was also observed that a better coupling to the cavity could be achieved, with a much lower percentage of input power reflected. This could also hint that once the input power is sufficient to couple to the entire gas load, any further increase in applied power does not affect the dissociation fraction. Even this high overall dissociation fraction is due to the discharge being stabilised to a "good" temperature, which played the dominant role. It is important to note that, in calculating the degree of dissociation, a constant electron temperature and density were used in the rate coefficients across different applied powers, assuming that increased coupled power does not contribute to a change in plasma parameters. This was considered a sufficiently reasonable assumption, since, from general Langmuir probe measurements in literature [40] [41], the electron temperature does not change with different powers applied, while the electron density increases with increasing power, but the effect seen was not significant.

2.6.3 Uncertainties in the Dissociation fraction

This section will quantify the uncertainties in the calculated dissociation fraction arising from uncertainty in the plasma parameters in our system, under different assumptions. For comparison, the variation of the degree of dissociation of a single emission spectrum is plotted against different parameters, and the spectrum chosen is the one with the highest dissociation fraction when the discharge temperature was swept, labelled "A" in Fig. 2.1. Firstly, the main parameters of the plasma to be known within the assumption of direct excitation channel were namely, the electron temperature (T_e), electron density (n_e), Vibrational temperature of the ground state of hydrogen molecule ($T_{vib}(X^1)$), and the rotational temperature of the vibrational levels of $d^3\Pi_u$ state (T_{rot}). The dependence of T_{rot} appearing in the intensities through the $f_v(T_{rot})$ extrapolation factors could be eliminated if the intensity from the "Integrate fit (P, Q, R)" or "Integrating" method is

²⁰No good data was retrieved below 10 W since it was hard to ignite the plasma below these powers and even after ignition the plasma was incredibly unstable, efforts to impedenace match cavity resulted in plasma extinguishing.

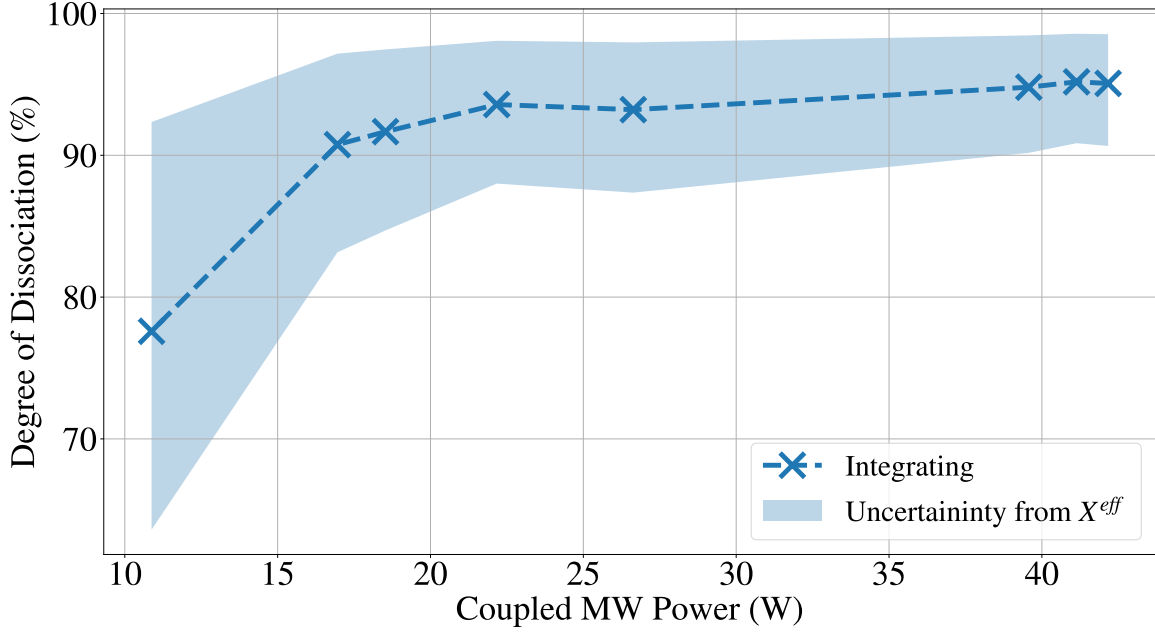


Figure 2.19: Change in degree of dissociation when the MW power applied to the cavity was varied. For each applied power, the data with the most power coupled was chosen. Fulcher intensities are taken by only "Integrating", and the rate coefficients for Fulcher emission used are from the MCCC cross-sectional database within the direct excitation channel of the molecular H_2 model in Yacora. Additionally, due to unknown values of plasma parameters, the rate coefficients used for calculations were as follows: the curve uses $T_e = 7.85 \text{ eV}$, $n_e = 5 \times 10^{17} \text{ m}^{-3}$ and $T_{vib}(X^1) = 5000 \text{ K}$, whereas the shaded region represents the spread of the curves when the plasma parameters vary. In particular, the upper boundary corresponds to $T_e = 3.79 \text{ eV}$, $n_e = 5 \times 10^{18} \text{ m}^{-3}$ and $T_{vib}(X^1) = 1000 \text{ K}$, while the lower boundary corresponds to $T_e = 10 \text{ eV}$, $n_e = 5 \times 10^{16} \text{ m}^{-3}$ and $T_{vib}(X^1) = 12000 \text{ K}$.

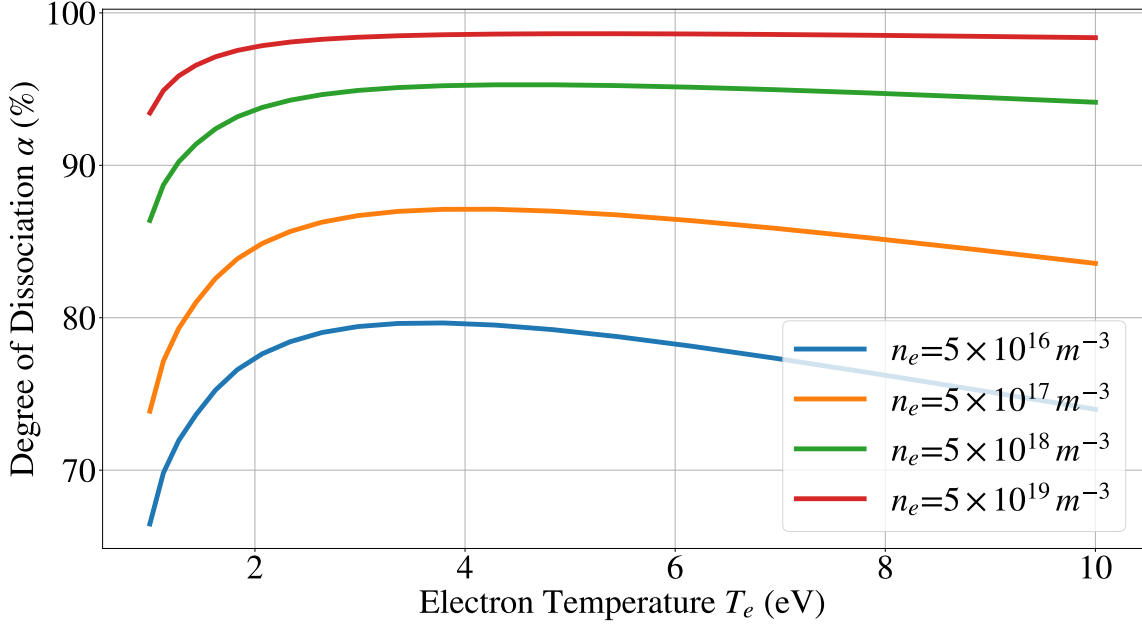


Figure 2.20: Variance in Degree of Dissociation of a single plasma spectrum (labelled spectrum A previously), with change in electron temperature and density, when the cross-sectional database for the H_2 model in Yacora was chosen to be MCCC [23]. Dummy values of $T_{H_2} = 500 \text{ K}$ and $T_{vib}(X^1) = 5000 \text{ K}$ are used.

used. Additionally, the direct excitation channel rate coefficients are independent of T_{gas} and hence, of T_{rot} . Next, the vibrational temperature of ground state ($T_{vib}(X^1)$) enters into the degree of dissociation through both the intensity extrapolation factor $f_v(T_{vib})$ and the rate coefficients through the effective Fulcher transition probability ($A_{Fulcher}^{eff}$). Both dependencies together lead to only a discrepancy of $\pm 0.5\%$ in the dissociation fraction when $T_{vib}(X^1)$ was varied between 1000 K to 12000 K (optimized value from the P, Q, R fit $T_{vib}(X^1) \approx 11,800 \text{ K}$), and hence can be deemed irrelevant for our purposes.

One of the biggest uncertainties arises from the unknowns of T_e and n_e . The variation of the degree of dissociation due to the change in rate coefficients from Yacora in the T_e and n_e range corresponding to low-pressure and low-temperature plasmas is shown in Fig. 2.20. In these, since MW plasmas lie on the higher end of the T_e range [21], we could potentially ignore the steeper variance near 1 eV, and can treat the degree of dissociation to be mostly independent of T_e , but the value of n_e can lead to a discrepancy of $\approx 30\%$. In principle, this mostly appears as a constant offset 2.19 and hence could be neglected for a relative measurement.

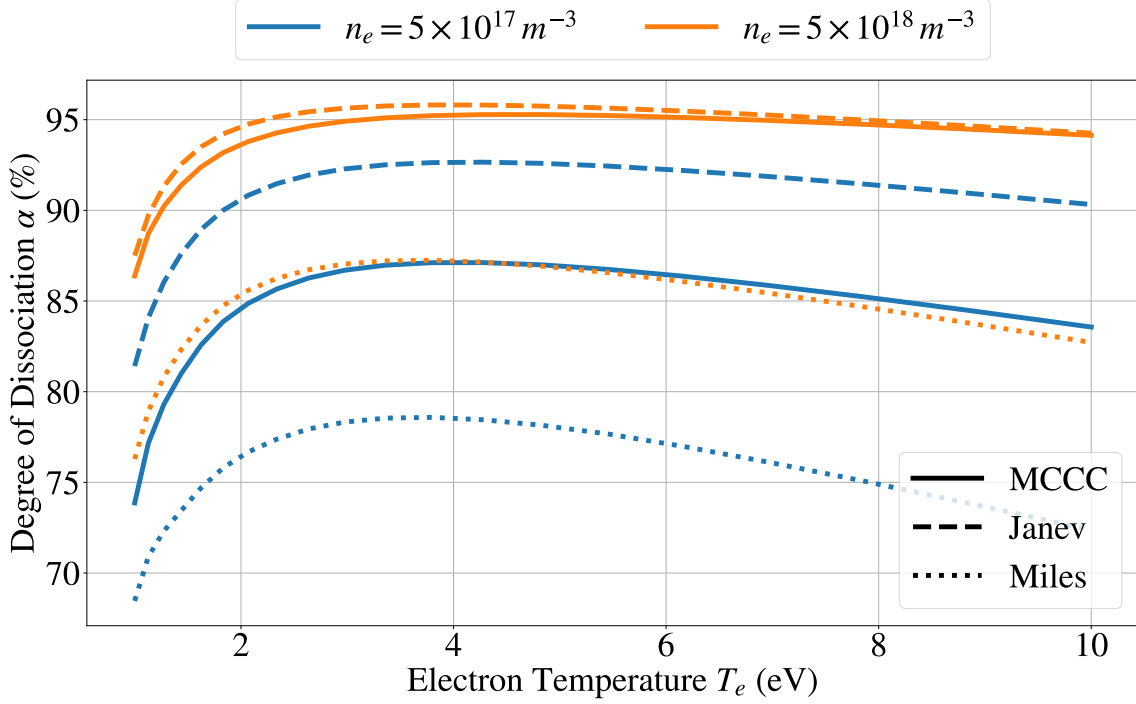


Figure 2.21: Variance in the degree of dissociation of a single plasma spectrum (labelled spectrum A in Fig. 2.18), with change in electron temperature and density, with the 3 collisional cross-sectional databases available in H_2 Yacora model: Miles[35], Janev [36] and MCCC [23]. Only two values of n_e are shown for representative purposes. Dummy values of $T_{H_2} = 500 \text{ K}$ and $T_{vib}(X^1) = 5000 \text{ K}$ used.

Another source of uncertainty is the collisional cross-sectional database used by Yacora to calculate rate coefficients for Fulcher emissions. Fig. 2.21 shows different databases and their effect on the degree of dissociation. Just the choice of cross-sectional database can change the degree of dissociation by $\pm 10\%$ for the same value of T_e and n_e . Additionally, when additional channels of Quenching (Q), Dissociative Attachment (D) and Charge Exchange (C) are added in the calculation of Fulcher rate coefficients in Yacora, the value of dissociation fraction becomes completely uncertain, and its dependence on the plasma parameters also changes in comparison to the case of only the direct excitation channel (See Sec. 2.4.2).

All these uncertainties in plasma theory rendered the experimental uncertainties in recorded intensities, reproducibility, etc., completely moot, and determining an absolute value of the dissociation fraction was outside the scope of this thesis. Nevertheless, the analysis of optical emission spectroscopy could be used for the comparison of different discharge operations, by using only a relative value of the dissociation fraction.

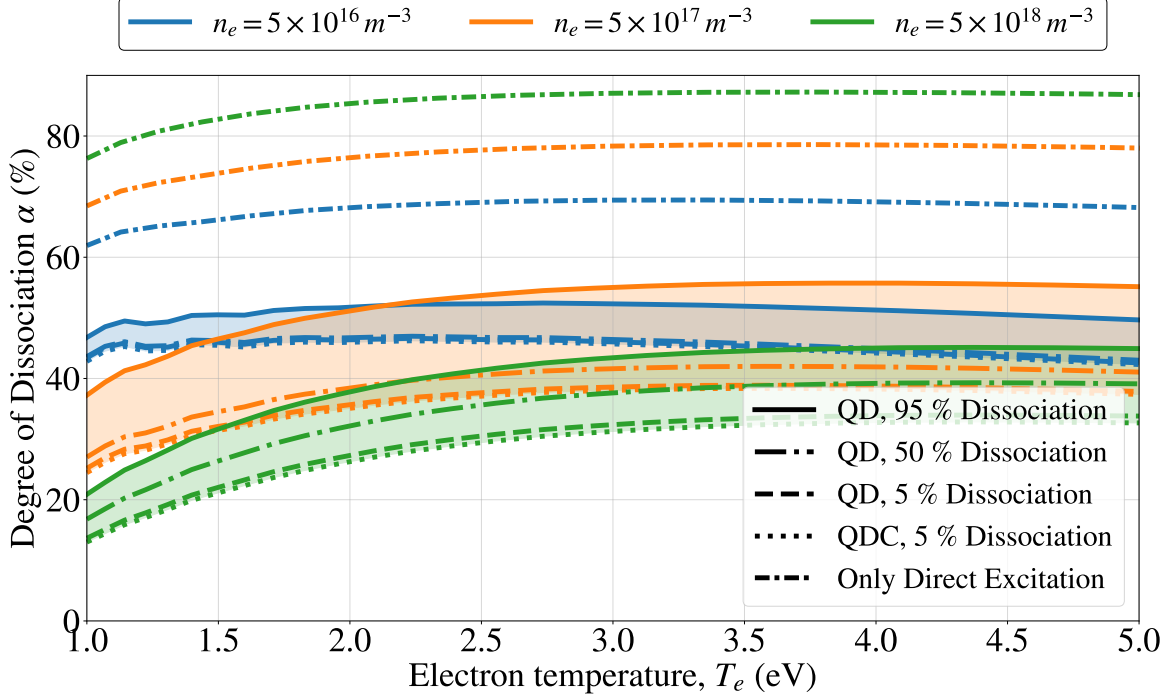


Figure 2.22: Plot depicting the uncertainties introduced to the degree of dissociation due to theory uncertainty in the value of the effective rate coefficients ratio. The top 3 lines show the degree of dissociation by including only the direct excitation channel for molecular hydrogen, at different electron densities and temperatures. Additionally, when Quenching (Q) and Dissociative Attachment (D) Channels are added, the degree of dissociation progressively falls down with increasing electron density and additionally depends on the absolute value of molecular hydrogen population (n_{H_2}), and hence for representative purposes, the spread due to n_{H_2} is plotted in terms of 3 values of degree of dissociation (95 %, 50 % and 5 %) corresponding to 3 values of n_{H_2} (2×10^{20} , 3×10^{21} and $8 \times 10^{21} m^{-3}$). All of these values correspond to a single plasma emission spectrum, labelled "A" in Fig. 2.18, and all include only the Fulcher Intensity from the "Integrating" method with a $T_{vib}(X^1) = 5000 K$ used for extrapolation.

Chapter 3

The Wire Detector

The conclusion from the plasma optical emission spectroscopy was that, even if a general trend can be established with different parameters, it is not possible to completely decouple and control all parameters. From the general experience with the plasma discharge, it was observed that even with exactly reproduced conditions of measurable parameters, the plasma dissociation fraction was not reproducible; hence, a defined optimised set of these parameters could not be used to achieve the best atomic hydrogen population in the beam¹. Also, the analysis proved to be a qualitative rather than a quantitative, as significant uncertainties in the degree of dissociation arose from unknown theory of plasma dynamics. In addition, even if a very high degree of dissociation was achieved at the plasma, there was no way to determine the fraction of that atomic hydrogen population sustained till it reaches the PCV, after passing through the Teflon tube and the cold nozzle. Hence, it was decided to develop an in situ calorimetric wire detector for atomic hydrogen to reliably measure the degree of dissociation and, potentially, use it to optimise the hydrogen discharge operation as well. In principle, to gain greater freedom and control over the parameters and to allow comparison and improvement of different methods for hydrogen beam characterisation, a new vacuum chamber setup with a hydrogen discharge similar to that of the main experimental setup was developed.

3.1 Hydrogen Discharge Test Setup

A schematic of the new vacuum chamber with the test discharge setup is shown in Fig. 3.1. Due to limitations in port sizes and available turbo pumps, the background vacuum pressure was only close to 5×10^{-4} mbar, measured with Pfeiffer MPT200 Pressure Gauge. Since the discharge could be operated at these pressures, for the first trials, this was deemed enough for measurements. The hydrogen dissociator was similar to the one in the main spectroscopic chamber. Hydrogen was introduced into a Pyrex tube with a dosing valve, after which the Evenson MW Cavity from Opthos Instruments was installed, and 30 W power was applied at 2.45 GHz. The H_2 flow was at ≈ 0.37 mln/min measured with the Bronkhost EL-Flow Select Mass Flowmeter before the dosing valve. The pressure in the discharge for good operation was about 1-3 mbar, measured with

¹Nevertheless, some insight on how to operate the discharge to get the good degree of dissociation from experience can be found in App. A.2

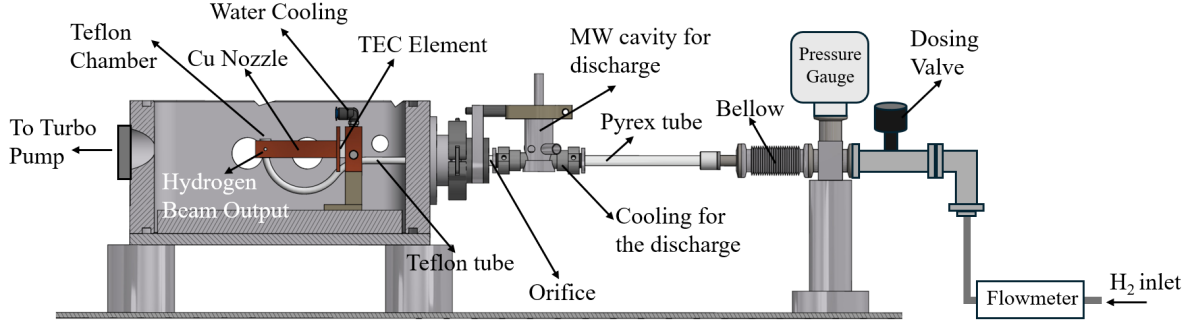


Figure 3.1: A schematic drawing of the test discharge setup, with the MW discharge and a copper nozzle. In contrast to a cryostat in the main chamber, the nozzle was water-cooled and temperature-controlled with a TEC element.

the Pfeiffer PKR251 Compact Full Range gauge. The cavity was impedance-matched by monitoring the input and reflected power with a bidirectional coupler. A Teflon tube was used to guide the atomic hydrogen from the plasma to the nozzle. To emulate the cooled nozzle used in the main spectroscopic setup, a water-cooled nozzle with a copper sink and temperature control via a TEC element was used. We expected that a temperature-controlled nozzle would help regulate the beam gas temperature and remove any residual heat from the plasma. Also, the nozzle acts as a barrier, separating the detection region from radiative heating by the hot plasma by redirecting the beam. Additionally, it was expected to have a better contrast in the atomic and molecular hydrogen beam signal on the wire if the molecular gas was cooled. With the Peltier, the nozzle can be cooled down to maximally to -20°C . The nozzle temperature was measured with an HP34401A multimeter as a voltage reading. Additionally, a small "Teflon chamber" was added to the box inside the Copper nozzle, where the Teflon tube fits and the hydrogen beam enters, to prevent recombination of incoming atomic hydrogen on the copper surface. From previous particle tracing simulations in [2], the mean number of wall collisions for the atomic hydrogen inside the nozzle was calculated to be 60; which was deemed enough for good thermalisation of the beam with the nozzle from the observations in [20]. For analysis, the hydrogen flow, the input and reflected MW power from the cavity, and the temperature of the discharge tube were all converted to analog voltages and were simultaneously logged using an Arduino.

3.2 Principle of the Wire Detector

Atomic hydrogen has a very high recombination efficiency on clean metallic surfaces. The recombination releases energy of $2E_{rec} = 4.6$ eV per molecule (or 7.136×10^{-19} J). In contrast, the kinetic energy of a single hydrogen molecule at 298.15 K is 1.0291×10^{-20} J. Hence, when atomic hydrogen recombines on a very thin wire with a high temperature coefficient of resistivity, it changes the wire's temperature and, consequently, its re-

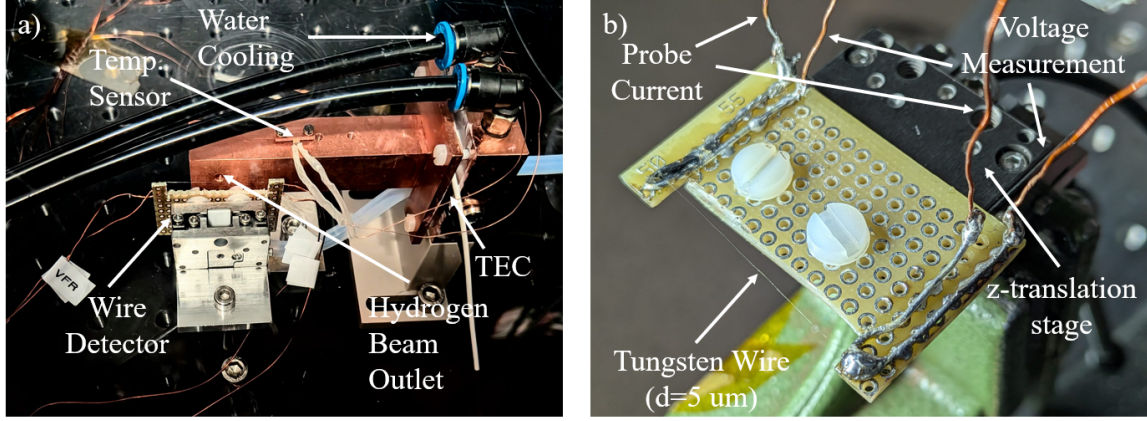


Figure 3.2: a) The nozzle with the wire detector in the test discharge setup. The nozzle is temperature-controlled with a TEC element. The wire detector on a translation stage sits 0.5 cm away from the nozzle, in the path of the hydrogen beam. b) A closer look at the wire detector: the PCB is screwed onto the z-translation stage, and the Tungsten wire is soldered at the top. Two Kapton wires, soldered at both ends, supply current and measure voltage for the "Four-Wire" resistance measurement.

sistance, which can be measured with high precision using the "Four-Wire" resistance measurement. Simply put, if

$$R_0 = \frac{\rho l}{A} \quad (3.1)$$

is the resistance of the wire of length l , area A and density ρ at the reference temperature T_{ref} , then

$$R(T) = R_0 + R_0 \alpha_\rho (T - T_{ref}) \quad (3.2)$$

where α_ρ is the temperature coefficient for resistivity for the particular material. For precise resistance measurement with the four-wire method, 2 wires carry a probe current through the wire under test, and 2 separate wires measure the voltage drop across it. This conveniently removes the contribution of any lead or contact resistances up to the wire itself, leading to a very precise value for the wire under test. A schematic of the circuit is shown in Fig. 3.3.

For our wire detector design, a Tungsten wire from Goodfellow Advanced Materials with a few μm thickness (5 to 25 μm) was strung taut across the atomic hydrogen beam and soldered to a PCB. Additionally, 2 Kapton wires were soldered to the Tungsten wire on both sides of the PCB to apply the probe current and to measure the voltage drop across it. An HP34401A Multimeter was used for this, which applied a probe current of $I_{probe} = 1 \text{ mA}$ in the highest-resolution setting of 0.5 mΩ. The PCB was mounted on a z-translation stage to scan the atomic hydrogen beam vertically. The PCB was thermally and electrically isolated from the translation stage. We expected a longer wire to be more

beneficial due to its higher resistance; hence, we used the maximum length so that the wire detector could fit through a KF40 vacuum flange. The wire detector was installed at a distance of 5mm from the nozzle. Such a detector can, in principle, be used not just to detect atomic hydrogen but to measure its actual population and eventually to map the profile of the hydrogen beam [42]. This requires a precise decoupling of all sources of heating and cooling that affect the wire's resistance to extract the atomic hydrogen heating signal.

3.3 The Atomic Hydrogen Beam

The typical atomic hydrogen beam in our spectroscopic setup is produced by igniting the plasma discharge with a molecular hydrogen flow of 0.37 mln/min . Since plasma emission spectroscopy could not provide a accurate value for the degree of dissociation, in the initial steps, the old calculations were assumed to be correct. Hence, the dissociation in the plasma region was considered to be much more than 90 %, considering the discharge to be very similar to the design in [20]. From the number of wall collisions during transport of the beam from the plasma source to the nozzle, which was calculated to be approximately 8000 in [19], it is sufficient to thermalise the gas to the temperature of the Teflon tube. Combining this with the recombination coefficient $\gamma_{\text{Teflon}} \approx 10^{-4}$, the degree of dissociation was calculated to have dropped to as low as 12% till the atomic hydrogen reaches the nozzle, with a clean Teflon tube² [19]. From particle tracing simulations done in [2], 99.6 % of the atoms undergo more than one wall collision inside the nozzle, which is also enough for the beam to completely thermalise with the nozzle. Hence, the beam temperature was henceforth considered the same as the nozzle temperature. Additionally, using the previous attempts of measurement of atomic hydrogen population using the H_2^+ peak of the RGA [2][19], the expected degree of dissociation in the beam after escaping the nozzle was assumed to be 5%. Due to lower pumping speeds, the pressure in the chamber was $5 \times 10^{-4}\text{ mbar}$, which rose to $1.5 \times 10^{-2}\text{ mbar}$ during discharge operation with the standard flow. To understand the background and beam gas dynamics at this high vacuum pressure, calculations of the mean free path and Knudsen number were performed. The mean free path (λ) of a particle of a diameter d in a gas at temperature T , pressure p , is given by

$$\lambda = \frac{k_b T}{\sqrt{2} \pi d^2 p} \quad (3.3)$$

from which we can obtain the Knudsen number (Kn), which indicates the gas flow regime, by comparing the mean free path of the particles with the characteristic length of the chamber.

$$Kn = \frac{\lambda}{L_{\text{char}}} \quad (3.4)$$

where L_{char} is the characteristic length.

²Note that this does not consider the cold nozzle.

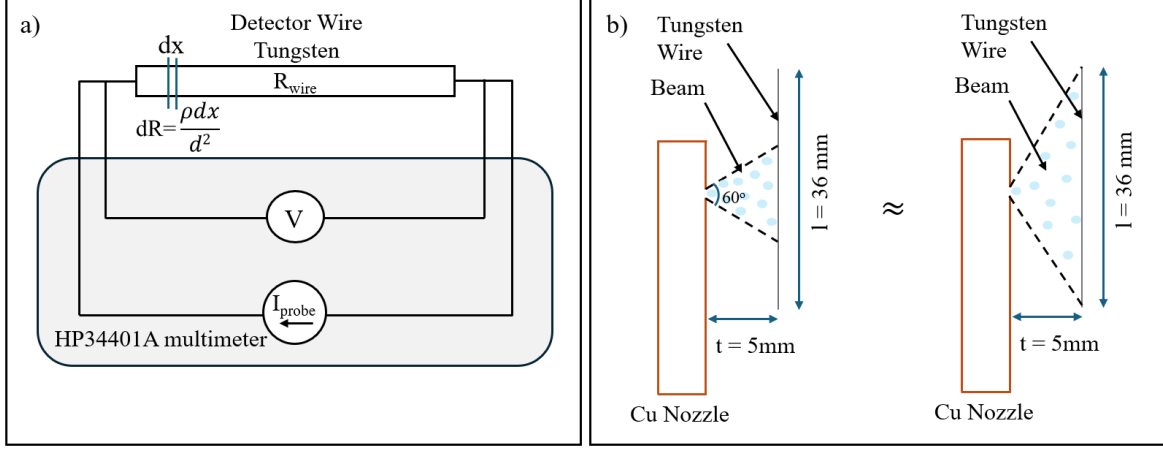


Figure 3.3: a) Schematic circuit of the four-wire resistance measurement method, and the representation of an element dx for which heat load simulations were performed. b) Schematic diagram of the assumed uniform beam distribution on the wire. The left image shows the real length of the wire hit by the beam for an opening angle of 60° . The total number of particles from the beam hitting the wire in this configuration was calculated and was assumed to be hitting the whole wire, as shown on the right, hence reducing the flux while keeping the number of particles on the wire constant. This was a simplification used in simulations.

For the background gas, which is mainly H_2 , we find that with a pressure of 1.5×10^{-2} mbar and temperature 298.15 K, the mean free path is $\lambda_{H_2} = 0.0074$ m, and with the characteristic length of chamber walls to the wire $L_{char} = 16.5$ cm, we get $Kn = 0.0448$ which is very close to the continuum regime of flow. But, the mean free path of background gas (λ_{H_2}), although very close, is greater than the distance of the nozzle to the wire (t), which is the characteristic length of the beam with respect to the wire.

$$t = 5\text{ mm} < \lambda_{H_2}^{bkg} = 7.4\text{ mm} \quad (3.5)$$

Hence, even though low, there was still a finite probability that the beam gas particles would reach the wire, and not all of the beam would be completely diffused in the background gas at the wire position.

Now, the flow rate of hydrogen used in the experiment is $f_{H_2} = 0.37$ mln/min, which corresponds to a particle flow rate of $\dot{N} = 1.66 \times 10^{17}$ molecules/sec. From calculations in [19], the hydrogen beam impinging from the nozzle is in a molecular flow regime, and hence an ideal effusive beam can be considered coming out of the nozzle. This means that the particle density in the beam will follow a cosine law. For simplicity, the initial simulations in this thesis were conducted assuming a uniform particle density distribution in the beam. Due to the proximity of the wire to the nozzle, only a part of the length of the wire is actually hit by the beam. The length of the wire hit by the beam $l_{eff} = 5.77\text{ mm} \ll l_{wire} \approx 36\text{ mm}$ for an opening angle of 60° for the beam.

But simulations could not produce an analytical solution when a sharp box function was applied to the beam transversal length. Hence, the simulations in this thesis were carried out assuming the beam impinged over the entire wire length while conserving the total number of particles hitting the wire. This means, for the 60° opening angle, the flux of the beam at the position of the wire is

$$\begin{aligned}\phi_{H_2}^{true} &= \frac{\dot{N}_{H_2}}{A_{beam}} \\ &= \frac{\dot{N}_{H_2}}{\pi(t \cdot \tan(\theta))^2}\end{aligned}\quad (3.6)$$

where t is the distance of the wire from the nozzle and $\pi(t \cdot \tan(\theta))^2$ is the area of the beam at that position, with half-opening angle of $\theta = 30^\circ$. With this flux, the total particles hitting the wire become,

$$\dot{N}_{wire} = \phi_{H_2}^{true} \cdot d \cdot l_{eff} \quad (3.7)$$

If this total number of particles is assumed to be hitting the whole wire and not just the l_{eff} transversal length, the flux becomes,

$$\begin{aligned}\phi_{H_2} &= \frac{\dot{N}_{wire}}{d \cdot l} \\ &= \frac{\phi_{H_2}^{true} \cdot l_{eff}}{l}\end{aligned}\quad (3.8)$$

This essentially reduced the flux of the beam impinging on the dx element of the wire, without affecting the total number of molecules of hydrogen hitting the wire. This will affect the temperature distribution along the length of the wire, but the mean values of temperature and resistance will remain the same.

3.4 Sources of resistance change of the wire detector

The main challenge of the wire detector is extracting the atomic hydrogen-induced heating from the bath of various heating and cooling sources on the wire, using a single metric: the wire resistance. To get an idea about the relative strength of different heat loads on the wire, as done in [43] for the recombination, the heat loads were analysed with conduction along the wire to solve the differential equation for a temperature distribution along the wire. This section will describe all heat components expected to affect the wire resistance, the simulation of those heat loads and the effects of different parameters on the wire resistance.

As a convention, all heat sources, i.e., those that impart heat to the wire, are considered positive, and heat sinks that cool the wire are considered negative. For thermalisation along the wire, to make the differential equation analytically solvable, linearising assumptions were made for the heat loads with respect to temperature. With such approximations, all heat loads (in W) produced a differential equation of a common form as below:

$$T''(x) + aT(x) + b = 0 \quad (3.9)$$

The general solution to this equation, within reasonable approximations from real positive physical constants ($a > 0$), is given by

$$T(x) = -\frac{b}{a} + C_1 \cos(\sqrt{a}x) + C_2 \sin(\sqrt{a}x) \quad (3.10)$$

For a particular solution,

$$\left. \frac{dT}{dx} \right|_{x=0} = 0 \quad (3.11)$$

was used as a boundary condition for symmetry along the wire length, and the wire endpoints were considered to be thermalised with the wire holder, so that they are always at room temperature.

$$T(l/2) = T(-l/2) = 298.15 \text{ K} \quad (3.12)$$

With the above boundary conditions and the particular values of a and b for each heat-load case, a temperature distribution along the wire was obtained. This distribution was averaged over the length of the wire to calculate a mean temperature change, and hence a mean resistance change.

$$\begin{aligned} \Delta T_{mean} &= \frac{1}{l} \int_{-l/2}^{l/2} \Delta T(x) dx \\ \Delta R_{mean} &= R_0 \alpha_\rho \Delta T_{mean} \end{aligned} \quad (3.13)$$

Additionally, the simulation was carried out for different combinations of heat loads that were not individually controllable in the experimental setup. Note that all these calculated temperature and resistance changes are after the heat loads have stabilised and an equilibrium is reached, since this calculation did not include any time evolution, and hence essentially only calculates the steady state value³.

Also, all these calculations were done for two wire configurations, Wire-1 ($l_1 = 36 \text{ mm}$, $d_1 = 25 \text{ } \mu\text{m}$) of resistance $R_1 = 4.017 \text{ } \Omega$ and Wire-2 ($l_2 = 37.7 \text{ mm}$ and $d_2 = 5 \text{ } \mu\text{m}$) of resistance $R_2 = 107.523 \text{ } \Omega$, but only relevant calculations are shown below.

3.4.1 Conduction along the wire

Assuming a simple 1-dimensional approximation for the length of the wire, we can use Fourier's law of heat transfer to calculate the total heat load on an infinitesimal element

³In principle, the heat loads can also be time evolved in steps using the heat capacity of the wire, as done in [44].

of the wire due to conduction with its neighbouring elements. Hence, the heat flux in W/m^2 is given

$$q = -k\nabla T \quad (3.14)$$

where k is the thermal conductivity of the wire. Since it is a very thin wire and conduction along the length is the dominant contribution, we will only consider the temperature gradient along the wire (x-direction). So, the total heat load on an infinitesimal length dx of the wire:

$$\begin{aligned} Q &= A \cdot (q(x) - q(x + dx)) \\ &= \left[-k \frac{dT}{dx} \Big|_{x=x_0} - \left(-k \frac{dT}{dx} \Big|_{x=x_0+dx} \right) \right] \cdot \pi \frac{d^2}{4} \\ &= \left[-k \frac{dT}{dx} \Big|_{x=x_0} + k \frac{dT}{dx} \Big|_{x=x_0} - k \frac{d^2T}{dx^2} dx \right] \cdot \pi \frac{d^2}{4} \\ &= -k \frac{d^2T}{dx^2} \cdot \pi \frac{d^2}{4} dx \end{aligned} \quad (3.15)$$

where d is the diameter of the wire. The temperature dependence of the thermal conductivity of tungsten was neglected.

3.4.2 Probe Current

For the four-wire resistance measurement, 2 wires carry a probe current of $I_{probe} = 1 \text{ mA}$ ⁴ through the wire under test. Due to the wire's extremely small thickness, this current also changes the wire's resistance. The electric heating from the probe current on a particular infinitesimal element dx of the wire is:

$$Q = I_{probe}^2 \left(\frac{\rho_R dx}{\pi \left(\frac{d}{2} \right)^2} \right) \quad (3.16)$$

For simulation of the temperature distribution, we can equate the conduction on the element dx to the probe current heating:

$$\begin{aligned} -k \frac{d^2T(x)}{dx^2} \cdot \pi \frac{d^2}{4} dx &= I_{probe}^2 \left(\frac{\rho_R dx}{\pi \left(\frac{d}{2} \right)^2} \right) \\ &= I_{probe}^2 \left(\frac{\rho_0 (1 + \alpha_R T(x)) dx}{\pi \left(\frac{d}{2} \right)^2} \right) \end{aligned} \quad (3.17)$$

As mentioned previously, solving this differential equation and averaging over the length gives us a mean ΔT from the probe current. For the wire of $25 \text{ } \mu\text{m}$ diameter, this

⁴As specified by the HP34401A multimeter.

becomes $\Delta T = 0.34001 \text{ K}$. From this, we can calculate the resistance change in the wire with a base resistance of $R_0 = 4.10696 \Omega$

$$\begin{aligned}\Delta R &= R_0 \alpha_p \Delta T = 0.00628 \Omega \\ R_{measured}^{Expected} &= R_0 + \Delta R = 4.11324 \Omega\end{aligned}\tag{3.18}$$

The measured value of the $25 \mu m$ diameter wire at a chamber pressure of 2×10^{-2} mbar was

$$R_{measured} \approx 4.301 \Omega\tag{3.19}$$

Similarly, for the $5 \mu m$ wire, the probe-current heating leads to a $\Delta T = 506.96 \text{ K}$. Since the base resistance for this wire is $R_0 = 107.523 \Omega$, the resistance change is

$$\begin{aligned}\Delta R &= R_0 \alpha_p \Delta T = 245.292 \Omega \\ R_{measured}^{Expected} &= R_0 + \Delta R = 352.815 \Omega\end{aligned}\tag{3.20}$$

The measured value of the $5 \mu m$ diameter wire at a chamber pressure of 3×10^{-6} mbar was

$$R_{measured} \approx 158.8 \Omega\tag{3.21}$$

We expect this large discrepancy between the measured and calculated resistance of the $5 \mu m$ wire, even at lower pressure, to be due to the significant radiative cooling of the wire by the chamber (See Table 3.1).

Since the probe current is a constant heat load, the temperature and resistance changes of the wire caused by it can be treated as a constant offset added to all other heat-load calculations. In principle, the probe current heating can never be decoupled from other heat loads. Especially for heat loads that depend on the wire temperature, probe current heating will play a significant role, as it inevitably heats the wire. Hence, for all subsequent heat loads, the simulation was carried out, including probe current heating, to account for the effect of the higher wire temperature. In addition, to ensure a fair comparison, the temperature change due to the heat load was calculated by subtracting the temperature distribution from the probe current heating. For example, for the atomic hydrogen recombinative heat load, the differential equation from

$$Q_{conduction} + Q_{Current} + Q_{Recomb} = 0\tag{3.22}$$

was solved and the resistance change by recombination was obtained by

$$\begin{aligned}\Delta T_{Recomb}(x) &= T_{Recomb+Current}^{sim}(x) - T_{Current}^{sim}(x) \\ \Delta T_{mean}^{Recomb} &= \frac{1}{l} \int_{-l/2}^{l/2} \Delta T_{Recomb}(x) \\ \Delta R_{mean}^{Recomb} &= R_0 \alpha_p \Delta T_{mean}^{Recomb}\end{aligned}\tag{3.23}$$

3.4.3 Cooling from the Beam Gas

The recombination heat produced by a pair of atomic hydrogen recombining on the wire is much higher than the kinetic energy of a hydrogen molecule. But, due to the low degree of dissociation that we expect in the hydrogen beam (5%, see Sec. 3.3), the cooling from the beam gas can potentially drown the atomic hydrogen heating signal. This beam gas cooling remains non-negligible even at room temperature because the probe current heats the wire above room temperature. Additionally, if the beam gas is cooled with the nozzle, this will lead to an increased cooling on the wire.

The total heat load on the wire from the gas directly comes from the kinetic energy of the gas:

$$Q = \dot{N}C\Delta T \quad (3.24)$$

where $\dot{N}$ is the rate of particles hitting the wire, and C is the heat capacity of the gas. The average fraction of this total gas kinetic energy in the free-molecular regime that can be transferred to the wire is given by an empirical factor called the accommodation coefficient (β). It is defined as the

$$\beta_E = \frac{E_{in} - E_{re}}{E_{in} - E_W} \quad (3.25)$$

where E_{in} is the incident energy flux of the particles, E_{re} is the reflected energy flux, and E_W is the reflected energy flux that would be achieved if the reflected particles were emitted in thermal equilibrium with the wire [45]. On account of this being an empirical factor, different measurements of hydrogen on tungsten provide different values, and hence, the values were considered to be $\beta_H^W = \beta_{H_2}^W = 1$ to calculate the maximum cooling effect.⁵ Hence, the total heat load from the beam gas becomes

$$Q_{total} = \dot{N}_{H_2}C_{H_2}(T_{wire} - T_{H_2}) + \dot{N}_HC_H(T_{wire} - T_{H_2}) \quad (3.26)$$

The number of particles hitting an infinitesimal amount dx of the wire is given by,

$$d\dot{N} = \phi_N d \quad (3.27)$$

where, ϕ_N is the flux of the particles, from Eq. 3.8 and d is the diameter of the wire. With the degree of dissociation (α),

$$\begin{aligned} d\dot{N}_{H_2} &= (1 - \alpha)\phi_{H_2}^{undissociated} d \, dx \\ d\dot{N}_H &= 2\alpha\phi_{H_2}^{undissociated} d \, dx \end{aligned} \quad (3.28)$$

Additionally, assuming the atomic hydrogen gas to be an ideal monoatomic gas and the molecular hydrogen gas to be an ideal diatomic gas,

$$\begin{aligned} C_{H_2} &= \frac{5k_B}{2} \\ C_H &= \frac{3k_B}{2} \end{aligned} \quad (3.29)$$

⁵A fitting of the simulation to a measurement for calibrating this coefficient can potentially be done, but is not done here due to time constraints.

Hence,

$$Q_{Beam} = (1 - \alpha)\phi_{H_2}^{undissociated} d dx \frac{5k_B}{2}(T_{wire} - T_{H_2}) + 2\alpha\phi_{H_2}^{undissociated} d dx \frac{3k_B}{2}(T_{wire} - T_{H_2}) \quad (3.30)$$

3.4.4 Recombination heat of Atomic Hydrogen

The most important heat load is the heat of recombination of atomic hydrogen on the wire. This, for an element dx on the wire

$$Q_{rec} = \eta_{rec} \frac{E_{rec}}{2} \phi_H d dx = \eta_{rec} \frac{E_{rec}}{2} 2\alpha\phi_{H_2}^{undissociated} d dx \quad (3.31)$$

where $2E_{rec}$ is the recombination energy when one hydrogen molecule is formed.

One of the most important uncertainties in the calculation of the recombination resistance change is the recombination efficiency η_{rec} . This consists of various probability factors that cause the wire's resistance to change due to recombination. This includes the sticking coefficient α_H , which represents the probability of atomic hydrogen getting stuck to the surface of the Tungsten wire, the adsorption coefficient γ_S , which represents the probability of the atomic hydrogen to get adsorbed in the surface for the recombination to occur, and finally the accommodation coefficient β_H , which represents the fraction of the released recombination energy that gets absorbed by the wire instead of being carried away by the formed H_2 [44] [45].

$$\eta_{rec} = \alpha_H \cdot \gamma_S \cdot \beta_H \quad (3.32)$$

Additionally, these coefficients were for a clean metal wire, and the coefficients are expected to drop significantly if the wire is not clean⁶. Hence, a humble approximation of $\eta_{rec} = 0.1$ was used in calculations, as a value close to the one from [45].

As in previous sections, we can average the temperature distribution over the wire length and thus obtain the expected change in resistance from a 5% dissociation fraction. For wire of $25 \mu m$ diameter, this becomes $\Delta T = 0.115 K$. From this, we can calculate the resistance change in the wire with a base resistance of $R_0 = 4.10696 \Omega$

$$\Delta R = R_0 \alpha_p \Delta T = 0.0021 \Omega \quad (3.33)$$

Similarly, for the $5 \mu m$ wire, recombination heating leads to a $\Delta T = 1.313 K$. Since the base resistance for this wire is $R_0 = 107.523 \Omega$, the resistance change is

$$\Delta R = R_0 \alpha_p \Delta T = 0.635 \Omega \quad (3.34)$$

⁶In an attempt to clean the wire, we applied increasing amounts of current until the Tungsten wire was glowing hot red, similar to a light bulb. We expected this to evaporate away all dirt and impurities.

3.4.5 Cooling from the Background Gas in the chamber

One unexpected cooling load on the wire was due to the background gas in the chamber. As mentioned before, the chamber background pressure was 5×10^{-4} mbar, and with the gas flow on at 0.37 ml/min, it increased to about 1.5×10^{-2} mbar. This led to a huge cooling just from the background gas thermalised to the chamber walls at room temperature on the wire at an elevated temperature due to probe current heating.

For estimating the cooling from the background gas, the flux of background H_2 gas on the walls, within the ideal gas approximation, is given by

$$\phi = \frac{P}{4k_B T} \sqrt{\frac{8k_B T}{\pi m_{H_2}}} \quad (3.35)$$

So, background cooling on an infinitesimal element dx of the wire becomes

$$\begin{aligned} Q_{bkg} &= \beta_{H_2} \pi d dx C_{H_2} \phi(P_{bkg}, T_{chamber})(T_{wire} - T_{chamber}) \\ &= \beta_{H_2} \pi d \cdot dx \frac{P_{bkg}}{4k_B T_{chamber}} \sqrt{\frac{8k_B T_{chamber}}{\pi m_{H_2}}} \frac{5k_B}{2} (T_{wire} - T_{chamber}) \end{aligned} \quad (3.36)$$

where β_{H_2} is the accommodation coefficient of H_2 gas on Tungsten wire, representing the fraction of energy delivered to the wire. For estimating the maximum cooling load, it was set to 1. This cooling load was solved with conduction and probe current to calculate a steady-state mean change in temperature and resistance. Additionally, the dependence of this mean resistance on the background gas pressure was simulated. Fig. 3.4 shows the background pressure dependence of the resistance change from this cooling, and a similar functional dependence was reproducibly measured experimentally while pumping or venting the chamber. A closer look at the curve revealed that the background pressure in the chamber with gas flow sits at a very steep point in the background pressure vs. resistance curve ($P_{bkg} = 1.5 \times 10^{-2}$ mbar), making the resistance very sensitive to small changes in background pressure. This was found to be the reason the atomic hydrogen signal was drowned out. Efforts to eliminate this to extract the atomic hydrogen signal are discussed in Sec. 3.6.

3.4.6 Radiative Cooling

The proximity of the wire to the cold nozzle (7 K in the main experiment) makes the wire highly sensitive to radiative cooling from the nozzle. Even when the nozzle is controlled at room temperature, since the wire is always at an elevated temperature due to the probe current, the nozzle radiation inevitably leads to a cooling effect on the wire.

For a gray body with emissivity $\epsilon < 1$, the Stefan-Boltzmann Law gives the radiative exitance (in W/m^2) as

$$M = \sigma \epsilon T^4 \quad (3.37)$$

where σ is the Stephan-Boltzmann constant. Multiplying the exitance by the surface area of the wire gives us the radiative cooling power. Since the wire is only 0.5 cm away

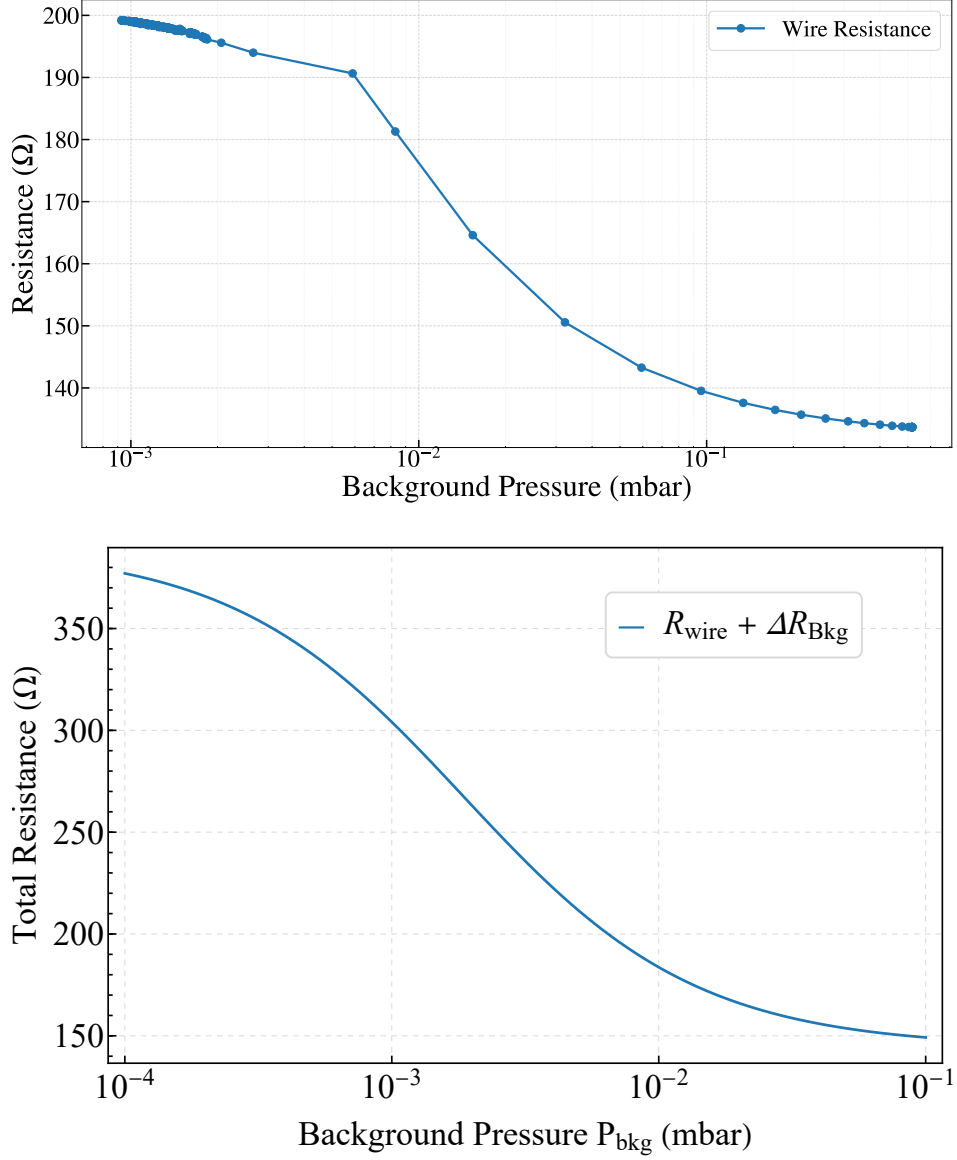


Figure 3.4: a) The figure shows the pressure dependence of resistance, measured reproducibly while pumping or venting the chamber. The dominant background gas cooling can be explained by the steep dependence of the curve at our operating chamber pressure when the discharge is on, $P_{bkg} = 1.5 \times 10^{-2}$ mbar. The kink observed was reproducibly present in the curve, potentially arising from another competing heat load, primarily the radiation cooling from the nozzle. Simulation to reproduce the kink did not work out due to the unknowns of emissivities. b) The simulated curve for the pressure dependence of steady state resistance after solving with conduction and probe current heat loads. Only the functional dependence matched the measurement, whereas the resistance and pressure values did not, potentially due to neglected heat loads and their crosstalk in the simple simulation. Both simulation and measurement are for Wire-2.

from the large nozzle, we assume that all background radiation on the wire is from the nozzle itself. Additionally, assuming the wire is in thermal equilibrium, the emissivity and absorptivity of the wire can be taken as equal ($\epsilon_W = \alpha_W^{rad}$) under Kirchhoff's law of Thermal Radiation. Hence, the radiative cooling load on an infinitesimal length dx of the wire is

$$Q_{rad}^{Nozz} = -\pi d dx \sigma \epsilon_W \epsilon_{Cu} T_{nozzle}^4 \quad (3.38)$$

Additionally, the wire element dx radiatively cools itself by

$$Q_{rad}^{wire} = \pi d dx \sigma \epsilon_W T_{wire}^4 \quad (3.39)$$

So, the total radiative cooling on an element dx of wire

$$Q_{rad} = \pi d dx \sigma \epsilon_W T_{wire}^4 - \pi d dx \sigma \epsilon_W \epsilon_{Cu} T_{nozzle}^4 \quad (3.40)$$

In a more accurate description, a fraction of the solid angle of the radiative cooling seen by the wire comes from the chamber walls at room temperature, which we could estimate with the approximation of the chamber as a black body emitter ($\epsilon = 1$) as

$$Q_{rad}^{Chamber} = -\pi d dx \sigma \epsilon_W T_{chamber}^4 \quad (3.41)$$

But since the nozzle is extremely close to the wire, in the final setup with a cold nozzle, the nozzle temperature will play the dominant role; hence, only the nozzle's radiative cooling was included in the preliminary calculations.

Even within these approximations, the uncertainty in the values of emissivity of a copper nozzle (ϵ_{Cu}) and that of the tungsten wire (ϵ_W) makes it difficult to precisely calculate the cooling power from radiation. Additionally, the radiative cooling from the nozzle is always coupled with the background gas cooling on the wire, which also includes different fractions due to thermalisation with the nozzle and the chamber walls, and contains another unknown, the accommodation coefficient (β_E). In principle, calibrative fitting to measured data could provide insight into removing the effects of both cooling factors simultaneously, but it will not be feasible to decouple them completely. Hence, to calculate the maximum cooling load, the emissivity $\epsilon_{Cu} = 1$ was considered⁷, and the value of $\epsilon_W = 0.32$ was taken from [46].

Additionally, for having an analytically solvable differential equation while solving with conduction, a linearising assumption was made by considering the wire temperature (T_{wire}) to be very close to the nozzle temperature (T_{nozzle}) and hence ignoring the higher order terms

$$\begin{aligned} Q_{rad} &= \pi d dx \sigma \epsilon_W T_{wire}^4 - \pi d dx \sigma \epsilon_W T_{nozzle}^4 \\ &= 4\pi d dx \sigma \epsilon_W T_{nozzle}^3 (T_{wire} - T_{nozzle}) \\ &\quad + O(T_{wire} - T_{nozzle})^2 \end{aligned} \quad (3.42)$$

⁷This is completely incorrect, since Copper, being a metal, is not at all a black body. Value of ϵ_{Cu} is generally 0.07 [46]. But at room temperature, anyway, the chamber radiative cooling will dominate, which can be considered a black body, and hence the value $\epsilon = 1$ is a fair assumption at room temperature.

Using this linear approximation, again the differential equation was solved to calculate the average temperature and resistance change⁸.

3.5 Comparison of heat loads

For a quick first trial, the aforementioned heat loads were solved for steady state, only with the conduction along the wire for an order-of-magnitude comparison. But a closer look revealed that, except for recombination heating from atomic hydrogen, all other heat loads depend on the wire temperature, which is essentially determined by probe-current heating. Since this heating is inevitable, in the second step, each of the heat loads was solved for steady state with conduction along the wire and the probe current heating to find a temperature distribution along the wire.

Fig. 3.5 shows the comparison of these simulated temperature distributions for the condition of our test discharge setup for the 25 μm wire and 5 μm wire, given by

$$\Delta T_{load}(x) = (T_{load+Curr}(x) - T_{Curr}(x)) \quad (3.43)$$

The calculated mean temperature and resistance changes after removing the effect of probe current heating are shown in Table 3.1, as given by Eq. 3.23. As we can see in both plots, the background gas cooling load remains the dominant factor in both wire configurations and is significantly higher than the signal of interest, the recombination heating. Additionally, we can see that the heat loads increase drastically when calculated for the lower diameter wire. This is because, due to increased base resistance, the probe current heating increases drastically, raising the wire temperature and, in turn, the cooling from the "temperature-dependent" cooling loads. It is important to note that this simulation does not include crosstalk between different heat loads due to their inherent dependence on the temperature difference with the wire. In the description closer to the experimentally observable effect, one should calculate a heat load and its temperature change by simulating a steady state of all heat loads with and without the heat load under consideration. The difference between the two should indicate the effective contribution of the heat load in crowding the signal.

⁸Note that the linearising assumption is more and more inaccurate, the colder the nozzle gets, and hence the differential equation has to be solved numerically without the approximation.

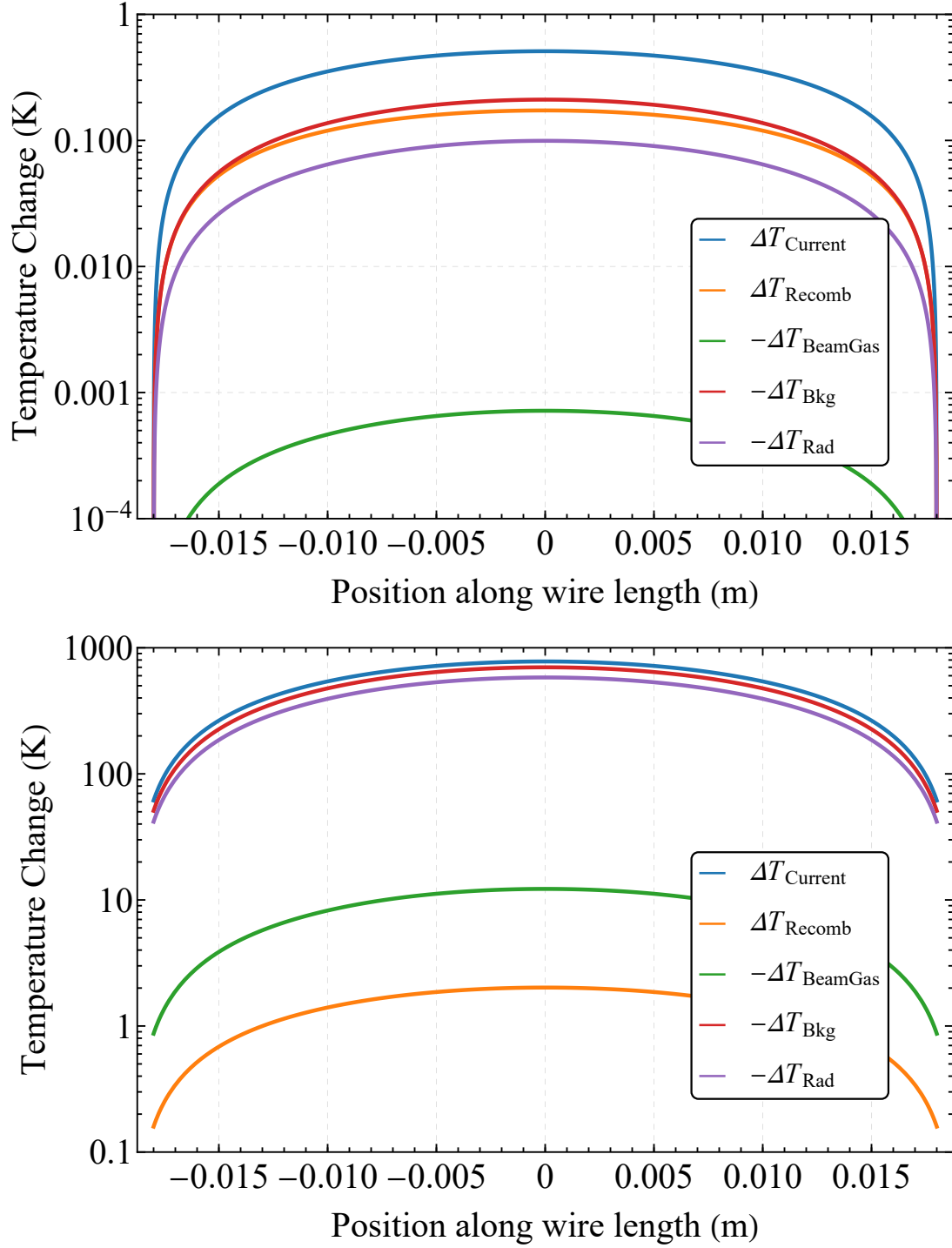


Figure 3.5: Calculation of heating rate from different heat loads by solving for steady state with conduction and probe current heating, after removing the probe current contribution, for a) wire of 25 μm diameter and b) wire of 5 μm diameter. Both calculations assume $P_{\text{bkg}} = 1.5 \times 10^{-2}$ mbar, $T_{\text{bkg}} = 298.15$ K, $T_{\text{nozzle}} = T_{\text{beam}} = 298.15$ K, degree of dissociation $\alpha = 0.05$ and H_2 flow of 0.37 ml/min. The dominant cooling rate of the background gas was evident in measurements.

Heat Loads	For 25 μm wire ($R_0 = 4.10696 \Omega$)		For 5 μm wire ($R_0 = 107.523 \Omega$)	
	ΔT_{avg} (K)	ΔR_{avg} (Ω)	ΔT_{avg} (K)	ΔR_{avg} (Ω)
Electric Heating ($I_{probe} = 1 mA$)	0.340011	0.00628	506.958	245.292
Cold beam Gas (flow= 0.37 mln/min and $\alpha = 0.05$, $T_{gas} = 298.15 K$)	-4.59×10^{-4}	-8.48×10^{-6}	-7.79	-3.77
Recombination of atomic H ($\eta_{rec} = 0.1$)	0.115	0.0021	1.313	0.635
Cooling from background gas while discharge on ($T_{gas} = 298.15 K$ and pressure at 1.5×10^{-2} mbar)	-0.135	-0.0025	-449.057	-217.277
Radiative cooling ($T_{nozzle} = 298.15 K$, $\epsilon_W = 0.32$, $\epsilon_{Cu} = 1$)	-0.0636	-0.0012	-371.679	-179.837

Table 3.1: Effect of different heat loads on the wire of diameter $d = 25 \mu m$ and $d = 5 \mu m$.

3.5.1 Effect of the Wire Diameter

To further investigate these heat loads, the effect of changing wire diameter on them was analysed. In general, decreasing the diameter increases the base resistance of the wire and, hence, the change in resistance due to recombination of atomic hydrogen. This was beneficial since it avoids the need to use very high multimeter resolution for wire measurements, and hence provides a more time-resolved measurement⁹. But this also leads to increased resistance change from all other heat loads. The effect of wire diameter on resistance change from different heat loads is shown in Fig. 3.6

Essentially, all heat loads are proportional to the wire's surface area $\pi d dx$, and the conduction along the wire is proportional to the wire's cross-sectional area $\pi(d/2)^2$. Hence, when solving for the steady state only with conduction, only the $1/d$ dependence remains. In contrast, the probe current heating is inversely proportional to the wire's cross-sectional area (through resistance $R \propto 1/d^2$), and hence, when solved with

⁹That is why the wire was changed from 25 μm to 5 μm thickness, since the 25 μm wire produced a resistance change with atomic hydrogen, barely above the least count of the multimeter.

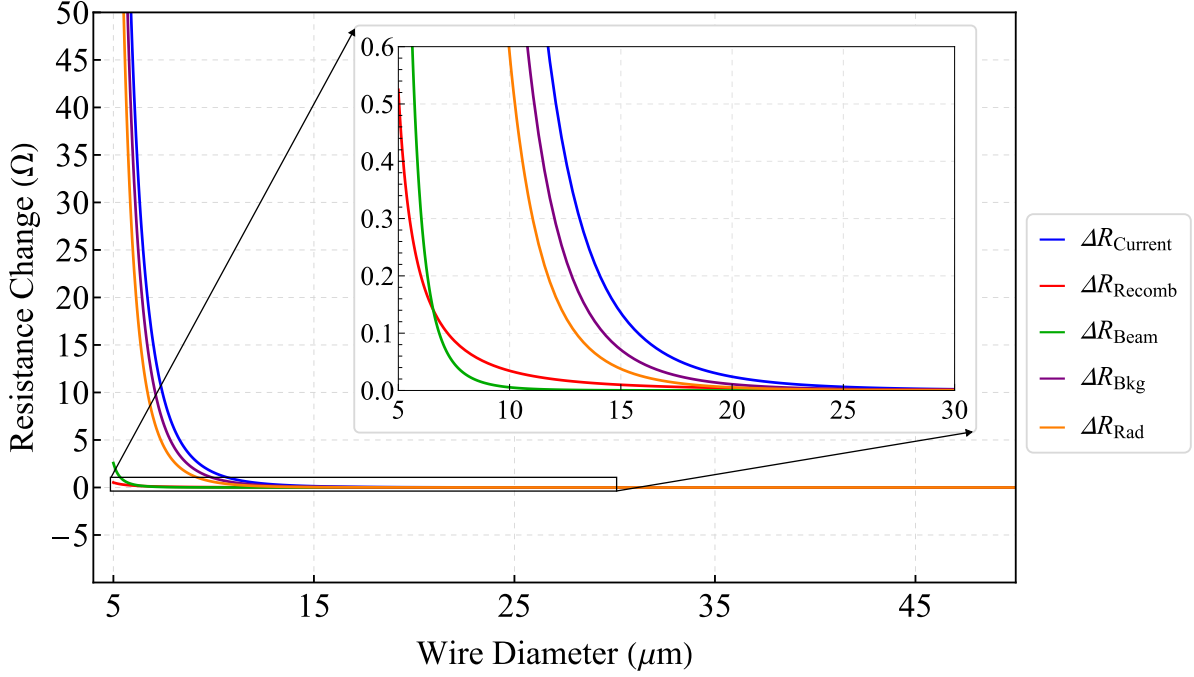


Figure 3.6: Effect of the changing wire diameter on the steady state resistance change of each heat load. We can see that, due to the $1/d^4$ relation of probe current heating, it rises sharply at low diameters, and hence so do all heat loads that depend on wire temperature. On the other hand, recombination heating does not depend on the wire resistance and hence remains at a comparatively low value.

conduction, leads to a sharp dependence on the wire diameter of the order $\propto 1/d^4$.

$$\begin{aligned}
 Q_{\text{conduction}} &= -k \frac{d^2 T(x)}{dx^2} \cdot \pi \frac{d^2}{4} dx \propto d^2 \\
 Q_{\text{rad}} &= \pi d dx \sigma \epsilon_W (T_{\text{wire}}^4 - T_{\text{nozzle}}^4) \propto d \\
 Q_{\text{Current}} &= I_{\text{probe}}^2 \left(\frac{\rho_0 (1 + \alpha_R T(x)) dx}{\pi (\frac{d}{2})^2} \right) \propto \frac{1}{d^2} \\
 Q_{\text{Recomb}} &= \eta_{\text{rec}} E_{\text{rec}} \alpha \phi_{H_2}^{\text{undissociated}} d \cdot dx
 \end{aligned} \tag{3.44}$$

This can be seen in the sharply rising $\Delta R_{\text{current}}$ curve in Fig. 3.6. This increase leads to a sharp rise in the wire's temperature, as reflected in the increased ΔR for different heat loads when solved by including the probe-current heating. In contrast, ΔR_{recomb} has a much more gradual slope and stays at a very low value even at $d = 5 \mu\text{m}$ due to it being independent of the wire temperature.

3.5.2 Interplay between Recombination heating and Beam Gas cooling

Essentially, if other parameters are well controlled and stabilised, the only active heat loads on the wire during discharge operation are beam gas cooling and recombinative heating. To this end, $Q_{beamgas}$, Q_{recomb} and $Q_{Current}$ were solved together with the conduction to simulate the total resistance change and investigate the dependence of different parameters on this resistance change, most importantly the dissociation fraction (α) and the temperature of the beam gas (T_{H_2}).

Initially, we expected that cooling the beam gas would result in a greater contrast in the wire-resistance signal when comparing discharge-on and discharge-off signals. This is why, in the first measurements, the nozzle was actively cooled with a peltier to $-20^\circ C$. But in a low-dissociation regime, the cooled beam proved disadvantageous when measuring the discharge-on signal. This means the beam gas cooling completely drowned the atomic hydrogen recombinative heating, and both discharge on and off cases produced a cooling effect.

$$Q_{NODischarge} = -Q_{H_2}^{KE}$$

$$Q_{Discharge} = Q_H^{Recomb} - (1 - \alpha)Q_{H_2}^{KE} - Q_H^{KE} \quad (3.45)$$

To investigate this, the interplay between the dissociation fraction α and the temperature of the beam gas T_{H_2} was analysed. Fig. 3.7 shows the changing calculated average resistance change (See Eq. 3.13), with the dissociation fraction (α), for different gas temperatures (T_{gas}) from 5 K to 298 K. At room temperature, we can see a positive resistance change even at very low dissociation fractions, but as the gas gets colder, the minimum required dissociation fraction for a positive ΔR to be seen, increases. For a general beam measurement, sweeping the wire across the beam should produce a heating on the wire for a quick proof of atomic hydrogen. At 5 K, to see such a positive change of resistance from the atomic hydrogen beam, we need a minimum of $\approx 15\%$ dissociation fraction in the beam. It is important to note that this is a rough calculation and does not represent the true dynamics. In reality, a 5 K beam gas will also mean the addition of radiative cooling from a 5 K nozzle, as well as cooling from some part of the background gas thermalised with the nozzle, but essentially these changes can be well stabilised and removed. We can additionally observe that with the 5 μm diameter of the wire, we need very high dissociation fractions to overcome the high cooling load by the beam gas on the highly heated wire from the probe current.

3.6 Discharge On-Off measurement

Since the background pressure in the chamber was a very large cooling load, it was decided to perform a on-off measurement by first stabilising all parameters, and then periodically turning the discharge on and off. Such a measurement should allow us to

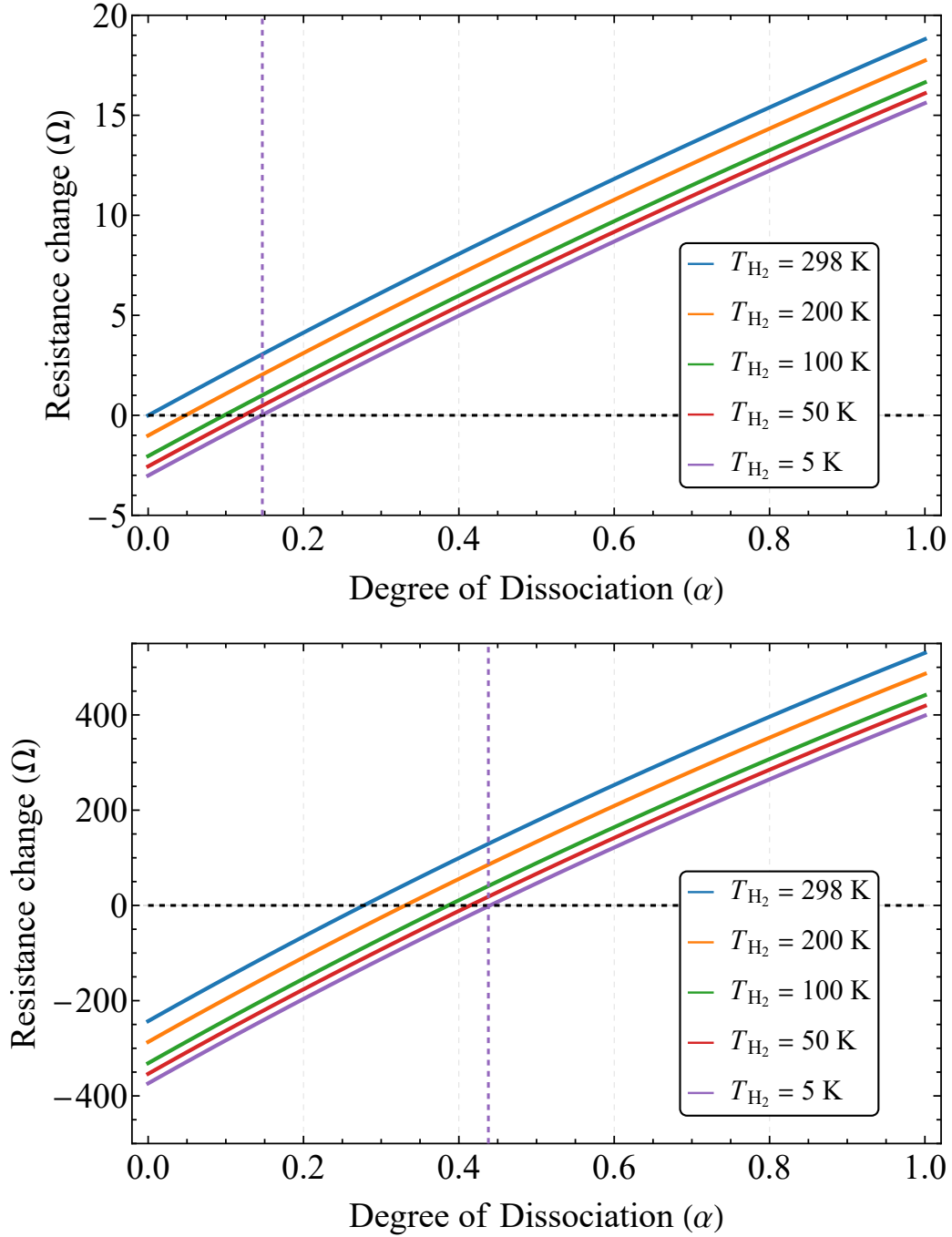


Figure 3.7: Mean simulated steady state resistance change by solving $Q_{Conduction}$, $Q_{Current}$, Q_{Recom} and $Q_{Beamgas}$ together, for different degree of dissociations α and beam gas temperatures T_{H_2} for wire diameter of $d = 25 \mu m$ in a) and $d = 5 \mu m$ in b). Regions below zero resistance change represent the drowning of the atomic hydrogen recombination signal by beam-gas cooling and effectively produce a cooling on discharge-on. Simulation at the normal flow of 0.37 mln/min. As seen in the figure, a smaller-diameter wire, since it is heated much more due to the probe current, will cool much more to the beam gas, and hence a higher dissociation fraction is required for the beam to produce a positive change in resistance.

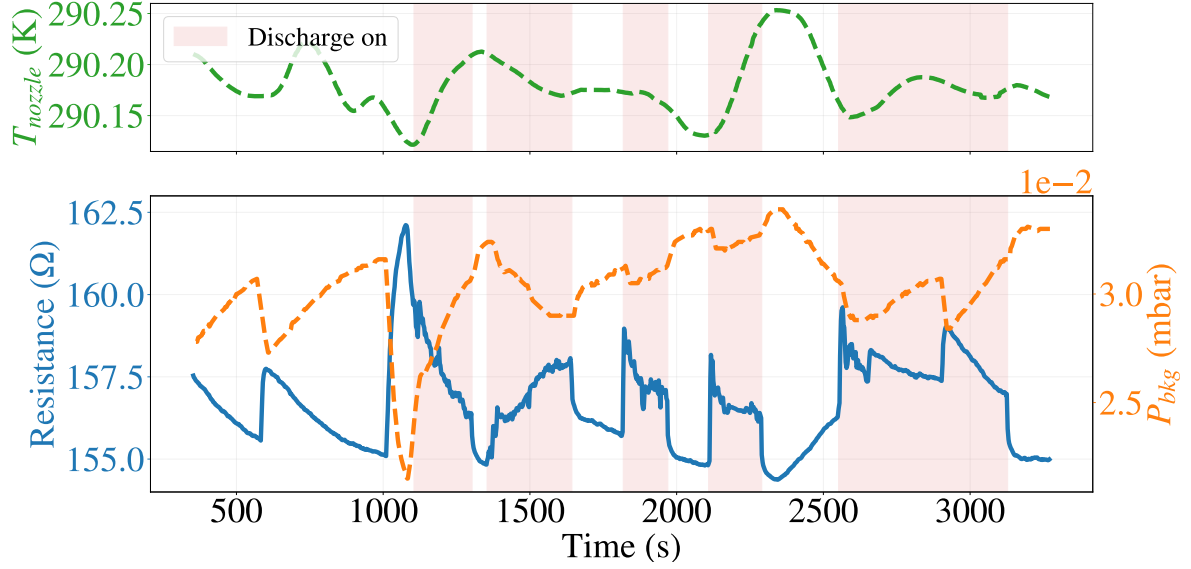


Figure 3.8: Figure shows a time sweep of different parameters in which a discharge on-off measurement for atomic hydrogen signal extraction was performed. The resistance consistently increases at plasma ignition, suggesting the presence of atomic hydrogen, but a corresponding drop in background pressure casts uncertainty on the origin of the increase in resistance.

conclusively observe atomic hydrogen recombinative heating, provided we have a sufficient dissociation fraction, and should reproducibly show only the effect of the beam. Fig. 3.8 shows such a discharge-on-off measurement in search of an atomic hydrogen signal. We can clearly see a characteristic jump in wire resistance every time the discharge is ignited. But the measurement of background pressure revealed that a corresponding dip in background pressure is present every time the discharge is turned on¹⁰, which leads to a reduced cooling on the wire, which could potentially also be the source of the jump in resistance. This put uncertainty on the presence of the atomic hydrogen signal. Hence, an effort was made to correct the resistance signal to completely eliminate the effect of background pressure.

Correcting for background gas cooling

An attempt was made to correct the measured resistance by removing the expected effect of background cooling on the resistance. As discussed in Sec. 3.4.5, the simulated and measured curves of pressure dependence of the resistance showed the same functional dependence, even though the exact values did not match. In principle, to get a simplified

¹⁰The background pressure drop every time the discharge was ignited, was previously also mentioned in [19], the source of which is unknown.

description, the background gas cooling is given by

$$Q_{bkg} = \pi d dx \frac{P_{bkg}}{4k_B T_{chamber}} \sqrt{\frac{8k_B T_{chamber}}{\pi m_{H_2}}} \frac{5k_B}{2} (T_{wire} - T_{chamber})$$

$$Q_{bkg} \propto P_{bkg} \cdot T_{wire} \quad (3.46)$$

In the simplest approximation, assuming the wire is at a steady state and hence all heat loads are constant, we can write

$$T_{wire} \propto \frac{1}{P_{bkg}}$$

$$R_{wire} \propto \frac{1}{P_{bkg}} \quad (3.47)$$

which agrees well with the functional dependence of the curve measured and simulated (See Fig. 3.4). In principle, such a $1/P_{bkg}$ dependence easily fits the measured P_{bkg} vs Resistance data, but because this was an approximation, the optimised fit parameters were not reproducible across different measurement iterations. Hence, the data from Fig. 3.8, with the discharge off, was used for calibration. Fitting a $1/P_{bkg}$ curve to this data, the effect of background pressure on resistance was extracted, as shown in Fig. 3.9. This extracted resistance curve was subtracted from the measured resistance in both the on- and off-discharge regions, using the known background pressure values, to remove the effect of background-gas cooling.

$$R_{fit} = \frac{a}{P_{bkg}^{measured}} + c$$

$$R_{corrected} = R_{measured} - R_{fit} \quad (3.48)$$

Fig. 3.10 shows such a corrected resistance measurement. We can see that the resistance values when the discharge is off are nearly zero after correction, since the dominant determinant of resistance was the background pressure. But there is still a characteristic increase in resistance every time the discharge is turned on, indicating that this is atomic hydrogen recombinative heating. One issue with this correction is that, even though the pressure dependence was removed, the background gas cooling also depends on the temperature difference with the wire, which was not accounted for. Essentially, every time discharge is turned on, the presumed recombination heating increases the wire temperature. With this increased temperature difference with the wire, we still see the background gas cooling, evident from the declining resistance after the first jump upon turning on the plasma, which then settles to a new value. Additionally, negative corrected resistance values during sharp pressure changes indicate that the elimination method cannot be considered entirely foolproof.

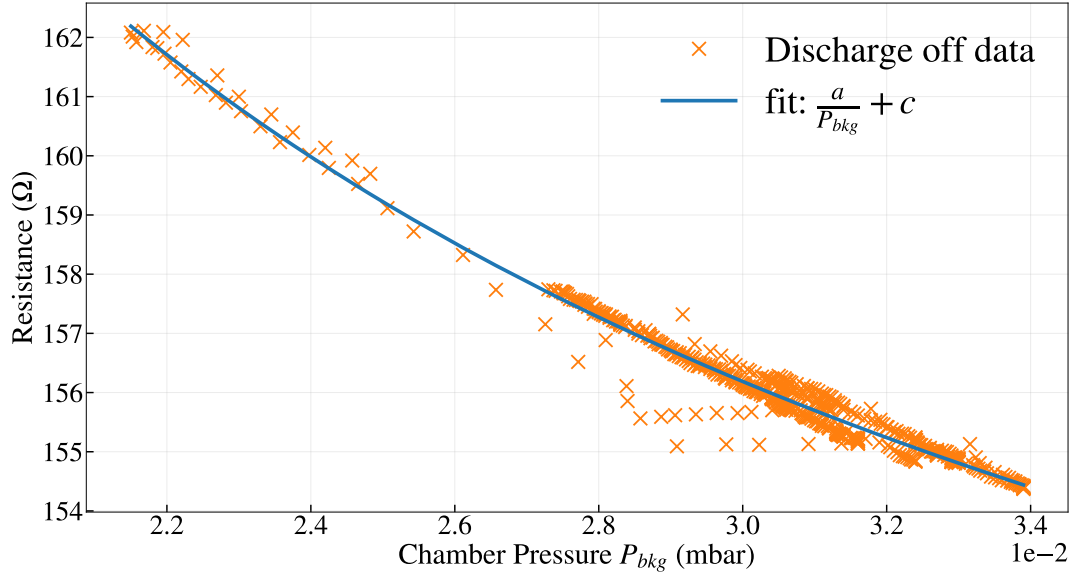


Figure 3.9: Plot of resistance of the discharge-off regions of the on-off measurement in Fig. 3.8 with respect to the measured background pressure, fitted with a $a/P_{bkg} + c$ curve, for eliminating the dependence.

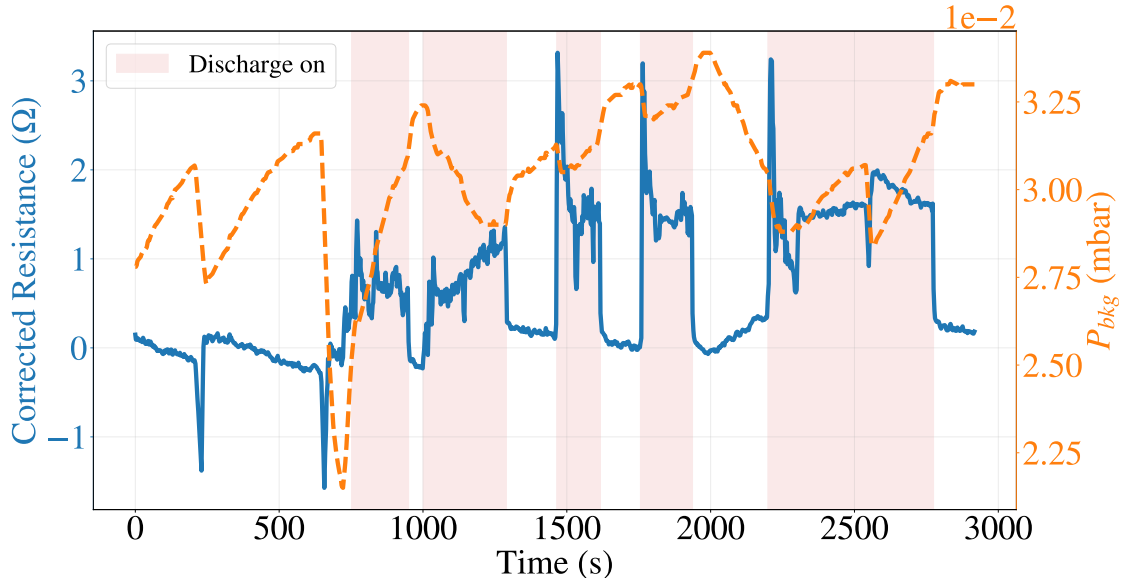


Figure 3.10: The time sweep of the corrected resistance after removing the effect of background gas cooling from the fitting mentioned above. We can see the characteristic jumps in resistance on every discharge ignition remain after correction, indicating the presence of atomic hydrogen. The sharp decrease in resistance after the first jump is the residual temperature dependence of the background cooling effect in the recombinatively heated wire.

3.7 New chamber with lower Background pressure

In general, the ex post facto fitting and elimination were successful for one measurement, but it was better to completely remove the background gas cooling on the wire to obtain a clear atomic hydrogen signal. For attaining a lower background pressure, the vacuum system was rebuilt. In the rebuild, in addition to the lower background pressure, other improvements of the system were made. Firstly, we reduced the nozzle size to minimise unnecessary radiative cooling. We also blocked the nozzle's beam output on the back side and installed another wire detector to mirror the front wire detector. Potentially, this back-wire detector could serve as a normalisation signal to remove the effects of background cooling and nozzle radiation cooling on the front wire, thereby enabling the extraction of only the "beam" heat-load effects.

With the increased pumping speed, we can achieve a background pressure of 10^{-6} mbar, rising to 1.5×10^{-5} mbar with normal gas flow at the discharge, which is very close to the main spectroscopic chamber configuration. With this background pressure, the mean free path of the background gas while the discharge is on, is

$$\lambda_{H_2}^{bg} = 7.4 \text{ m} \gg t = 5 \text{ mm} \quad (3.49)$$

which is much larger than the characteristic length of nozzle to the wire (t), and hence virtually no interference of the background gas with the beam can be assumed.

A quick calculation of the heat loads reveals a significant decrease in the background cooling. Fig. 3.11 shows the comparison, where the background cooling load remains comparable to the atomic hydrogen heating signal. It is important to note that the background cooling is highly overestimated with all gas kinetic energy going to the wire ($\beta_E = 1$) and the recombination heating is underestimated by a humble η_{rec} and α assumption.

Nevertheless, for the first trial, it was decided to reproduce the discharge on-off measurement of atomic hydrogen, like Fig. 3.8. Fig. 3.12 shows a measurement of turning the discharge on and off after stabilising the H_2 flow, discharge pressure, and all other parameters. A very clear, almost 1Ω increase in wire resistance is observed every time the discharge is ignited. We still see the background pressure drop every time the plasma is ignited, but the drop is not significant enough to cause a observed change in resistance (A change in background gas pressure of $\Delta P_{bg} = 0.5 \times 10^{-5}$ mbar leads to a resistance change of only $\approx 6 \text{ m}\Omega$). Additionally, we do not observe the rapid drop in resistance after the first jump, as seen in the previous measurements at higher background pressure Fig. 3.8. The slight step in resistance before the jump on plasma ignition was correlated with turning on the MW power. It is also important to see that the flow of hydrogen is virtually constant for the whole measurement, the discharge pressure only slightly rises due to increased gas temperature in the plasma and much smaller fluctuations are seen in the nozzle temperature, compared to the previous measurement in Fig. 3.8, due to lower background pressure. No systematic correlation is observed in any parameter that would invalidate the presence of atomic hydrogen, thereby confirming the successful operation of the wire detector.

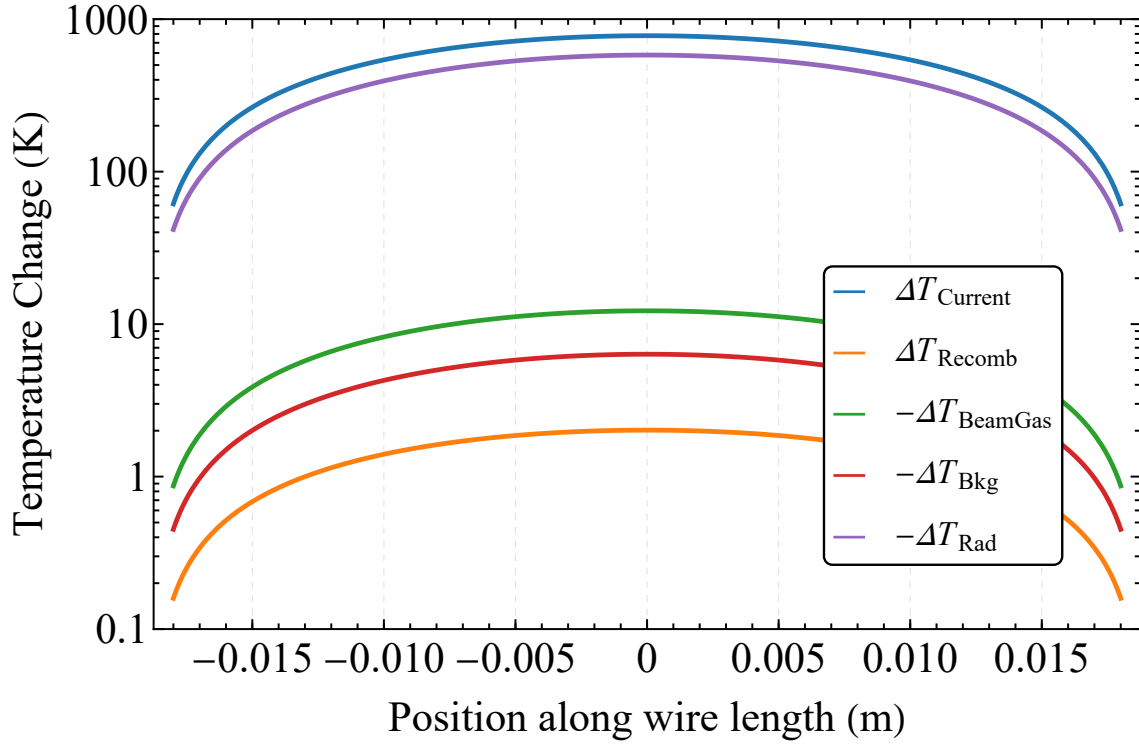


Figure 3.11: Calculation of heating rate from different heat loads by solving for steady state with conduction and probe current heating, after removing the probe current contribution, for the wire of $5\ \mu\text{m}$ diameter. The calculations assume $P_{bkg} = 1.5 \times 10^{-5}$ mbar, $T_{bkg} = 298.15\ \text{K}$, $T_{nozzle} = T_{beam} = 298.15\ \text{K}$, degree of dissociation $\alpha = 0.05$ and H_2 flow of $0.37\ \text{mln/min}$. The previously dominant cooling rate of the background gas is significantly reduced.

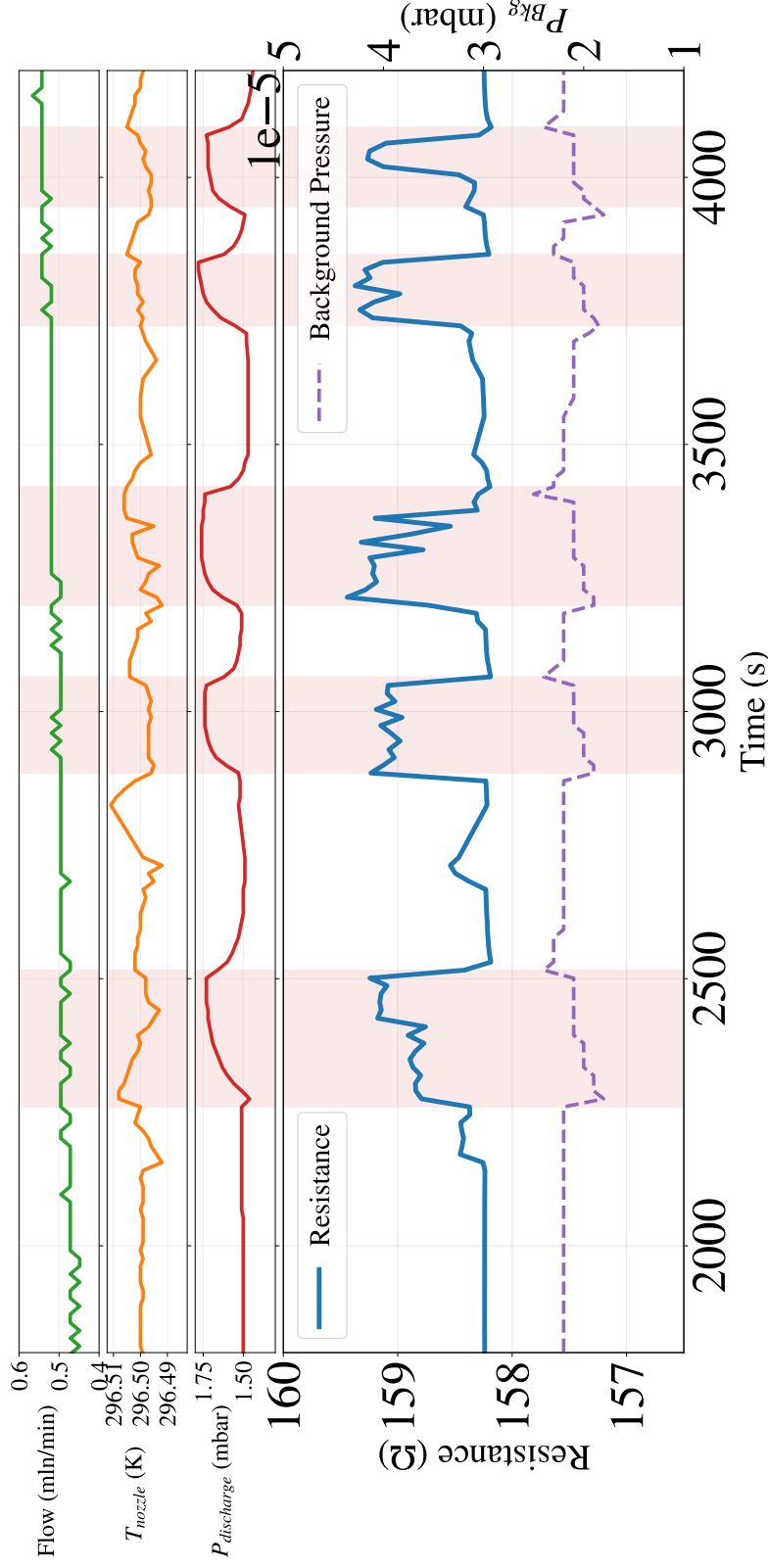


Figure 3.12: Time sweep of resistance with discharge turned on and off periodically, showing characteristic jump in resistance from the atomic hydrogen recombination heating, every time the discharge is ignited. The data is taken using a $5\mu m$ diameter wire in the lower-background-pressure chamber configuration. The plot shows simultaneous measurements of background pressure (P_{bkg}), discharge pressure (P_{dis}), H_2 flow, and nozzle temperature (T_{Nozz}) along with the resistance of the wire. The systematic dip in P_{bkg} is insufficient to account for the increase in resistance. Additionally, the flow of H_2 was virtually constant for the whole measurement, and only slight jumps are observed in the P_{dis} on plasma ignition due to increased gas temperature. All parameters were stabilised before starting the plasma on-off measurement. The characteristic $\approx 1\Omega$ jump every time the plasma is ignited conclusively proves the presence of atomic hydrogen on the wire.

Chapter 4

Conclusion and Outlook

This thesis presents different efforts to quantify the population of the atomic hydrogen beam used for precision spectroscopy of the 1S-3S transition. First of these is the optical emission spectroscopy of the plasma source, used to dissociate molecular hydrogen into atomic hydrogen, to relate the emission intensities of the Balmer- γ line of atomic hydrogen and the Fulcher- α band of molecular hydrogen to their respective populations. This is followed by a thorough investigation of the plasma dynamics to systematically account for the appropriate effects in calculating the degree of dissociation. The precise calculation of many plasma parameters was beyond the scope of this thesis, leading to large uncertainties in the degree of dissociation in the hydrogen plasma. Nevertheless, a qualitative analysis of various operational parameters of the plasma discharge was conducted for a better understanding of the hydrogen discharge for the highest atomic hydrogen population. Additionally, due to uncertainty in the percentage of the atomic hydrogen population in the beam that survives till the PCV, a Calorimetric wire detector was developed for in situ detection of atomic hydrogen. This detects atomic hydrogen by its recombination on the wire surface, and the resulting energy release causes a change in the wire's resistance. To this end, a Vacuum chamber with a test discharge setup was built for wire detector testing. To systematically decouple the different heat loads on the wire, simulations were carried out to estimate strengths and expected changes in resistance under different experimental conditions, and to understand the dependence of different parameters. Eventually, this thesis presents preliminary evidence of atomic hydrogen on the wire, based on discharge-on/off measurements.

Now, efforts are underway to quantitatively extract the atomic hydrogen signal across different experimental conditions and to develop a miniaturised prototype that could be installed in the main spectroscopic chamber beside the PCV. Additionally, using the two tools for quantifying atomic hydrogen, we plan to take comparative measurements to observe a correlated signal in both to confirm good discharge operation and the presence of hydrogen atoms near the PCV.

Appendix A

Hydrogen Discharge Operational Observations

A.1 Additional Emissions observed

Sodium line

One of the most peculiar behaviours observed during the operation of the discharge was the Na-D line at 589 nm appearing in the spectrum, as also observed previously in [15]. This Sodium D-line, as shown in Fig. A.1, was observed to grow rapidly as the discharge tube temperature increased, for example, when cooling was completely turned off. The sodium line disappeared when cooling was turned on again, even though the temperature of the tube was still relatively high. Also, even if the temperature was stabilised to relatively high values (15 to 20°C-ish), the sodium line did not appear. We expect this to be from some sodium impurities present in the Pyrex discharge. Upon a sharp increase in the discharge tube temperature, the Pyrex tube begins to break down and release material, with sodium emitting the Na-D line. If strong enough, the presence of the Na-D line is observable with the naked eye, since the plasma fluorescence starts turning orange instead of the bright pink colour. The presence of the Na-D line is regarded as an indicator of damage to the discharge tube.

Nitrogen emissions

It was observed that when the discharge gets too cold, especially before ignition of the plasma, a lot of molecular nitrogen bands and lines were observed in the spectrum, sometimes completely burying the hydrogen Balmer emission in them. This is shown in Fig. A.2. Sometimes, just turning the cooling off momentarily, we could retrieve the hydrogen spectrum and get rid of the nitrogen lines. We expect this because when the discharge gets too cold, the residual N_2 in the chamber accumulates at the coldest point, causing the discharge to contain a large amount of N_2 . This suspected "cryo-cooling" can sometimes be reversed by turning off the cooling, flushing the chamber with a lot of hydrogen, and pumping it out. The Nitrogen spectrum was also sometimes observed when the discharge was ignited immediately after the discharge tube was changed. Presence of nitrogen lines makes the plasma appear a dull whitish pink colour instead of the normal bright magenta.

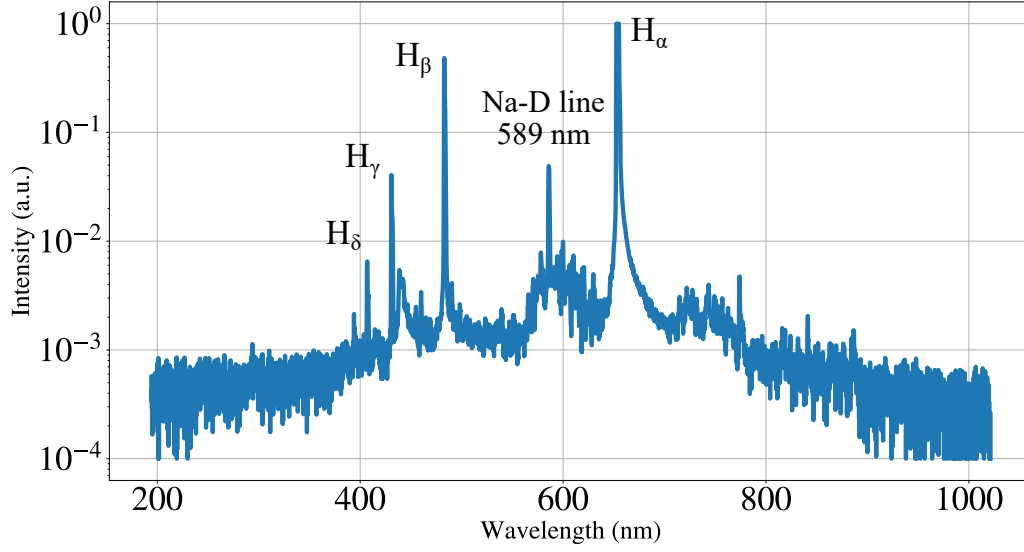


Figure A.1: The Na-D line at 589 nm observed in the plasma emission spectrum when the discharge tube temperature rose very suddenly, probably from the impurities of sodium in the Pyrex tube.

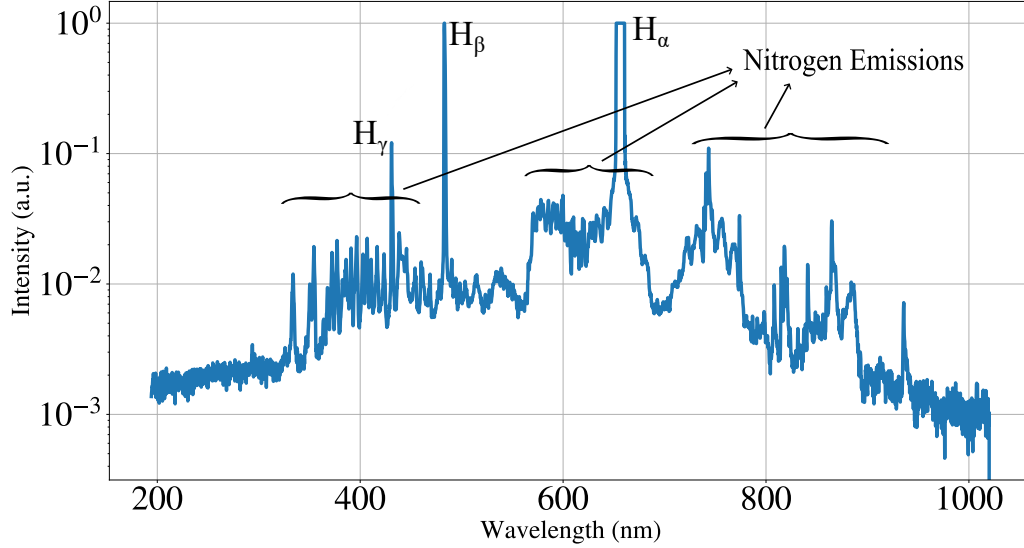


Figure A.2: Nitrogen emission bands observed in the plasma emission spectrum, sometimes even crowding the hydrogen signals, as can be seen from the distorted Fulcher band in the spectrum. These mainly arise at very low discharge temperatures or in an iteration right after opening the chamber.

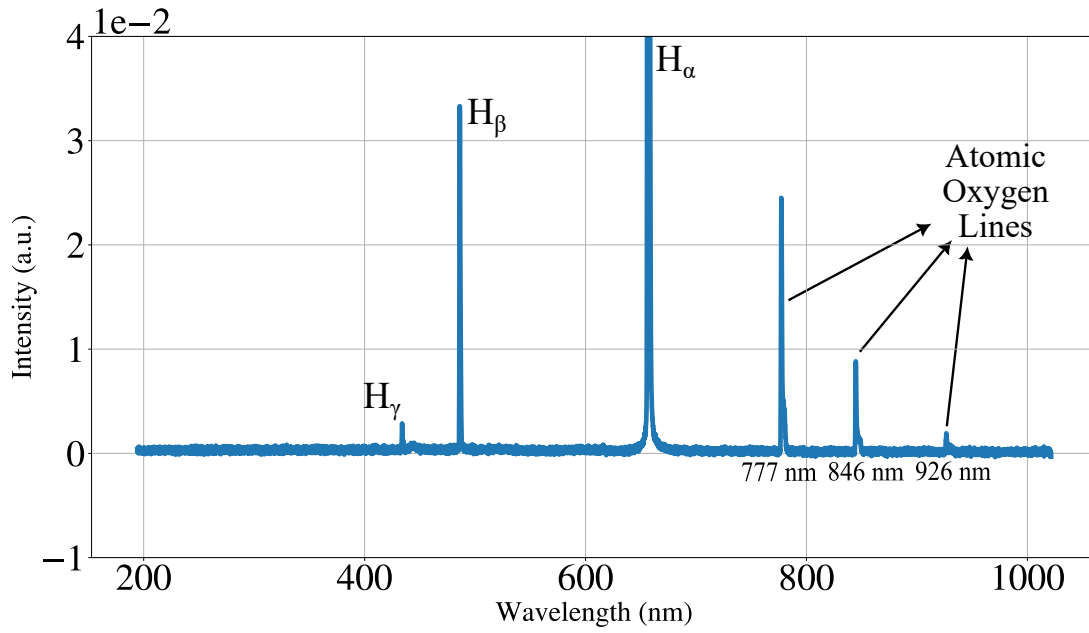


Figure A.3: 3 Atomic oxygen lines at 777 nm, 846 nm, and 926 nm, observed in the discharge spectrum, probably indicating leakage of some air in the discharge.

Oxygen Emission lines

In some discharge operations, 3 oxygen atomic lines at 777 nm, 846 nm, and 926 nm were observed. These lines were initially observed in the 1S-3S hydrogen experiment during discharge, when the discharge tube temperature was stabilised at a high value, and were seen to grow as the temperature was raised further. Additionally, in the 2S-6P hydrogen experiment's discharge, after properly pumping out the entire discharge before ignition, the oxygen lines were absent from the spectra. But in some iterations of the discharge operation, pumping the discharge out did not remove the oxygen lines, likely due to air leaking into the discharge from an untight vacuum compression fittings between the tube and the vacuum system.

Argon cooling

In an attempt to make a reliable cooling and temperature measurement setup for the discharge, Argon was tried in place of Nitrogen for cooling the discharge. Argon, having a lower volumetric heat capacity than nitrogen, thermalised with the discharge faster, and provided a better estimate of the discharge tube temperature. That is why controlling the flow of Argon provided much better control over the discharge tube's temperature and could reproducibly heat or cool the tube. But the much lower heat capacity also meant that the temperature was highly susceptible to small changes in flow and plasma heat and hence was hard to stabilise. Additionally, the boiling point of argon (87.3 K) is higher than that of nitrogen (77.4 K), causing argon to liquefy when passed through

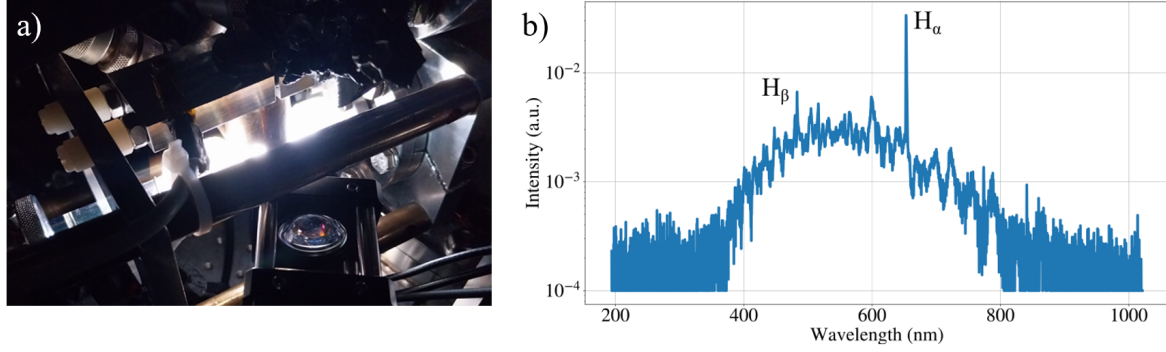


Figure A.4: a) White glowing plasma observed in the discharge. b) Spectrum of the white glow, a broad pedestal below the hydrogen signature emissions.

liquid nitrogen and essentially preventing its use for cooling.

White Glowing Plasma

During certain discharge operations, the plasma was observed glowing white. The white glowing plasma and the spectrum of the glow are shown in Fig. A.4. This was observed either at the start, when ignited, after which the hydrogen spectrum slowly grew above the broad white spectrum, or when the hydrogen flow was reduced in the discharge or pumped out, after which the hydrogen spectrum slowly disappeared, and a white pedestal appeared underneath sustaining the plasma without any hydrogen flow. The source of this white glow is unknown. An effort to fit the Planck radiation function to the spectrum was unsuccessful.

A.2 General Procedure of Operating the Discharge

For optimal performance of the discharge, it was observed that the following step-by-step procedure to turn it on worked the best:

- With the bottle closed, open all valves and pump out all residual hydrogen in the tubings, to be able to attain stability in the flow while operation.
- Start the H_2 , with the defined flow, so as not to have any nitrogen in the discharge.
- Apply MW Power to the cavity.
- Start cooling N_2 for the discharge, and immerse the cooling apparatus in the liquid nitrogen canister.
- Without much delay after the cooling has started, ignite the discharge with the Tesla coil. To ignite, the "with plasma" cavity resonance must be close enough to the applied frequency; hence, the cavity must be tuned to approach this resonance.

Appendix A Hydrogen Discharge Operational Observations

Note that this cannot be seen in the reflected power when there is no plasma. But as we get closer to the "with plasma" resonance, we start seeing an unsustained pinkish glow in the discharge, which lasts only for the duration of the spark.

- After igniting, match the impedance of the cavity with the tuning stubs by monitoring the reflected power from the cavity.
- Control the cooling air flow to stabilise the temperature of the discharge

Appendix B

Supplementary Molecular and Plasma Physics Concepts

B.1 Energy Levels of a Diatomic Molecule

Energy of any state of a diatomic molecule can be split into 3 separate contributions: namely, electronic (E_{el}), rotational ($G(v)$) and vibrational energy ($F_v(J)$),

$$T_{el} = E_{el}(p) + G(v) + F_v(J) \quad (\text{B.1})$$

where p , v and J are the principal (electronic), rotational and vibrational quantum numbers, respectively. T_{el} signifies the energy minimum (the equilibrium distance) of the potential curve of the bound electronic state, and the vibrational and rotational terms are fixed by the internuclear distance from the minimum of the curve. The vibrational term is calculated by considering a harmonic oscillator with anharmonic corrections:

$$G(v) = \hbar[w_e(v + 1/2) - x_e w_e(v + 1/2)^2 + y_e w_e(v + 1/2)^3 - \dots]; v = 0, 1, 2, 3.. \quad (\text{B.2})$$

First term is the harmonic term with w_e as the eigenfrequency of the oscillator in state p , and the further terms are anharmonic corrections with x_e and y_e as the anharmonic constants.

- Anharmonic corrections reduce the energy gap as v rises.
- Vibrational potential curves are asymmetric, the mean internuclear distance increases with v , and the curves flatten out.
- For very high v states, the internuclear distance is large, and at one point the atomic H potential energy curves cross the molecular one; hence, vibrationally excited molecule dissociates into $\text{H}(1\text{S}) + \text{H}(2\text{S})$: this is called predissociation. This is why the number of vibrational states in an electronic state is finite, unlike electronic states in an atom.

The rotational energy term comes from assuming a non-rigid vibrating rotator (excluding the influence of vibration):

$$F(J) = hc[B_e J(J + 1) - D_e J^2(J + 1)^2 \pm \dots]; J = 0, 1, 2.. \quad (\text{B.3})$$

where B_e is the rotational constant and D_e is the centrifugal distortion constant, which depend on the electronic state. These are modified as follows to include the effect of vibration:

$$B_v = B_e - \alpha_e(v + 1/2) + \dots \quad (\text{B.4})$$

$$D_v = D_e + \beta_e(v + 1/2) + \dots \quad (\text{B.5})$$

where α_e and β_e are molecular constants. These energy formulations were used to calculate all Fulcher transition wavelengths and to determine the Boltzmann population distributions defined by energy levels. All molecular constants are taken from the NIST database [47], and are listed below. Derivations of these energy formulations can be found in [22].

State	T_e (cm^{-1})
$X^1\Sigma_g^+$	0
$d^3\Pi_u$	112700
$a^3\Sigma_g^+$	95936

Table B.1: Minimum Electronic energy for states of molecular hydrogen

State	w_e (cm^{-1})	w_ex_e (cm^{-1})	w_ey_e (cm^{-1})
$X^1\Sigma_g^+$	4401.21	121.33	0
$d^3\Pi_u$	2371.58	66.27	0.88
$a^3\Sigma_g^+$	2664.83	71.65	0.92

Table B.2: Vibrational energy constants for states of molecular hydrogen

State	B_e (cm^{-1})	α_e (cm^{-1})	γ_e (cm^{-1})	D_e (cm^{-1})	β_e (cm^{-1})
$X^1\Sigma_g^+$	60.853	3.062	0	0.0471	0
$d^3\Pi_u$ ($v = 0$)	30.364	1.545	0	0.0191	0
$d^3\Pi_u$ ($v > 0$)	30.364	1.545	0	0	0
$a^3\Sigma_g^+$ ($v = 0$)	34.216	1.671	0	0.0216	0
$a^3\Sigma_g^+$ ($v > 0$)	34.216	1.671	0	0	0

Table B.3: Rotational energy constants for states of molecular hydrogen

B.2 Channels in Yacora model for atomic hydrogen

Channel Name	Reactions included
Effective excitation channel	Spontaneous radiative transition $H(q) \xrightarrow{A(q,p)} H(p < q) + h\nu$
	Electron impact excitation $H(q) + e^- \xrightarrow{X_{qp}} H(p > q) + e^-$
	Electron impact de-excitation $H(q) + e^- \xrightarrow{X_{pq}} H(p < q) + e^-$
Recombination channel	Radiative recombination $H^+ + e^- \xrightarrow{\beta(p)} H(p) + h\nu$
	Three-body recombination $H^+ + 2e^- \xrightarrow{\alpha(p)} H(p) + e^-$
Dissociative recombination channel	$H_2^+ + e^- \rightarrow H(p) + H(1)$
	$H_3^+ + e^- \rightarrow H(p) + H_2$
Dissociative excitation channel	$H_2 + e^- \rightarrow H(p) + H(1) + e^-$
Mutual neutralisation channel	$H_x^+ + H^- \rightarrow H(p) + H(1)$
Electron impact ionisation	$H(q) + e^- \xrightarrow{S_p} H^+ + 2e^-$

Table B.4: Channels in the Yacora model for Atomic Hydrogen [27]

B.3 Franck Condon Factors

For optically allowed transitions, the vibrationally resolved transition probabilities $A_{v'v''}^{p'p''}$ are proportional to the dipole matrix element,

$$A_{v'v''}^{p'p''} \propto \left| \int_0^\infty \psi_{v'}^{*p'}(r) D_{el}(r_{el}) \psi_{v''}^{p''}(r) dr \right|^2 \quad (\text{B.6})$$

where, $D_{el}(r_{el})$ is the transition dipole moment. Here, the electronic transition dipole moment depends only on the electronic coordinates, while the wavefunctions depend on both electronic and nuclear coordinates. Under the Born-Oppenheimer Approximation, because of mass ratios, electronic transitions, occurring on much faster timescales, can be considered separate from nuclear motion. Building upon this, in the Franck-Condon Approximation, we can consider the nuclear coordinates to be fixed when an

electronic transition occurs. This essentially means that with the electronic transition probability, the change of vibrational state is just proportional to the overlap of the vibrational wavefunctions of the states involved, and hence transitions with the most overlap of vibrational states are preferred. Mathematically, this means that we can separate the electronic and nuclear coordinates in the above Eq. B.6 and hence essentially the electronic wavefunction and vibrational wavefunctions. The overlap integral of the vibrational wavefunction comes out of the electronic coordinates integral, which are called the Franck-Condon factors.

$$q_{v''v'}^{p''p'} = \left| \int \psi_{v''}^{p''}(r_{nuclear}) \psi_{v'}^{p'}(r_{nuclear}) dr_{nuclear} \right|^2 \quad (\text{B.7})$$

For electronic transition with two states of similar equilibrium nuclear distance, the transition conserves the vibrational population distribution and hence the Franck-Condon matrix is diagonal, e.g, for $d^3\Pi_u$ to $a^3\Sigma_g^+$ spontaneous emission (the Fulcher transition). But for states with different equilibrium nuclear distance, the Franck-Condon matrix is non-diagonal, which means that in the electronic transition, vibrational population is redistributed. This is the case for the $X^1\Sigma_g^+$ to $d^3\Pi_u$ electron impact excitation, and hence we carry out a population distribution projection as done in 2.4.1. The Franck Condon factors used for this projection were taken from [31] and are listed below. A heat plot is also shown to represent the non-diagonal nature of the matrix. More information on the Franck Condon approximation can be found in [22] as well as on the projection used can be found in [32].

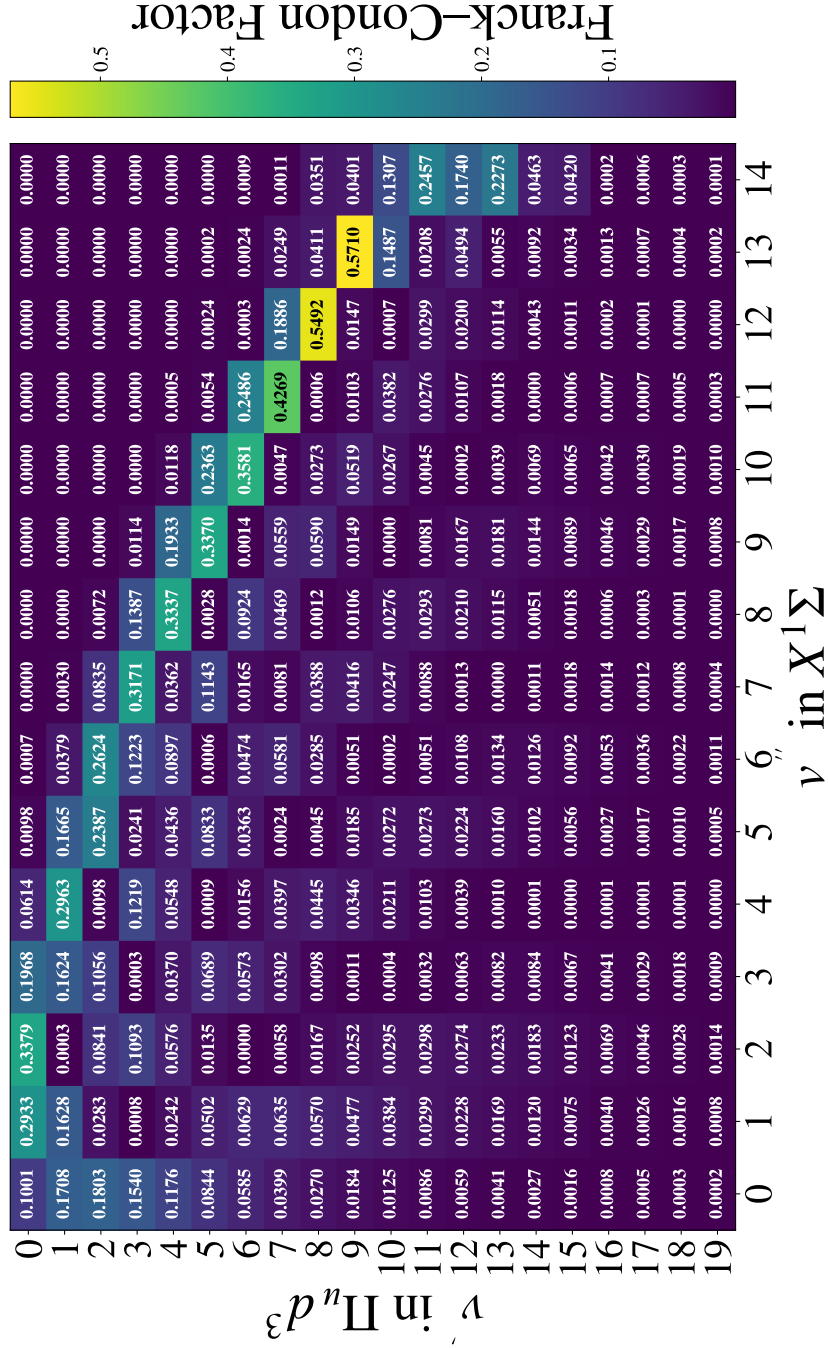


Figure B.1: Franck Condon Factors for the $X^1\Sigma_g^+ \rightarrow d^3\Pi_u$ transition in molecular hydrogen, used for vibrational population projection in Sec. 2.4.1, taken from [31].

v' / v''	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
0	0.1001	0.2933	0.3379	0.1968	0.0614	0.0098	0.0007	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
1	0.1708	0.1628	0.0003	0.1624	0.2963	0.1665	0.0379	0.0030	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2	0.1803	0.0283	0.0841	0.1056	0.0098	0.2387	0.2624	0.0835	0.0072	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
3	0.1540	0.0008	0.1093	0.0003	0.1219	0.0241	0.1223	0.3171	0.1387	0.0114	0.0000	0.0000	0.0000	0.0000	0.0000
4	0.1176	0.0242	0.0576	0.0370	0.0548	0.0436	0.0897	0.0362	0.3337	0.1933	0.0118	0.0005	0.0000	0.0000	0.0000
5	0.0844	0.0502	0.0135	0.0689	0.0009	0.0833	0.0006	0.1143	0.0028	0.3370	0.2363	0.0054	0.0024	0.0002	0.0000
6	0.0585	0.0629	0.0000	0.0573	0.0156	0.0363	0.0474	0.0165	0.0924	0.0014	0.3581	0.2486	0.0003	0.0024	0.0009
7	0.0399	0.0635	0.0058	0.0302	0.0397	0.0024	0.0581	0.0081	0.0469	0.0559	0.0047	0.4269	0.1886	0.0249	0.0011
8	0.0270	0.0570	0.0167	0.0098	0.0445	0.0045	0.0285	0.0388	0.0012	0.0590	0.0273	0.0006	0.5492	0.0411	0.0351
9	0.0184	0.0477	0.0252	0.0011	0.0346	0.0185	0.0051	0.0416	0.0106	0.0149	0.0519	0.0103	0.0147	0.5710	0.0401
10	0.0125	0.0384	0.0295	0.0004	0.0211	0.0272	0.0002	0.0247	0.0276	0.0000	0.0267	0.0382	0.0007	0.1487	0.1307
11	0.0086	0.0299	0.0298	0.0032	0.0103	0.0273	0.0051	0.0088	0.0293	0.0081	0.0045	0.0276	0.0299	0.0208	0.2457
12	0.0059	0.0228	0.0274	0.0063	0.0039	0.0224	0.0108	0.0013	0.0210	0.0167	0.0002	0.0107	0.0200	0.0494	0.1740
13	0.0041	0.0169	0.0233	0.0082	0.0010	0.0160	0.0134	0.0000	0.0115	0.0181	0.0039	0.0018	0.0114	0.0055	0.2273
14	0.0027	0.0120	0.0183	0.0084	0.0001	0.0102	0.0126	0.0011	0.0051	0.0144	0.0069	0.0000	0.0043	0.0092	0.0463
15	0.0016	0.0075	0.0123	0.0067	0.0000	0.0056	0.0092	0.0018	0.0018	0.0089	0.0065	0.0006	0.0011	0.0034	0.0420
16	0.0008	0.0040	0.0069	0.0041	0.0001	0.0027	0.0053	0.0014	0.0006	0.0046	0.0042	0.0007	0.0002	0.0013	0.0002
17	0.0005	0.0026	0.0046	0.0029	0.0001	0.0017	0.0036	0.0012	0.0003	0.0029	0.0030	0.0007	0.0001	0.0007	0.0006
18	0.0003	0.0016	0.0028	0.0018	0.0001	0.0010	0.0022	0.0008	0.0001	0.0017	0.0019	0.0005	0.0000	0.0004	0.0003
19	0.0002	0.0008	0.0014	0.0009	0.0000	0.0005	0.0011	0.0004	0.0000	0.0008	0.0010	0.0003	0.0000	0.0002	0.0001

 Table B.5: Franck–Condon matrix for $X^1\Sigma_g^+(v'') \rightarrow \Pi_u d^3(v')$.

Appendix C

Spectrometer Calibration

In the course of the plasma emission spectroscopy, two spectrometers were used to acquire spectra of the plasma. The following post-processing of the spectra was performed for calibration.

The Thorlabs CCS200 and CCT11 Spectrometers both came with wavelength calibration, but it was observed that the calibration gradually worsened over time during use. Hence, all data taken with the spectrometers was wavelength-calibrated *ex post facto* before analysis by shifting the wavelengths by a "wavelength drift", always calculated relative to the Balmer- γ line.

Additionally, the Spectrometer needed to be intensity-calibrated due to wavelength-dependent grating efficiency and the CCD response of the spectrometer. The Fig. C.1 shows the grating efficiency of the CCS200 spectrometer grating and its CCD response obtained from the manufacturer. Hence, the recorded spectra were intensity-corrected in the analysis to exclude the effects of these two dependencies. This correction showed that the Balmer- γ intensity was slightly underestimated before correction and, in turn, the degree of dissociation. A similar calibration was also done with the grating efficiency of the CCT11 spectrometer.

Additionally, for extracting the point spread function of the spectrometer, the spectrum of Coherent Verdi V-18 was recorded with both spectrometers, and a Gaussian was fitted to them. Fig. C.3 shows the recorded spectra and fitted Gaussians affirming the assumption of a Gaussian point spread function in the intensity fitting procedure in Sec. 2.5.

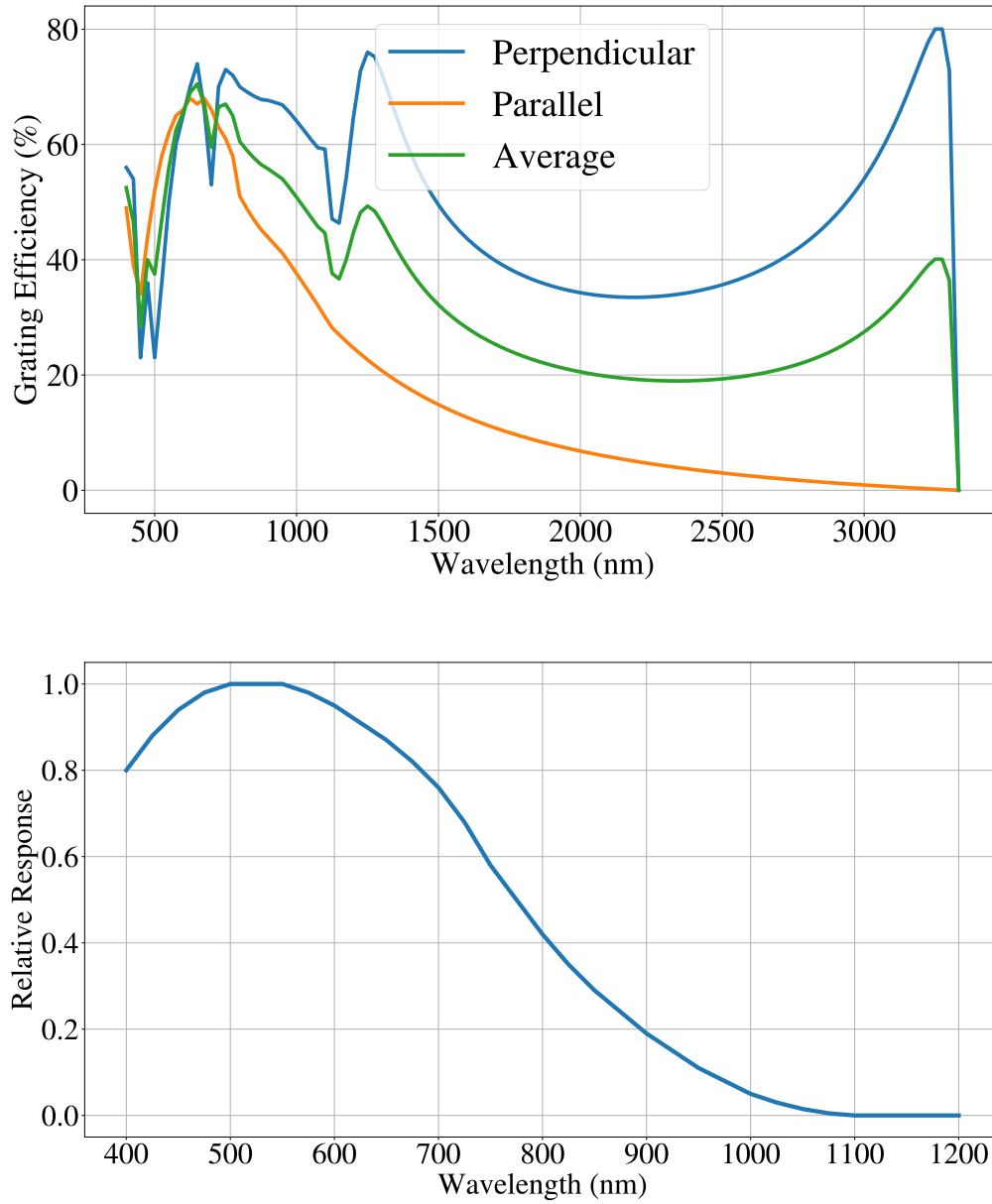


Figure C.1: a) Grating efficiency of the CCS200 Spectrometer with wavelengths. b) Wavelength-dependent spectral response of the CCD of the CCS200 spectrometer.

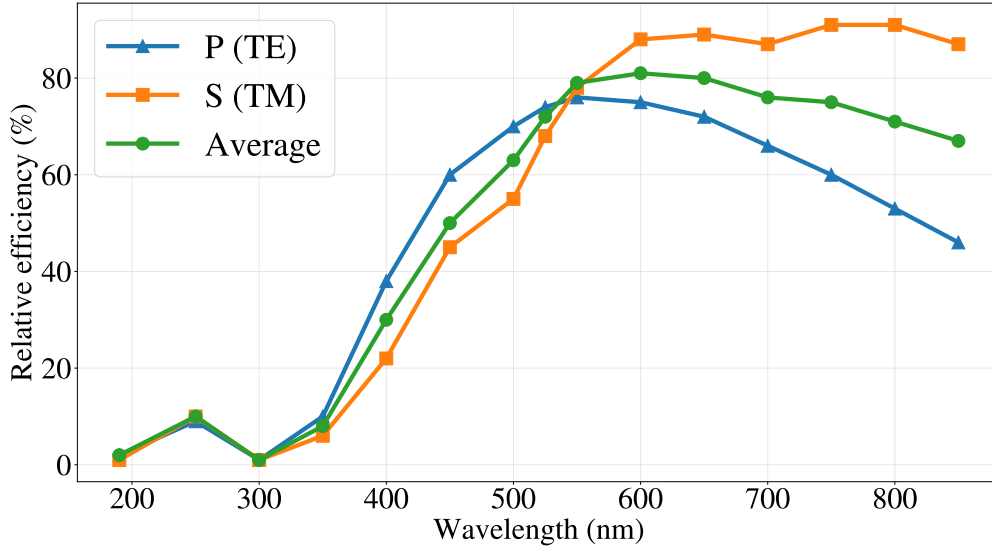


Figure C.2: Grating efficiency of the CCT11 Spectrometer with wavelengths.

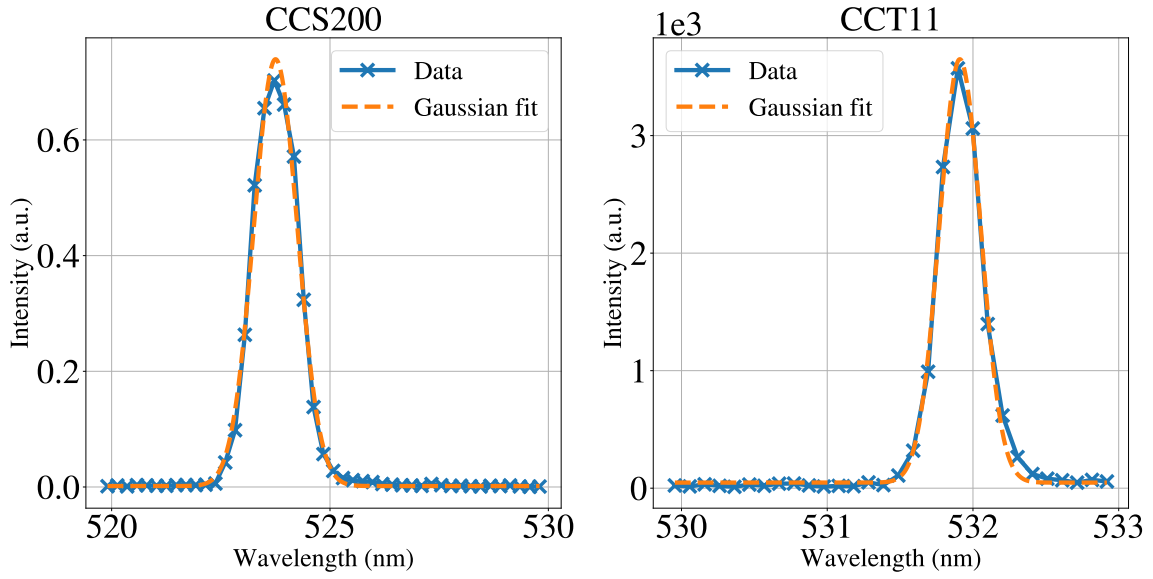


Figure C.3: Spectrum of output of Coherent Verdi V-18 at 532 nm with a) Thorlabs CCS200 Spectrometer, and b) Thorlabs CCT11 Spectrometer and a Gaussian fit to both, in an attempt to find the point spread function of the spectrometers.

Appendix D

Operational Parameters of the Discharge and the Wire Detector

D.1 Hydrogen Discharge parameters

Parameter	Symbol	Value	Unit
Hydrogen flow	f_{H_2}	0.37	mln/min
Discharge Pressure	$P_{discharge}$	$\approx 1\text{-}3$	mbar
Microwave Power applied		30-40	W
Microwave Frequency		≈ 2.45	GHz
Orifice size		200-400	μm
Discharge tube temperature	$T_{discharge}$	-30 to -10 ¹	°C
Vibrational Temperature of ground state of H_2	$T_{vib}(X^1)$	≈ 11000 ²	K
Rotational Temperature of $d^3\Pi_u$	$T_{rot}(v = 1)$	$\approx 300\text{-}500$ ³	K
Neutral gas temperature	$T_{gas} = T_H = T_{H_2}$	≈ 800 ⁴	K

Table D.1: Parameters of Hydrogen Discharge

¹From [19] and analysis in this thesis.

²This is the value from the P, Q, R fit carried out in Sec. 2.5.2. Very consistently, the fits optimized to a value of $T_{vib}(X^1)$ 11,000 to 12,000 K, which did not show any consistent change while sweeping experimental parameters. This value eventually is only relevant in determining the degree of dissociation within $\pm 5\%$, see Sec. 2.6.3.

³The general optimised value for the first vibrational band from fits. The value did vary while sweeping the parameters, but no consistent dependence was observed; hence, it was probably a result of uncertainty in the fitting.

⁴From projecting an average value of $T_{rot}^{d^3}(v = 1) \approx 400\text{K}$ using Eq. 2.27

D.2 Wire Detector Parameters

Parameter	Symbol	Typical Value	Unit
Wire length	l_1	36	mm
	l_2	37.7	mm
Wire Diameter	d_1	25	μm
	d_2	5	μm
Base Resistance	R_1	4.10696	Ω
	R_2	107.523	Ω
Distance from nozzle to wire	t	5	mm
Probe Current	I_{probe}	1	mA
Degree of dissociation	α	5	%
Opening angle of the beam	2θ	60°	$^\circ$
Transversal length of the beam ⁵	l_{eff}	5.77	mm
Flux of beam at the wire position ⁶	$\phi_{H_2}^{undissociated}$	6.33×10^{21}	molecules/ $m^2 \cdot \text{sec}$
Number of molecules hitting the wire	$\dot{N}_{H_2}(d_1)$	9.14×10^{14}	molecules/sec
	$\dot{N}_{H_2}(d_2)$	1.82×10^{14}	molecules/sec

Table D.2: Parameters used for the Wire detector

⁶True value considering an opening angle of 60°

Appendix E

List of Physical Constants

Constant	Symbol	Value	Unit	Source
H Recombination energy	$2E_{rec}$	4.46	e.V./ molecule	[48]
		7.239×10^{-20}	J/molecule	
Density of W at $20^\circ C$	ρ_W	19254	kg/m^3	[49]
Thermal Conductivity of W	k_W	173	$W/m \cdot K$	[50]
Resistivity of W	$\rho_{R,W}$	5.6×10^{-8}	$\Omega \cdot m$	[51]
Temp. Coeff. of Resistivity	$\alpha_{\rho,W}$	4.5×10^{-3}	/K	[51]
Emissivity of W filament	ϵ_W	0.32		[46]
Emissivity of Cu @ 20°	ϵ_{Cu}	0.07		[46]
H Recombination Coefficient on W	η_{rec}	0.1		[45]
Boltzmann Constant	k_B	1.38×10^{-23}	$m^2 kgs^{-2} K^{-1}$	
Stephan-Boltzmann Constant	σ	5.67×10^{-8}	$Wm^{-2} K^{-4}$	
Speed of light	c	5.67×10^{-8}	$Wm^{-2} K^{-4}$	
Planck's Constant	h	6.626×10^{-34}	$J \cdot sec$	
Avogadro's Number	N_A	6.022×10^{23}	mol^{-1}	

Table E.1: Physical Constants used for the wire detector calculations

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Declaration of Authorship

I hereby declare that this thesis is my own work. All direct or in direct sources referred to in the text are cited in the references. This thesis was not previously presented to another examination board and has not been published.

Place: Garching, Munich

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Date: April 30, 2026

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